P802.1DD/Draft 1.1

March 11, 2025

Draft Standard for Local and metropolitan area networks—

Resource Allocation Protocol

Unapproved draft, prepared by the

Time-Sensitive Networking (TSN) Task Group of IEEE 802.1

Sponsored by the

LAN/MAN Standards Committee
of the
IEEE Computer Society

This and the following cover pages are not part of the draft. They provide revision and other information for IEEE 802.1 Working Group members and participants in the IEEE Standards Association ballot process, and will be updated as convenient. New participants: Please read these cover pages, they contain information that should help you contribute effectively to this standards development project. Blank pages allow for the addition of cross-references to changed text without forcing renumbering of all pages in the draft. Pages are numbered from 1 (including cover pages) for the convenience of reviewers whose PDF viewers do not easily accommodate different numbering sequences. Pages will of course be renumbered prior to publication.

The text proper of this draft begins with the Title page.

Editors' Foreword

Throughout this document any notes with a Cyan background (as in this paragraph) are temporary, inserted by the Editors for a variety of purposes. They will be removed prior to SA Ballot and are not part of the normative text. These cover pages will be edited, but will be retained so that information for the stages of SA ballot can be included (including, if appropriate, cross-references to text changed in the course of SA balloting). To avoid changes to page numbering between final WG ballot and initial SA Ballot, the number of pages in this set of cover pages should remain unchanged, with pages intentionally left blank marked as such. The records of participants in the development of the standard will be added at an appropriate time.

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All participants in the standardization activities of IEEE 802.1 should be aware of the Working Group Policies and Procedures, and the fact that they have obligations under the IEEE Patent Policy, the IEEE Standards Association (SA) Copyright Policy, and the IEEE SA Participation Policy. For information on these policies see 1.ieee802.org/rules/ and the slides presented at the beginning of each of our Working Group and Task Group meeting.

As part of our IEEE 802® process, the text of the PAR (Project Authorization Request) and CSD (Criteria for Standards Development) of each project is reviewed regularly to ensure their continued validity. The PAR is summarized in these cover pages and links are provided to the full text of both PAR and CSD. A vote of "Approve" on this draft is also an affirmation that the PAR and CSD for this project are still valid.

Comments on this draft are encouraged. NOTE: All issues related to IEEE standards presentation style, formatting, spelling, etc. are routinely handled between the 802.1 Editor and the IEEE Staff Editors prior to publication, after balloting and the process of achieving agreement on the technical content of the standard is complete. Readers are urged to devote their valuable time and energy only to comments that materially affect either the technical content of the document or the clarity of that technical content. Comments should not simply state what is wrong, but also what might be done to fix the problem.

Full participation in the work of IEEE 802.1 requires attendance at IEEE 802 meetings. Information on 802.1 activities, working papers, and email distribution lists etc. can be found on the 802.1 Website:

http://ieee802.org/1/

Use of the email distribution list is not presently restricted to 802.1 members, and the working group has a policy of considering comments from all who are interested and willing to contribute to the development of the draft. Individuals not attending meetings have helped to identify sources of misunderstanding and ambiguity in past projects. The email lists exist primarily to allow the members of the working group to develop standards, and are not a general forum. All contributors to the work of 802.1 should familiarize themselves with the IEEE patent policy and anyone using the email distribution list will be assumed to have done so. Information can be found at http://standards.ieee.org/db/patents/

Comments on this draft may be sent to the 802.1 email exploder, to the Editor, or to the Chairs of the 802.1 Working Group and Time-Sensitive Networking (TSN) Task Group.

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All participants in IEEE standards development have responsibilities under the IEEE patent policy and should familiarize themselves with that policy, see
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As part of our IEEE 802 process, the text of the PAR and CSD (Criteria for Standards Development, formerly referred to as the 5 Criteria or 5C's) is reviewed on a regular basis in order to ensure their continued validity. A vote of “Approve” on this draft is also an affirmation by the balloter that the PAR is still valid.

Draft development

During the early stages of draft development, 802.1 editors have a responsibility to attempt to craft technically coherent drafts from the resolutions of ballot comments and from the other discussions that take place in the working group meetings. Preparation of drafts often exposes inconsistencies in editor’s instructions or exposes the need to make choices between approaches that were not fully apparent in the meeting. Choices and requests by the editors’ for contributions on specific issues will be found in the editors’ Introduction to the current draft and at appropriate points in the draft.

The ballot comments received on each draft, and the editors’ proposed and final disposition of comments on working group drafts, are part of the audit trail of the development of the standard and are available, along with all the revisions of the draft on the 802.1 website (for address see above).

During the early stages of draft development the proposed text can be moved around a great deal, and even minor rearrangement can lead to a lot of ‘change’, not all of which is noteworthy from the point of the reviewer, so the use of automatic change bars is not very effective. In early drafts change bars may be omitted or applied manually, with a view to drawing the readers attention to the most significant areas of change. Readers interested in viewing every change are encouraged to use Adobe Acrobat to compare the document with their selected prior draft. Note that the FrameMaker change bar feature is useless when it comes to indicating changes to Figures.

iPAR (Project Authorization Request) and CSD

Extracts from the PAR, as approved by IEEE NesCom 26th September 2024:

https://development.standards.ieee.org/myproject-web/public/view.html#pardetail/11709

and the CSD (Criteria for Standards Development):

https://www.ieee802.org/1/files/public/docs2024/dd-CSD-0724-v01.pdf

PAR Scope:

This standard specifies protocols, procedures, and managed objects for resource allocation in bridged local area networks for dynamic creation and maintenance of data streams. This standard supports control signaling through data paths and through separate control paths in support of centralized control. This standard makes provisions for backward compatibility with the Stream Reservation Protocol specified in IEEE Std 802.1Q.

PAR Need for the Project:

Successful deployment of current protocol has revealed the need to accommodate a larger number of streams and resource allocation on simultaneously available redundant paths than can be accommodated by the current in-band signaling mechanisms.

CSD Broad market potential:

The original version of the Stream Reservation Protocol (SRP) specified in IEEE Std 802.1Q has been successfully and widely accepted by the professional audio/video, consumer, and automotive markets as an essential tool to realize automatic stream setup with dynamic resource allocation. The success of SRP has expanded the requirements on that protocol beyond that capability.

Individuals affiliated with multiple vendors and users for industrial automation, professional audio/video, automotive and other systems requiring a protocol to signal the resource reservation along the end-to-end paths of streams for time-sensitive applications will participate in the development of the project.

CSD Distinct Identity:

No existing IEEE 802 standard provides dynamic resource allocation that supports the required performance and capabilities.

This standard expands bridged network capabilities in the area of resource allocation. The rate of addition of other new capabilities to existing IEEE 802.1 standards means that it is more practical to specify expanded resource allocation capabilities in a separate standard focused on that topic. The proposed project replaces the previous IEEE P802.1Qdd Draft Standard for Local and Metropolitan Area Networks — Bridges and Bridged Networks — Amendment: Resource Allocation Protocol.

CSD Technical Feasibility:

The proposed resource allocation methods are similar in principle to those of the Stream Reservation Protocol (SRP) specified in IEEE Std 802.1Q.

There is a considerable body of experience in supplying data streams with guarantees for quality of service parameters such as latency, latency variation, or bandwidth. Mechanisms needed for this project are widely used by other protocols already, e.g., IEEE 802.1Q SRP for use in bridged networks and the Resource Reservation Protocol (RSVP, IETF RFC 2205, and IETF RFC 2750) for routers and hosts that use the Internet Protocol.

CSD Economic Feasibility:

The well-established balance between infrastructure and attached stations will not be changed by the proposed standard.

All the above cost factors are well-known from the existing resource allocation protocols and registration database distribution methods.

Introduction to the current draft

This introduction is not part of the draft, and will be revised for SA ballot. A set of cover pages will be retained for use during SA ballot.

D1.1

D1.0 was not used for TG ballot but as editor's draft.- D1.1 is an editor's draft as well with following changes:

Fixes some inconsistencies of D1.0

Adds backward compatibility (RAP/MSRP Gateway) skeleton structure.

Restructures Token Bucket TSpec clause.

D1.0

D1.0 is the first draft of P802.1DD and used for the first Task Group ballot.

Due to project change (P802.1DD replacing P802.1Qdd), this draft has been created based on the contents as well as the comment resolution results of P802.1Qdd/D0.9 but using a template for stand-alone IEEE 802.1 standards. In addition to the changes required to fit the new format, such as rearrangement of clauses and reworked references to other standards, the major changes from P802.1Qdd D0.9 include the following:

- Clause 5.Conformance reworked to define requirements for three types of RAP instance and ten types of RAP systems.
- Clause 6.1RAP overview added to give a brief overview of RAP.
- In Clause 6.3.1RAP architecture, added text and figures for describing the architecture of RAP/MSRP Gateway, an example of RAP Proxy systems, and RAP control of Bridging and FRER functions.
- Clause 6.3.7Traffic specification usage rules and types added to define TSpec types and usage rules.
- In all the state machines, replaced the use of primitives in conditions of state machine transitions with the use of event pointer variables as described in Clause 6.2.4Event pointer variables.
- Clause 8.YANG data model for RAP updated to reflect the changes made in Clause 7.Resource Allocation Protocol (RAP) management.

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Abstract: This standard specifies protocols, procedures, and managed objects for resource allocation in bridged local area networks for dynamic creation and maintenance of data streams. This standard supports control signaling through data paths and/or through separate control paths in support of centralized control. This standard makes provisions for backward compatibility with the Stream Reservation Protocol specified in IEEE Std 802.1Q.

Keywords: Bridged Network, Frame Replication and Elimination for Reliability, FRER, IEEE 802.1CB™, IEEE 802.1CS™, IEEE 802.1DD™, IEEE 802.1Q™, Link-local Registration Protocol, LRP, resource allocation, Resource Allocation Protocol, RAP, Time-Sensitive Networking, TSN, Virtual Bridged Local Area Network, VLAN Bridge

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Participants

<<The following lists will be updated in the usual way prior to publication>>

At the time this standard was completed, the IEEE 802.1 working group had the following membership:

Glenn Parsons, *Chair*
Jessy Rouyer, *Vice Chair*
Janos Farkas, *Time-Sensitive Networking Task Group Chair*
Martin Mittelberger, *Editor*

The following members of the individual balloting committee voted on this standard. Balloters may have voted for approval, disapproval, or abstention.

A.N. Other

<<The above lists will be updated in the usual way prior to publication>>

When the IEEE-SA Standards Board approved this standard on <dd> <month> <year>, it had the following membership:

Chair
Vice-Chair
Past Chair
Secretary

*Member Emeritus

<<The above lists will be updated in the usual way prior to publication>>

Introduction

This introduction is not part of IEEE Std 802.1DD-20XX, IEEE Standard for Local and metropolitan area networks—Resource Allocation Protocol

This standard specifies profiles of IEEE 802.1 Time-Sensitive Networking (TSN) and IEEE 802.1 Security standards for aerospace onboard bridged IEEE 802.3 Ethernet networks.

This standard was first published as IEEE Std 802.1DD-20XX.

This standard contains state-of-the-art material. The area covered by this standard is undergoing evolution. Revisions are anticipated within the next few years to clarify existing material, to correct possible errors, and to incorporate new related material. Information on the current revision state of this and other IEEE 802 standards may be obtained from

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1. Overview

The standardization of Ethernet communication technology in IEEE Std 802.3™, specifying transmission over the physical media of individual Local Area Networks (LANs), and in IEEE Std 802.1Q™, specifying Bridges that interconnect IEEE 802® LANs,[1] has facilitated widespread deployment of networks that connect significantly more end stations, with significantly greater bandwidth, and at significantly reduced cost compared to prior technology. All these metrics have been improved by several orders of magnitude—reducing costs through the multi-vendor provision of common components (bridges, end station interfaces, integrated circuit and circuit designs, connectors, and software) for a wide range of network applications.

The use of Ethernet communication technology in networks with high-reliability and deterministic latency requirements is further supported by Time-Sensitive Networking (TSN) provisions in IEEE Std 802.1Q, IEEE Std 802.1AS, IEEE Std 802.1CB, IEEE Std 802.1CS, and the security provisions in IEEE Std 802.1AE and IEEE Std 802.1X. The provisions in these standards can be used in various ways, and include options that address different network requirements and parameters that vary by network and application scale. Network design, time to deploy, and component development, selection, validation, and configuration for a particular network can all benefit from consistent choices, across similar networks and network applications, of the provisions, parameters, and options specified in the relevant standards.

This standard specifies protocols, procedures, and managed objects to support the Resource Allocation Protocol (RAP). RAP is designed as an application of the Link-local Registration Protocol (LRP) specified by IEEE Std 802.1CS to provide automatic and dynamic stream setup with resource allocation for time-sensitive applications in need of bounded latency, zero-congestion loss and high availability and reliability.

1.1 Scope

This standard specifies protocols, procedures, and managed objects for resource allocation in bridged local area networks for dynamic creation and maintenance of data streams. This standard supports control signaling through data paths and through separate control paths in support of centralized control. This standard makes provisions for backward compatibility with the Stream Reservation Protocol specified in IEEE Std 802.1Q.

1.2 Purpose

<No purpose is provided in PAR>

[1] IEEE and IEEE 802 are registered trademarks in the U.S. Patent & Trademark Office, owned by The Institute of Electrical and Electronics Engineers, Incorporated.

1.3 Introduction

To enable decentralized resource allocation for the streams that require using the IEEE 802.1 Time-Sensitive Networking (TSN) features, such as the QoS functions defined by IEEE Std 802.1Q and the seamless redundancy techniques defined by IEEE Std 802.1CB, and to leverage the performance and scalability of the LRP defined by IEEE Std 802.1CS, this standard specifies protocols, procedures, and managed objects for an Resource Allocation Protocol (RAP). RAP provides interoperability with the Multiple Stream Registration Protocol (MSRP) specified in IEEE Std 802.1Q and can also operate in proxy systems to support centralized control. To this end, it

a) Specifies the requirements for RAP implementations declaring conformance to this standard.
b) Defines the principles of RAP operation, including the following.
    1) RAP architecture
    2) RA class and RA advertisement
    3) Talker Announcement and Listener Announcement
    4) Resource allocation constraints
    5) Traffic specification usage rules and types
    6) Stream Rank and reservation importance
    7) Resource allocation for seamless redundancy
    8) Relationship of RAP to MSRP
c) Defines LRP protocol elements for RAP.
d) Defines RAP attributes and TLV encoding
e) Specifies the RAP Endpoint Service Interface (RESI) and the operation of the RAP Endpoint.
f) Specifies the RAP Participant Service Interface (RPSI) and the operation of the RAP Participant.
g) Specifies the operation of the RAP Propagator.
h) Specifies the operation of the RAP/MSRP Converter.
i) Defines managed objects for management of RAP.
j) Defines YANG data models for RAP.
k) Provides examples of resource allocation using RAP.

1.4 Reference conventions

This standard makes frequent references to specific subclauses in several other IEEE 802.1 standards. To avoid repeating the full names of these referenced standards, the following conventions are used:

— A reference to “subclause x.y in IEEE Std 802.1Q-2022” is of the form: “[Q]:x.y”.
— A reference to “subclause x.y in IEEE Std 802.1CS-2020” is of the form: “[CS]:x,y”.
— A reference to “subclause x.y in IEEE Std 802.1CB-2020” is of the form: “[CB]:x,y”.
— A reference to “subclause x.y in IEEE Std 802.1AB-2016” is of the form: “[AB]:x,y”.
— A reference to “subclause x.y in IEEE Std 802.1AC-2016” is of the form: “[AC]:x,y”.
— A reference to “subclause x.y in this standard” is of the form: “x,y”.

2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used, so each referenced document is cited in the text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

IEEE Std 802®, IEEE Standard for Local and metropolitan area networks: Overview and Architecture.[2,3]

IEEE Std 802.1AB™, IEEE Standard for Local and metropolitan area networks—Station and Media Access Control Connectivity Discovery.

IEEE Std 802.1AC™, IEEE Standard for Local and metropolitan area networks—Media Access Control (MAC) Service Definition.

IEEE Std 802.1CB™, IEEE Standard for Local and metropolitan area networks—Frame Replication and Elimination for Reliability.

IEEE Std 802.1CS™, IEEE Standard for Local and metropolitan area networks—Link-local Registration Protocol.

IEEE Std 802.1Q™, IEEE Standard for Local and metropolitan area networks—Bridges and Bridged Networks.

IEEE Std 802.3™, IEEE Standard for Ethernet.

IETF RFC 7950, The YANG 1.1 Data Modeling Language, August 2016.

IETF RFC 8343, A YANG Data Model for Interface Management, March 2018.

[2] IEEE publications are available from The Institute of Electrical and Electronics Engineers (https://standards.ieee.org/).
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3. Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.[4]

This standard makes use of the following term defined in IEEE Std 802:[5]

- end station
- frame
- station

This standard makes use of the terms defined in IEEE Std 802.1CS:

- Controlled system
- Native system
- Portal
- Proxy system
- record
- target port
- local target port
- LRP application
- neighbor target port

This standard makes use of the terms defined in IEEE Std 802.1CB:

- Compound Stream
- Member Stream

This standard makes use of the following terms defined in IEEE Std 802.1Q:[6]

- Bridge
- Bridge Port
- Bridged Network
- Edge Control Protocol (ECP)
- latency
- Listener
- Multicast
- Port
- Port State
- redundant trees
- Stream
- StreamID
- stream reservation (SR) class
- Talker
- time-sensitive stream
- traffic specification (TSpec)
- traffic class
- Virtual Local Area Network (VLAN)
- Virtual Local Area Network (VLAN) Bridge component
- Virtual Local Area Network (VLAN) Bridged Network

[4] *IEEE Standards Dictionary Online* is available at https://dictionary.ieee.org.

[5] This definition is from IEEE Std 802-2014. Notes to definitions are particular to this standard.

[6] These definitions are from IEEE Std 802.1Q-2022. Notes to definitions are particular to this standard.

The following terms are specific to this standard:

FRER-capable RAP Bridge: A RAP Native Bridge or a RAP Controlled Bridge that is FRER-capable.

FRER-capable RAP Listener: A RAP Native end station or a RAP Controlled end station acting as a Listener that is FRER-capable.

FRER-capable RAP Talker: A RAP Native end station or a RAP Controlled end station acting as a Talker that is FRER-capable.

resource allocation (RA) class: A priority of a traffic class or a set of traffic classes whose bandwidth and queue resources can be reserved for streams using the Resource Allocation Protocol (RAP).

Resource Allocation Protocol (RAP): A protocol providing resource reservation for streams.

RAP Bridge Proxy: A RAP Proxy for Bridges.

RAP Controlled Bridge: A Virtual Local Area Network (VLAN) Bridge operating as a Controlled system from a RAP Proxy.

RAP Controlled end station: An end station operating as a Controlled system from a RAP Proxy.

RAP Controlled system: A RAP Controlled end station or a RAP Controlled Bridge.

RAP End instance: A RAP instance residing in a RAP Native end station or offloaded from a RAP Controlled end station to a RAP End Station Proxy.

RAP End Station Proxy: A RAP Proxy for end stations.

RAP instance: An application instance of the Link-local Registration Protocol (LRP) that provides RAP functionality for an end station or Bridge.

RAP Native Bridge: A Virtual Local Area Network (VLAN) Bridge operating as a Native system using the Resource Allocation Protocol (RAP).

RAP Native end station: An end station operating as a Native system using the Resource Allocation Protocol (RAP).

RAP Native system: A RAP Native Bridge or a RAP Native end station.

RAP Proxy: A Proxy system using the Resource Allocation Protocol (RAP).

RAP Relay instance: A RAP instance residing in a RAP Native Bridge or offloaded from a RAP Controlled Bridge to a RAP Bridge Proxy.

RAP/MSRP Gateway: A Virtual Local Area Network (VLAN) Bridge that supports Multiple Stream Registration Protocol (MSRP) on one or more Bridge Ports and supports Resource Allocation Protocol (RAP) on other Ports, and provides functionality for conversion between these two protocols.

RAP/MSRP Gateway instance: A RAP instance residing in a RAP Gateway. for a RAP Gateway. providing translation functionality for conversion between the Multiple Stream Registration Protocol (MSRP) and the Resource Allocation protocol (RAP).

stream: A unidirectional flow of data from one Talker to one or more Listeners that require transmission with a bounded latency.

NOTE—The term “stream”, as used in this standard, is in line with the term “time-sensitive stream” defined in [Q].

4. Abbreviations and acronyms

The following abbreviations and acronyms are used in this standard: [7]

ATS	Asynchronous Traffic Shaping
AV	audio/video
CBS	committed burst size
CFM	Connectivity Fault Management
CID	Company ID [8]
CQF	cyclic queuing and forwarding
CRC	Cyclic Redundancy Check
EISS	Enhanced Internal Sublayer Service
FDB	Filtering Database
FRER	Frame Replication and Elimination for Reliability (IEEE Std 802.1CB)
IP	Internet Protocol
IPv4	Internet Protocol version 4
IPv6	Internet Protocol version 6
ISS	Internal Sublayer Service
LAN	Local Area Network
LLC	Logical Link Control
LLDP	Link Layer Discovery Protocol (IEEE Std 802.1AB)
LLDPDU	Link Layer Discovery Protocol Data Unit
LRP	Link-local Registration Protocol (IEEE Std 802.1CS)
LRP-DS	Link-local Registration Protocol–Database Synchronization
LRP-DT	Link-local Registration Protocol–Data Transport
LRPDU	Link-local Registration Protocol Data Units
MAC	Medium Access Control
MAD	MRP Attribute Declaration
MMRP	Multiple MAC Registration Protocol
MRP	Multiple Registration Protocol
MRTs	Maximally Redundant Trees
MSRP	Multiple Stream Registration Protocol
MVRP	Multiple VLAN Registration Protocol
OUI	organizationally unique Identifier
PCP	Priority Code Point
ISIS-PCR	Intermediate System to Intermediate System with Path Control and Reservation extensions
PICS	Protocol Implementation Conformance Statement
PSFP	Per-Stream Filtering and Policing
QoS	Quality of Service
R-TAG	Redundancy tag
RA	resource allocation
RAP	Resource Allocation Protocol
RESI	RAP Endpoint Service Interface
RPSI	RAP Participant Service Interface
RTID	Resource Allocation Class Template Identifier
SRP	Stream Reservation Protocol
TCP	Transmission Control Protocol

[7] The abbreviations listed include those defined in standards referenced by this standard and used in this standard.

[8] See https://standards.ieee.org/develop/regauth/tut/eui.pdf.

TLV	Type, Length, Value
TSpec	traffic specification
TSN	Time-Sensitive Networking
UNI	User Network Interface
VID	VLAN Identifier
VLAN	Virtual Local Area Network
YANG	Yet Another Next Generation [9]

[9] YANG is best viewed as a name, not an acronym.

5. Conformance

This clause specifies mandatory and optional capabilities provided by conformant implementations of this standard.

5.1 Requirements terminology

For consistency with existing IEEE and IEEE 802.1™ standards, requirements are expressed using the following terminology: [10]

a) ***Shall*** is used for mandatory requirements.
b) ***May*** is used to describe implementation or administrative choices ("may" means "is permitted to," and hence, "may" and "may not" mean precisely the same thing).
c) ***Should*** is used for recommended choices (the behaviors described by "should" and "should not" are both permissible but not equally desirable choices).

Protocol Implementation Conformance Statements (PICS) reflect the occurrences of the words "shall," "may," and "should" within the standard.

The standard avoids needless repetition and apparent duplication of its formal requirements by using ***is***, ***is not***, ***are***, and ***are not*** for definitions and the logical consequences of conformant behavior. Behavior that is permitted but is neither always required nor directly controlled by an implementer or administrator, or whose conformance requirement is detailed elsewhere, is described by ***can***. Behavior that never occurs in a conformant implementation or system of conformant implementations is described by ***cannot***. The word ***allow*** is used as a replacement for the phrase "Support the ability for," and the word ***capability*** means "can be configured to."

5.2 Protocol Conformance Statement (PCS)

A claim of conformance to this standard attests to the implementation of a system or system component specified in this and referenced standards.

The supplier of an implementation that is claimed to conform to this standard shall provide the information necessary to identify both the supplier and the implementation, and shall complete a copy of the relevant PCS proforma provided in this standard together with the Protocol Implementation Conformance Statements (PICS) for the referenced standards, as identified in the PCS.

5.3 Conformant systems, components, and functionality

This standard specifies conformance requirements for the following components:

a) RAP Relay instance (5.4)
b) RAP End instance (5.5)
c) RAP/MSRP Gateway instance (5.6)

and for the following systems that include instances of the above components:

d) RAP Native Bridge (5.7)
e) RAP Controlled Bridge (5.8)
f) RAP Bridge Proxy (5.9)

[10] Originally derived from ISO/IEC style requirements, and consistent with the terminology specified in the ISO/IEC Directives Part 2:2021, Clause 7 (http://www.iec.ch/members_experts/refdocs).

g) FRER-capable RAP Bridge (5.10)
h) RAP Native end station (5.11)
i) RAP Controlled end station (5.12)
j) RAP End Station Proxy (5.13)
k) FRER-capable RAP Talker (5.14)
l) FRER-capable RAP Listener (5.15)
m) RAP/MSRP Gateway (5.16)

5.4 RAP Relay instance requirements

A RAP Relay instance shall comprise one RAP Participant (6.7) for each Port and a single RAP Propagator (6.8).

A RAP Relay instance implementation conformant to this standard shall:

a) Support the RPSI and the RAP Participant state machine as specified in 6.7.
b) Support the RAP Propagator state machine as specified in 6.8.
c) Support the Application Information TLV as specified in 6.4.4.
d) Support the RAP attributes and their TLV encodings as specified in 6.5.
e) Support the RAP management as specified in Clause 7.

5.5 RAP End instance requirements

A RAP End instance shall comprise a single RAP Endpoint (6.6) and a single RAP Participant (6.7).

A RAP End instance implementation conformant to this standard shall:

a) Support the RPSI and the RAP Participant state machine as specified in 6.7.
b) Support the RESI and the RAP Endpoint procedures as specified in 6.6.
c) Support the Application Information TLV as specified in 6.4.4.
d) Support the RAP attributes and TLV encodings as specified in 6.5.
e) Support the RAP management as specified in Clause 7.

5.6 RAP/MSRP Gateway instance requirements

A RAP/MSRP Gateway instance shall comprise one RAP Participant (6.7) for each Port that supports the operation of RAP, one RAP/MSRP Converter (6.9) for each Port that supports the operation of MSRP, and a single RAP Propagator (6.8).

A RAP /MSRP Gateway instance implementation conformant to this standard shall:

a) Support the RPSI and the RAP Participant state machine as specified in 6.7.
b) Support the RAP Propagator state machine as specified in 6.8.
c) Support the RAP/MSRP Converter as specified in 6.9.
d) Support the Application Information TLV as specified in 6.4.4.
e) Support the RAP attributes and TLV encodings as specified in 6.5.
f) Support the RAP management as specified in Clause 7.

5.7 RAP Native Bridge requirements

A RAP Native Bridge implementation conformant to this standard shall:

a) Include a single VLAN Bridge component conformant to [Q]:5.4.
b) Conform to the required and optional behaviors for a Native relay system ([CS]:5.6, [CS]:5.7).
c) Support Edge Control Protocol ([Q]:43) on each Port and associated management entities ([Q]:12.27).
d) Conform to the requirements of [CS]:6.9.1 and 6.4.3 for use of ECP as the LRP-DT data transport mechanism.
e) Include a single RAP Relay instance conformant to the requirements of 5.4.

5.8 RAP Controlled Bridge requirements

A RAP Controlled Bridge implementation conformant to this standard shall:

a) Include a single VLAN Bridge component conformant to [Q]:5.4.
b) Conform to the optional behaviors for a Controlled system ([CS]:5.10).

5.9 RAP Bridge Proxy requirements

A RAP Bridge Proxy implementation that conforms to this standard shall:

a) Conform to the required and optional behaviors for a Proxy system ([CS]:5.8, [CS]:5.9).
b) Include, for each RAP Controlled Bridge it proxies for, one RAP Relay instance conformant to the requirements of 5.4.

5.10 FRER-capable RAP Bridge requirements

A FRER-capable RAP Bridge implementation conformant to this standard, whether a RAP Native Bridge or a RAP Controlled Bridge, shall:

a) Conform to the required and optional behaviors for a FRER C-component ([CB]:5.15).
b) Support Active Destination MAC and VLAN Stream identification function ([CB]:6.6).

5.11 RAP Native end station requirements

A RAP Native end station conformant to this standard shall:

a) Conform to the required behaviors for a Native end system ([CS]:5.4).
b) Support the operation of the Applicant state machine ([CS]:8.3) and the Registrar state machine ([CS]:8.4).
c) Support Edge Control Protocol ([Q]:43) and associated management entities ([Q]:12.27).
d) Conform to the requirements of [CS]:6.9.1 and 6.4.3 for use of ECP as the LRP-DT data transport mechanism.
e) Include a single RAP End instance conformant to the requirements of 5.5.

5.12 RAP Controlled end station requirements

A RAP Controlled end station conformant to this standard shall:

a) Conform to the optional behaviors for a Controlled system ([CS]:5.10).

5.13 RAP End Station Proxy requirements

A RAP End Station Proxy implementation that conforms to this standard shall:

a) Conform to the required and optional behaviors for a Proxy system ([CS]:5.8, [CS]:5.9).
b) Include, for each RAP Controlled end station it proxies for, one RAP End instance conformant to the requirements of 5.5.

5.14 FRER-capable RAP Talker requirements

A FRER-capable RAP Talker implementation conformant to this standard, whether a RAP Native end station or a RAP Controlled end station, shall:

a) Conform to the required, recommended and optional behaviors for a Talker end system ([CB]:5.6, [CB]:5.7, [CB]:5.8).

5.15 FRER-capable RAP Listener requirements

A FRER-capable RAP Listener implementation conformant to this standard, whether a RAP Native end station or a RAP Controlled end station, shall:

a) Conform to the required, recommended and optional behaviors for a Listener end system ([CB]:5.9, [CB]:5.10, [CB]:5.11).

5.16 RAP/MSRP Gateway

A RAP/MSRP Gateway implementation conformant to this standard shall:

a) Include a single VLAN Bridge component conformant to [Q]:5.4.
b) Include a single RAP/MSRP Gateway instance conformant to the requirements of 5.6.
c) On each Port that supports the operation of RAP:
    1) Conform to the required and optional behaviors for a Native relay system ([CS]:5.6, [CS]:5.7).
    2) Support Edge Control Protocol ([Q]:43) and associated management entities ([Q]:12.27).
    3) Conform to the requirements of [CS]:6.9.1 and 6.4.3 for use of ECP as the LRP-DT data transport mechanism.
d) On each Port that supports the operation of MSRP, include a MSRP implementation for an end station conformant to the requirements and options of [Q]:5.18.3.

6. Resource Allocation Protocol (RAP)

6.1 RAP overview

RAP defines an application of the IEEE 802.1CS Link-local Registration Protocol (LRP) that provides resource allocation in bridged networks for dynamic creation and maintenance of data streams in needs of deterministic QoS (i.e., bounded latency and zero congestion loss) and high availability and reliability.

As an LRP application, RAP makes use of the service interface provided by LRP to manage connections and handle attribute declarations and registrations between neighboring devices, while RAP is responsible for processing and propagation of attributes across the network. Leveraging the flexibility of LRP systems, RAP can operate in Native systems as a distributed peer-to-peer protocol, or operate in Proxy/Controlled systems to support centralized control of end stations and Bridges. The architecture and the components for various types of RAP systems are defined in 6.3.1. The definitions of LRP protocol elements required for the operation of RAP are presented in 6.4.

Resource allocation relies on the capability of each Bridge to support one or more RA classes, each associated with a different priority used by the streams as the PCP value ([Q]:6.9.3) and providing a certain level of guaranteed QoS to each reserved stream. The definitions of RA class parameters and the concept of RA class domain are presented in 6.3.2.

As a signaling protocol, RAP performs three types of signaling process, each using a dedicated RAP attribute and serving a different purpose, as described in 6.3.3 for RA Advertisement, in 6.3.4 for Talker Announcement and in 6.3.5 for Listener Attachment. The definitions of these three RAP attributes and their encoding in the TLV format are provided in 6.5.

To facilitate decentralized latency and resource computation, RAP adopts the concept of “per-hop per-class guarantees” as the basic principle of resource allocation. This concept allows an examination of deterministic end-to-end QoS to break down into a hop-by-hop check for a set of Bridge-local constraints, as defined in 6.3.6. Used as an essential input for computation of latency bounds and buffer requirements at each hop, a per-hop adjustable Traffic Specification (TSpec), together with the original one generated by the Talker, is carried in each Talker Announcement, as described in 6.3.7.

RAP supports the concept of Stream Rank and reservation importance, as described in 6.3.8, in a manner similar to the MSRP in support of stream importance ([Q]:35.2.4.1). This allows the network, in case of lacking resources required for reserving a stream of higher importance, to remove some existing reservations associated with the streams of lower importance.

RAP also provides dynamic resource allocation to the streams in need of high availability by use of the Frame Replication and Elimination for Reliability (FRER) specified by IEEE Std 802.1CB, as described in 6.3.9. To achieve this, it uses a specific signaling mechanism to propagate the attributes in a set of redundant trees, e.g., Maximally Redundant Trees (MRTs) established using ISIS-PCR ([Q]:45). Upon successful reservation, in addition to the required resources allocated in each participating station, the required FRER functions are configured automatically at the intended locations.

Backward compatibility and interoperability of RAP with MSRP is supported through the use of special Bridges, termed RAP/MSRP Gateway, that operate as a protocol converter to interconnect MSRP-aware devices and RAP-aware devices.

6.2 Conventions

This subclause defines the conventions used in the specification of RAP.

6.2.1 State machine diagrams

The state machine diagrams defined in 6.7.3 for RAP Participant and in 6.8.2 for RAP Propagator use the conventions defined in [Q]:E with the following extensions:

a) All transitions and operations are atomic and finish without any progress of time, as soon as the associated conditions of transitions are met. The sole reason for steady states is that none of the conditions of any outgoing transition of a particular state is satisfied.
b) Conditions associated with transitions caused by invocation of service primitives are expressed using event pointer variables (6.2.4).
c) Service primitives that cannot be processed immediately are queued in their order of invocation for subsequent processing (i.e., no service primitive is lost).
d) In case the conditions of more than one transition are met simultaneously, the taken transition is arbitrarily chosen.

6.2.2 Pseudo-code

A C++ like pseudo-code is used where applicable in defining actions executed on entry to a state in state machine diagrams or when a procedure is invoked. The emphasis is on simplicity, clarity of specification and unambiguous description of the externally visible behavior. Efficiency (speed, memory usage, etc.) is left to the implementation (in software/firmware, hardware, combinations of the aforesaid, etc.).

6.2.3 Naming of parameters and variables

Table 6-1 shows the conventions used in naming RAP variables and parameters.

Table 6-1—Naming conventions for RAP parameters and variables

Format	Description	Applied to	Example
XxxXxx	upper camel case	parameters in Value fields of RAP TLVs/sub-TLVs	StreamId (6.5.3.1)
xxXxx	lower camel case with no prefix	variables local to a RAP component	helloTime (6.7.4.4)
tXxx	lower camel case with prefix ‘t’	variables local to a procedure or a state	tAttrReg (6.7.5.11)
rXxx	lower camel case with prefix ‘r’	parameters of request primitives	DECLARE_ATTR.request (rPortRef, rAttr) (6.7.2.1)
iXxx	lower camel case with prefix ‘i’	parameters of indication primitives	ATTR_REG.indication (iPortRef, iAttr) (6.7.2.3)
pXxx	lower camel case with prefix ‘p’	input parameters of procedures	checkHello (pHelloData) (6.7.5.3)
eXxx	lower camel case with prefix ‘e’	event pointer variables (6.2.4)	eFirstHello (6.7.4.15)

6.2.4 Event pointer variables

Event pointer is a data type used in the state machines defined by this standard to define the variables for use in conditions of the transitions caused by invocation of service primitives and for passing primitive parameters to the state machines.

A variable of this type is defined as being associated with a specific service primitive and is a pointer to a structure that contains one member variable for each parameter of the associated service primitive. It is set to a non-null value, when an associated service primitive is invoked, pointing to the set of parameter values carried in the service primitive. For example, a event pointer variable eXyz is associated with a service primitive with two parameters <A, B>. When eXyz is set, the value of A in the invoked service primitive can be read as eXyz.A. Upon completion of processing the invoked service primitive, the variable is reset by the state machine to NULL.

6.2.5 Array variables

RAP operates on a set of variables that have different scopes, such as per-Port, per-Port per-RA class, or per-Port per-Stream. Array variables are used to group the set of RAP variables that share the same scope together under a single container for convenience of reference in the pseudo-code.

For example, the following array notation

XYZ[<A, B>]

addresses an element in an array variable XYZ that is associated with a key that is expressed as <A, B>. It returns a reference to the addressed element if that element exists in the array, and a NULL value if that element does not exist. The asterisk character (‘*’) can be used as a wildcard for one or more elements of a key to address the subset of entries that match the remaining non-wildcarded key elements.

NOTE—The arrays used in this clause are analogous to associative arrays, a data type commonly used to store a collection of (key, value) pairs. The use of this data type only represents a way of organizing data inside a system. Thus, there is no explicit requirement that this be used in implementations.

Two functions, create and delete, are defined for use in pseudo-code, as follows:

a) Invocation of **`create`** `XYZ[<key>]` adds an element with the given key to the array XYZ and returns a reference to the added element.
b) Invocation of **`delete`** `XYZ[<key>]` removes the element (elements) that matches the given key from the array XYZ.

6.3 Principles of RAP operation

6.3.1 RAP architecture

RAP supports all three classes of systems as specified in IEEE Std 802.1CS Link-local Registration Protocol (LRP): Native systems, Controlled systems and Proxy systems. For the purposes of this standard, these systems can be further classified as shown in Table 6-2.

Table 6-2—RAP System classes

		RAP system classes	
LRP system classes	Native system	RAP Native system	RAP Native end station
			RAP Native Bridge[a]
	Controlled system	RAP Controlled system	RAP Controlled end station
			RAP Controlled Bridge
	Proxy system	RAP Proxy	RAP End Station Proxy
			RAP Bridge Proxy

[a] A RAP/MSRP Gateway can be regarded as a special RAP Native Bridge as it supports RAP operation on at least one Bridge Port.

A RAP instance is an LRP application that implements the RAP functionality. There are two types of RAP instances: RAP End instances for end stations and RAP Relay instances for Bridges.

There is a single RAP End instance or RAP Relay Instance residing in a RAP Native end station or RAP Native Bridge, respectively, executing the operation of RAP locally. For a RAP Controlled station, the operation of RAP is remotely executed at a given RAP Proxy. Conceptually, there is a single RAP End instance or RAP Relay instance offloaded by each RAP Controlled end station or RAP Controlled Bridge, respectively, onto a corresponding RAP Proxy.

In a RAP End instance or RAP Relay Instance, one RAP Participant (6.7) exists for each physical Port that is connected a full-duplex point-to-point medium, and communicates with the underlying LRP via the LRP-DS Service Interface (LSI, [CS]:10).

A RAP End instance has a single RAP End Point (6.6), which communicates with the RAP Participant via the RAP Participant Service Interface (RPSI, 6.7.2) and provides the RAP Endpoint Service Interface (RESI, 6.6.2) to the higher layer entities.

A RAP Relay instance has a single RAP Propagator (6.8), which communicates with the per-Port RAP Participants via the RPSI.

A RAP/MSRP Gateway instance comprises a single RAP Propagator (6.8), one RAP Participant (6.7) for each Port that supports the operation of RAP, and one RAP/MSRP Converter (6.9) for each Port that supports the operation of MSRP ([Q]:35). Each RAP/MSRP Converter in a RAP/MSRP Gateway instance communicates with the RAP Propagator via the RPSI and communicates with an underlying MSRP Participant via the Stream Registration Service Interface (SRSI) as defined in [Q:]:35.2.3.

Figure 6-1 illustrates the operation of RAP in a RAP Native end station and in a two-Port RAP Native Bridge, using the LRP-DT ECP mechanism ([CS]:6.9.1).

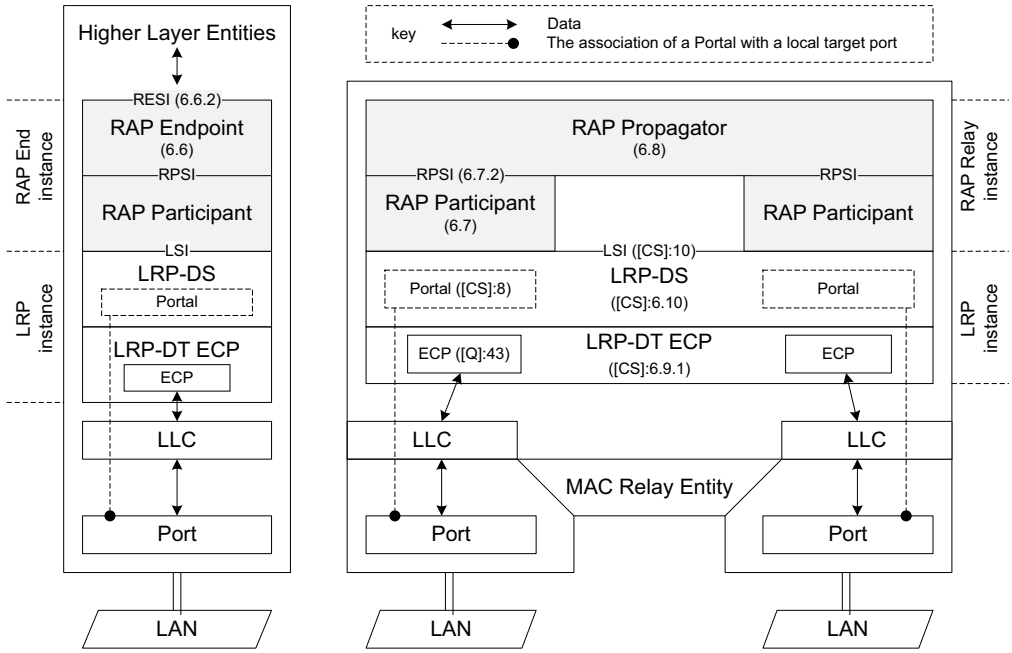
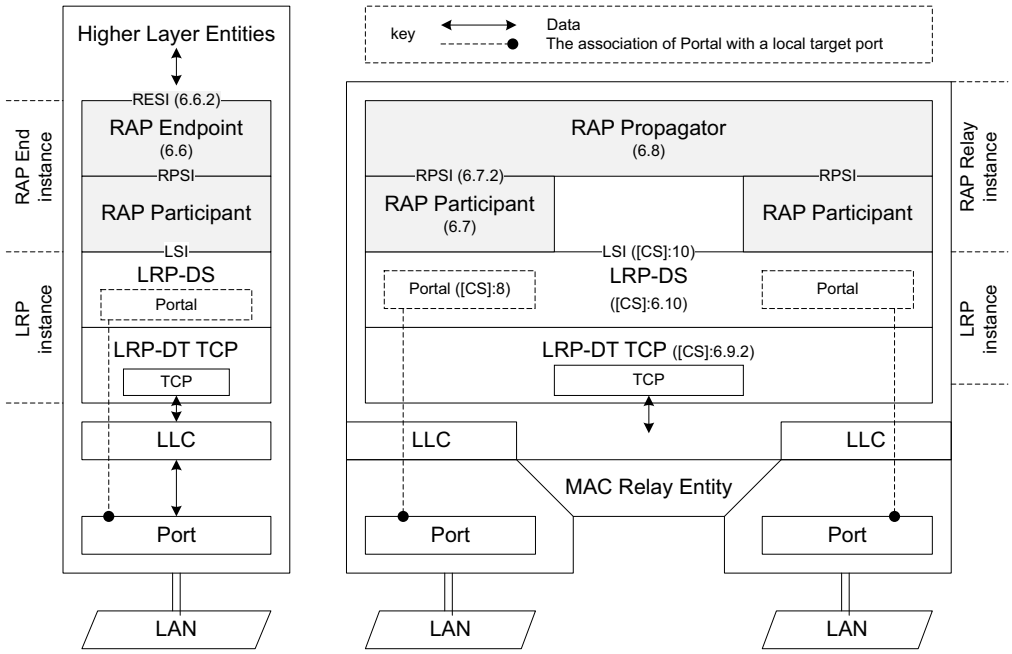


Figure 6-1—Operation of RAP in Native systems using ECP

Figure 6-2 illustrates the operation of RAP in a RAP Native end station and in a two-Port RAP Native Bridge, using the LRP-DT TCP mechanism ([CS]:6.9.2).



NOTE—TCP is typically connected to IP network through either LLC (IEEE Std 802) or a management Port ([Q]:8.3).

Figure 6-2—Operation of RAP in Native systems using TCP

Figure 6-3 illustrates the operation of RAP in a two-Port RAP/MSRP Gateway that supports RAP on one Port and MSRP on the other Port.

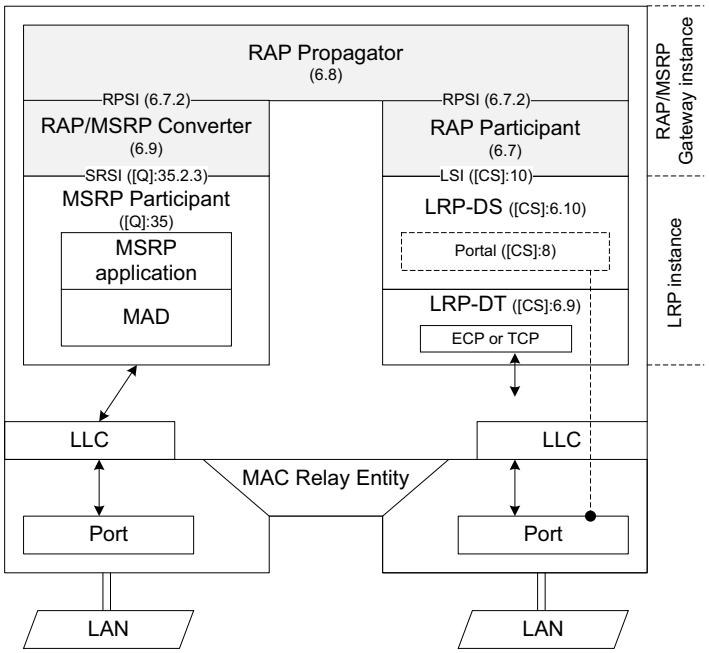


Figure 6-3—Operation of RAP in a RAP/MSRP Gateway

Figure 6-4 shows an example of operating RAP in proxy systems. In this example, a RAP Bridge Proxy proxies for all the Bridges in the network and operates as a central network controller to manage resource allocation for the whole network. There is also a RAP End station Proxy, which performs RAP and LRP operations on behalf of each end station only implemented as a RAP Controlled end station. The LRPDUs exchanged between two Proxy systems by using TCP represent the communications between each proxied end station and the network for the operations of RAP and LRP, meaning that control signaling could take a path different than that used for stream transmissions. As also shown in this example, a RAP Native end station participates in resource allocation through protocol exchanges with the RAP Bridge Proxy using TCP, appearing as if it were communicating with a neighboring RAP Native Bridge.

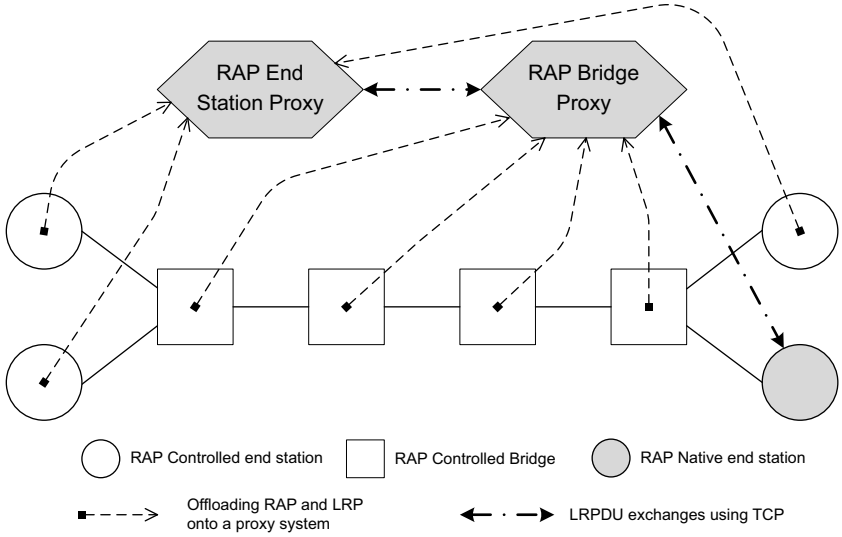


Figure 6-4—Example of RAP operation in Proxy systems

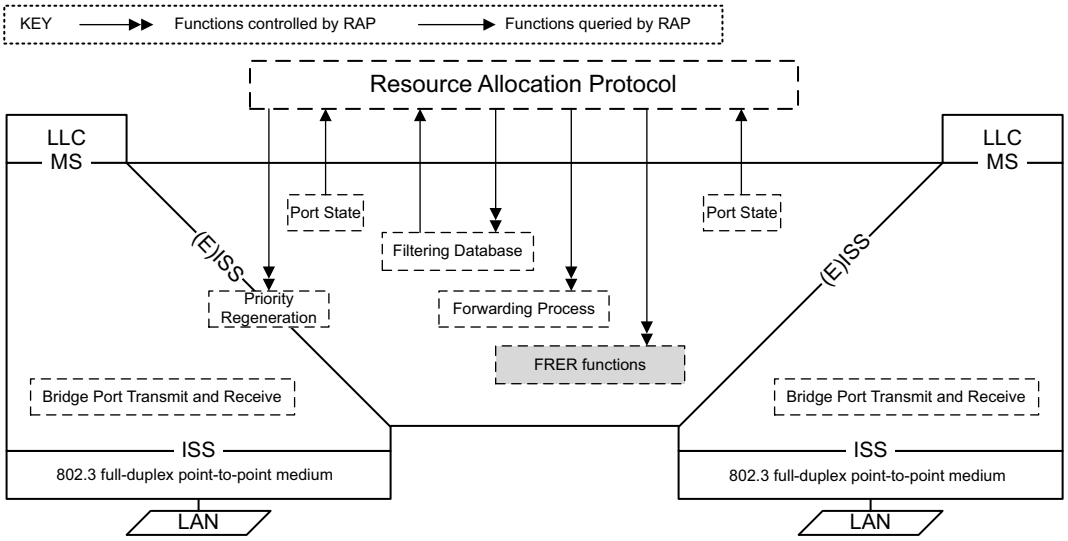


Figure 6-5—RAP control of Bridging and FRER functions

The relationship between RAP and various components and functions in Bridges is shown in Figure 6-5 and illustrated as follows:

- RAP relies on existing configurations of active topology ([Q]:7.3), VLAN topology ([Q]:7.4) and MAC address registration ([Q]:8.8.4) contained in Port State ([Q]:8.4) and the Filtering Database ([Q]:8.8) controlled by other relevant protocols and bridge management as specified in [Q] to perform Talker Announce propagation as described in 6.3.4.1.
- RAP controls the Priority Regeneration ([Q]:6.9.4) on each Bridge Port for the purpose as described in 6.3.2.4.
- RAP controls specific entries in the Filtering Database such as Dynamic Reservation entries ([Q]:8.8.7) for controlling the forwarding and filtering status of each stream on each Bridge Port corresponding to its reservation status.
- RAP controls configuration of specific Forwarding Process functions ([Q]:8.6), such as flow metering ([Q]:8.6.5), queuing frames ([Q]:8.6.6), queue management ([Q]:8.6.7) and transmission selection ([Q]:8.6.7), required for establishing and removing reservations for streams.
- RAP controls configuration of specific FRER functions in FRER-capable Bridges, as described in 6.3.9.1, required for establishing and removing reservations for streams using seamless redundancy.

6.3.2 RA class

An RA class represents a traffic class or a set of traffic classes in an end station or Bridge that uses a given transmission selection algorithm, in conjunction with other mechanisms (e.g., PSFP specified in [Q]:8.6.5.1, the traffic scheduling mechanisms specified in [Q]:8.6.8.4) if needed, and a resource reservation method, to provide a bounded latency and zero congestion loss for streams.

NOTE—An example of using more than one traffic class for a single RA class is an implementation of the cyclic queuing and forwarding shaper (CQF) using the approach as described in [Q]:T.2. In that case, two transmission queues operate in an coordinated manner and offer the same bounded latency to frames transmitted from either of the two queues.

6.3.2.1 RA class ID

The RA class ID is an unsigned integer in the range of 0 through 255 that identifies an RA class supported by an end station or Bridge.

NOTE—RA class ID is a numeric representation of the RA classes supported by a station. Among a set of RA classes on a given Bridge Port, one RA class that is assigned a numerically higher RA class ID than another RA class, is not necessarily mapped to a numerically higher traffic class than that RA class.

6.3.2.2 RA class priority

Each RA class supported by an end station or Bridge is associated with a unique priority value in the range 0 through 7, termed RA class priority. The RA class priority indicates the received priority of the frames which are to be mapped to the traffic class(es) with which that RA class is associated.

NOTE—Mapping of RA class priorities to traffic classes is a matter of port local configuration and can be managed using the Traffic Class Table ([Q]:12.6.3). This standard does not specify default priority to traffic class mapping for a system that supports RA classes.

Since there are eight values available for the RA class priority, a station can support up to seven RA classes, in order to ensure that there is at least one traffic class that can support non-stream traffic that is not subject to resource allocation, such as “best effort” traffic.

6.3.2.3 RA class template and RTID

An RA class template describes a specific method for making up an RA class, including the following:

— The shaping or scheduling mechanisms employed.
— The algorithms for computing latency bounds and resources.
— The encoding of the data in the RaClassTemplateDefinedData field (6.5.2.2.6).

An RA class template is identified by an RA Class Template Identifier (RTID), which encodes a 3-octet OUI or CID value identifying the organization that defines that template, followed by a 1-octet index allocated by that organization. The RA class templates currently defined by IEEE 802.1 are listed in Table 6-3.

Table 6-3—IEEE 802.1 RA class templates

RTID	Transmission Selection Algorithm
00-80-C2-00	Strict Priority ([Q]:8.6.8.1)
00-80-C2-01	ATS Transmission Selection ([Q]:8.6.8.5)
99-99-99-99	Credit-Based Shaper ([Q]:8.6.8.5)

Editor’s note: Use a proper RTID for CBS in Table 6-3

6.3.2.4 RA class domains and priority regeneration

An RA class domain is a connected subset of end stations and Bridges that support a common RA class (i.e. the same RA class ID, 6.3.2.1) and associate the same priority value with that RA class (i.e. the same RA class priority, 6.3.2.2). Within an RA class domain, any traffic associated with the RA class priority of that domain is treated as stream traffic and subject to resource allocation by RAP. The concept of RA class

domains also limits the scope of resource allocation in the sense that a stream can only be established for transmission within the domain of a given RA class.

An RA class domain boundary port is a Port of a Bridge that is part of a given RA class domain and is connected via a LAN to a neighboring station that is not part of that RA class domain. On such a Port, any traffic received with a priority identical to the RA class priority of that RA class domain is subject to priority regeneration ([Q]:6.9.4) according to the administrative values in RA Class Domain Boundary Priority Regeneration Table (7.7) such that the traffic is forwarded by the Bridge using another priority not associated with any RA class.

The detection of RA class domains as well as determining the location of RA class domain boundary ports is controlled by RA Advertisement (6.3.3). Figure 6-6 gives an example of multiple RA Class domains established in a single network, where three domains for RA class ID 1 and three for RA class ID 2 are separately depicted on the left-hand and the right-hand side of the figure, respectively.

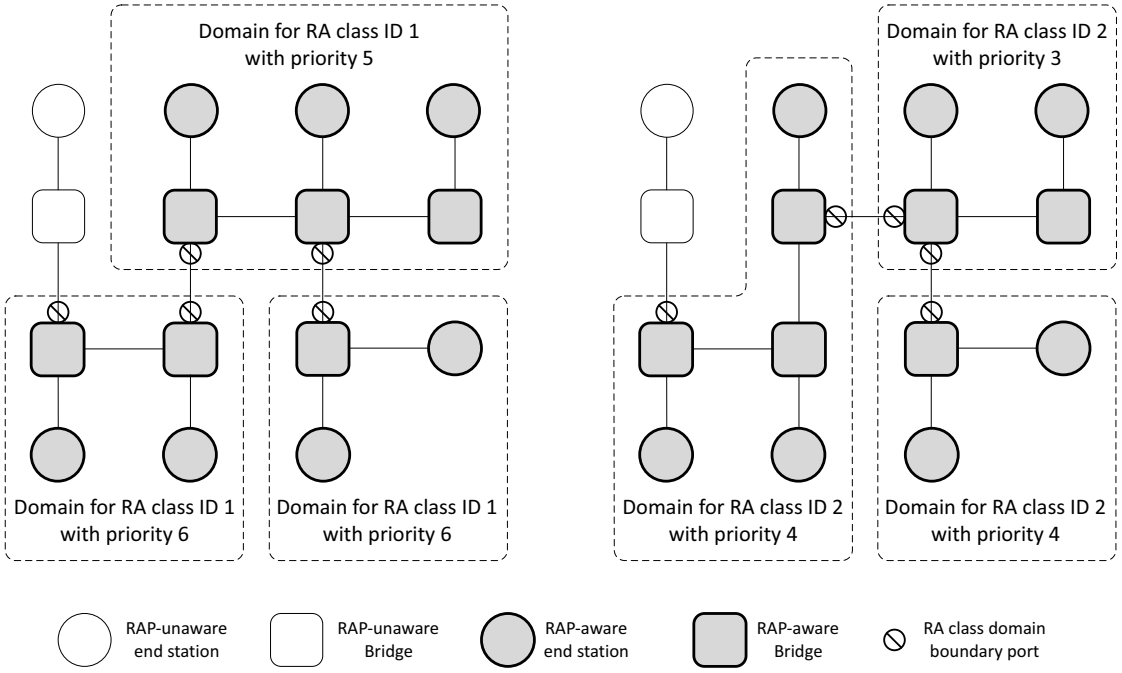


Figure 6-6—Examples of RA class domains and domain boundary ports

A RA Class domain can be connected to a SRP domain (see [Q], Clause 34.2) through a RAP/MSRP Gateway as shown in Figure XX. In this case, the RAP/MSRP Gateway acts as a RAP-aware end station in the RA domain and as a SRP-aware end-station in the SRP domain. The RA class boundary port will be then located inside the RAP/MSRP Gateway.

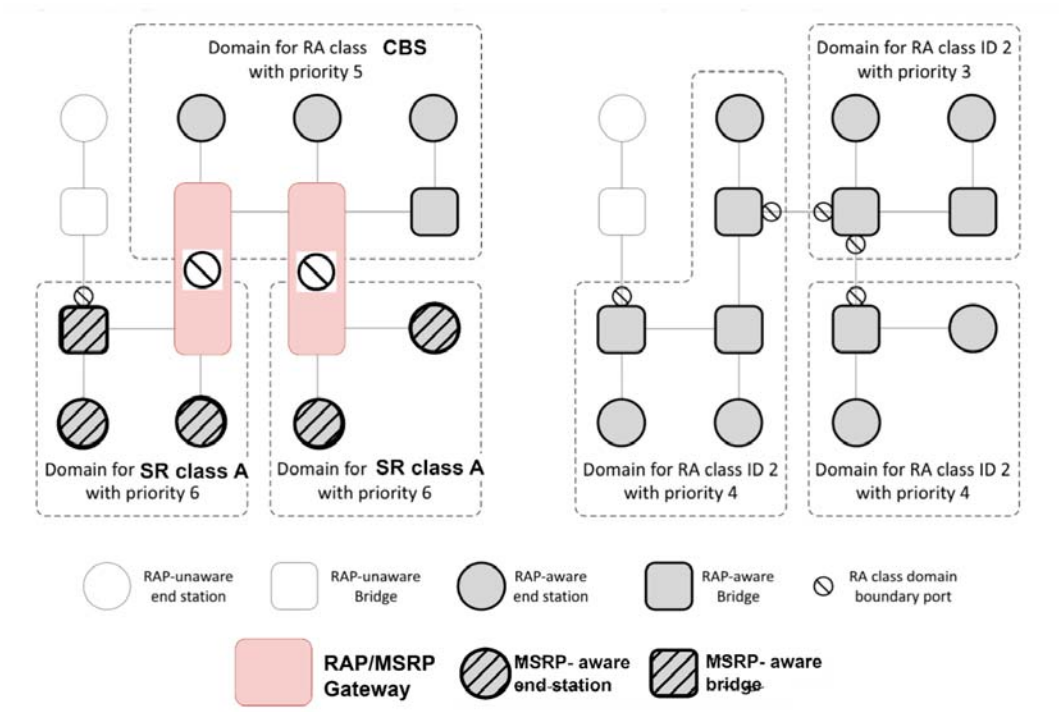


Figure 6-n - Example of RA-class domains with RAP/MSRP Gateways

6.3.3 RA Advertisement

RA Advertisement refers to declarations of the RA attribute (6.5.2) by a station to each neighboring station. The purposes of RA Advertisement are as follows:

— Advertisement of RA class domain configuration for detection of RA class domains and determination of the location of RA class domain boundary ports, as described in 6.3.2.4.

— End station learning of the RA class (6.3.2) and Redundancy Context (6.3.9.2) configurations in the network. To allow adaptive participation in resource allocation, an end station can support the ability to adjust its local parameters according to the information received from its neighboring Bridge and then declare its own RA attribute with the adjusted parameters.

— Advertisement of RA class template configuration. In the process of reserving a stream in a given RA class, the RA class template used by that RA class and corresponding parameter configuration in an upstream station is necessary information used by each downstream station in performing actions such as calculation of the worst-case latency of streams transmitted in that RA class from that upstream station.

The following terminology relating to RA Advertisement is used:

a) **RA declaration**: A declaration of the RA attribute on an end-station or Bridge Port.
b) **RA registration**: A registration of the RA attribute on an end-station or Bridge Port.
c) **RA deregistration**: The removal of an RA registration on an end-station or Bridge Port.

An RA registration received from a neighboring station on a Bridge Port is not propagated by the Bridge to declare that registered RA attribute to another neighboring on any of other Ports.

6.3.4 Talker Announcement

The Talker Announcement refers to the process of signaling the Talker Announce attribute (6.5.3) initially declared by the Talker of a stream across a Bridged network to potential Listeners. Each Talker Announcement is identified by a StreamId (6.5.3.1) and VID (6.5.3.5.3) as contained in the Talker Announce attribute.

The purposes of the Talker Announcement are as follows:

— Indication of a Talker's intention of supplying a stream.
— Advertisement of the stream's characteristics. The Talker Announce attribute contains the information needed by the network to identify the stream and determine the required resources on each Bridge along the stream path(s).
— Gathering of QoS information from the Bridges along each path such as the accumulated latency and reporting of gathered information to Listeners.
— Signaling of status information along each potential path about whether a reservation can be made, and if not the information about the cause and the location of the failure.

The following terminology relating to the Talker Announcement is used:

a) **Talker Announce declaration:** A declaration of the Talker Announce attribute on a Port.
b) **Talker Announce registration:** A registration of the Talker Announce attribute on a Port.
c) **Talker Announce deregistration:** The removal of a Talker Announce registration on a Port.
d) **Talker Announce propagation:** Propagation of a Talker Announce registration on an ingress Port of a Bridge to one or more other egress Ports of the Bridge, resulting in a Talker Announce declaration on each such egress Port.
e) **Talker Announce merge:** Merge of multiple Talker Announce registrations being propagated to a Bridge Port to result in a joint Talker Announce declaration on that Port.
f) **Originating Talker Announce registration:** A Talker Announce registration from which a given Talker Announce declaration results from is termed an originating Talker Announce registration of that Talker Announce declaration. A Talker Announce declaration can have more than one originating Talker Announce registration in the case of Talker Announce merge.

6.3.4.1 Talker Announce propagation

A set of rules use in Talker Announce propagation are defined in the following subclauses. These rules refer to specific Forwarding Process functions ([Q]:8.6) in VLAN Bridges, which can be configured by use of relevant protocols or management entities specified in [Q].

6.3.4.1.1 VLAN Context

Each Talker Announcement indicates one or more VLAN Contexts for use in Talker Announce propagation by Bridges. A VLAN Context in a Bridge is identified by a VID, termed VLAN Context ID, and corresponds to a VLAN topology formed by the set of Ports that includes each Bridge Port for which the following are true:

1) The Port is in the Forwarding state and part of the active topology that supports that VID.
2) The Port is a member of the member set ([Q]:8.8.10) for that VID.

Depending on the number of VLAN Contexts used, there are two types of the Talker Announcement, as follows:

a) **Single-Context Talker Announcement:** The Talker Announcement propagated in a single VLAN Context and used in resource allocation for transmission of a stream along a single path from the Talker to each Listener.

b) **Multi-Context Talker Announcement:** The Talker Announcement propagated in more than one VLAN Context and used in resource allocation for transmission of a stream with seamless redundancy along multiple paths from the Talker to each Listener.

One or more VLAN Contexts contained in a Talker Announce registration are used by a Bridge to determine a set of potential egress Ports in propagating a Talker Announce registration on an ingress Port. From a network-wide perspective, a specific VLAN Context used by a Talker Announcement constrains the extent of the Talker Announcement to a corresponding VLAN topology of the network.

NOTE—An appropriate VLAN configuration in the network for the set of VLAN Contexts used in resource allocation is required to control the reach of Talker Announcement. As one way to configure VLAN membership in Bridges for VLAN Contexts, each Listener desiring to receive a given Talker Announcement can use MVRP ([Q]:11.2) to issue an MVRP VLAN membership declaration for each VLAN Context ID to be used by that Talker Announcement. In such cases, the knowledge of the VLAN Context(s) concerned needs to be made available in advance to the Listener, e.g., by use of a specific higher layer protocol.

6.3.4.1.2 Stream DA Pruning

In addition to VLAN Context, Stream DA Pruning applies an additional constraint to Talker Announce propagation by reference to the status of MAC Address Registrations Entries ([Q]:8.8.4) in the FDBs of the Bridges. Stream DA Pruning can be enabled or disabled on a per-Bridge Port basis.

When Stream DA Pruning is enabled on a Port, a Talker Announce registration is not be propagated to that Port, if the DestinationMacAddress (6.5.3.5.1) value contained in the Talker Announce attribute is not found in the MAC Address Registration Entries for that Port.

NOTE—To propagate a Talker Announcement along a path containing one or more Bridges with Stream DA Pruning enabled on the egress port, a registration of the Stream Destination MAC Address used by the Talker Announcement on each such Port is required. As one way to register MAC addresses, a Listener on that path can use MMRP ([Q]:10) to issue an MMRP declaration for that MAC address. In such cases, the knowledge of the MAC address concerned needs to be made available in advance to the Listener, e.g., by use of a specific higher layer protocol.

6.3.4.1.3 Ingress Blocking

Ingress Blocking prohibits propagation of a Talker Announce registration on an ingress Port of a Bridge to any other egress Ports, if all of the following conditions are met:

a) Ingress Filtering ([Q]:8.6.2) is enabled on the Port.

b) The Port is not in the member set (8.8.10) associated with the VID (6.5.3.5.3) of the Talker Announce registration.

NOTE—In addition to the need for VLAN Context on egress Ports, there is also a need for an appropriate VLAN configuration on each potential Talker Announce registration Port in the network based on the required reach of Talker Announcement. For example, in case that it is not desired to block a Talker Announcement on an ingress Port of any Bridge with Ingress Filtering enabled, the Talker can use MVRP ([Q]:11.2) to issue an MVRP VLAN membership declaration for the VID concerned. B.4 shows another example of VLAN configuration in a case where ingress blocking a Talker Announcement on specific Bridge Ports is intended.

6.3.4.1.4 Talker Announcement across RA class domains

For the purpose of providing RAP allows Talker Announcement propagation across boundaries of RA class domains (6.3.2.4), but fails the Talker Announcement once it is propagated to a Bridge located outside the

RA class domain from which it originates, with the RAP Failure Code *CrossingDomainBoundary* defined in Table 6-10.

6.3.4.2 Talker Announce status

The Talker Announce status of a Talker Announce declaration made on a given Port is associated with an ordinary stream in the case of the Single-Context Talker Announcement, or a Compound or a Member Stream in the case of the Multi-Context Talker Announcement, to be transmitted on that Port, indicating whether the stream could be reserved from the Talker to that Port along a single path or one or more paths, respectively, in the VLAN Context(s) indicted in the Talker Announce declaration, as follows:

a) **Announce Success:** The stream can be reserved on the Port, as it has not encounter any problems on a single path from the Talker to that Port, if the stream is a ordinary stream, or on at least one path from the Talker to that Port, if the stream is a Compound Stream or Member Stream. The Announce Success status is represented by the absence of a Failure Information sub-TLV (6.5.3.10) in the Value field of the Talker Announce attribute TLV (6.5.3).

b) **Announce Fail:** The stream can not be reserved on the Port, as it has encounter problems on each path from the Talker to that Port. The Announce Fail status is represented by the presence of a Failure Information sub-TLV (6.5.3.10) in the Value field Talker Announce attribute TLV (6.5.3).

6.3.4.3 Talker Announce merge

Talker Announce merge can only occur in the Multi-Context Talker Announcement in resource allocation for seamless redundancy (6.3.9) and is associated with the FRER operation stream merging in FRER-capable Bridges or Listeners, as described in 6.3.9.5.

Talker Announce merge is performed in a FRER-capable Bridge or Listener among a set of Multi-Context Talker Announce registrations that are propagated to the same Port and contain the same StreamId (6.5.3.1) value, to result in a joint Multi-Context Talker Announce declaration on that Port for that StreamId and containing a merged set of VLAN Contexts comprising each one used in the propagation of those Talker Announce registrations.

6.3.5 Listener Attachment

Listener Attachment refers to the process of signaling the Listener Attach attribute (6.5.4) initially declared by one or more Listeners desiring to receive a stream across a Bridge network to the stream's Talker. Each Listener Attachment is identified by a StreamId (6.5.4.1) and VID (6.5.3.5.3) as contained in the Listener Attach attribute.

The purposes of Listener Attachment are as follows:

— Indication of the intention of one or more Listeners of receiving a stream.

— Triggering of resource allocations (or deallocation) on each Bridge along potential stream path(s).

— Signaling of reservation status hop-by-hop in an upstream direction from each participating Listener back to the Talker.

The following terminology relating to Listener Attachment is used:

a) **Listener Attach declaration**: A declaration of the Listener Attach attribute on a Port.

b) **Listener Attach registration**: A registration of the Listener Attach attribute on a Port.

c) **Listener Attach deregistration**: The removal of a Listener Attach registration on a Port.

d) **Listener Attach propagation**: Propagation of a Listener Attach registration from a Bridge Port to one or more other Bridge Ports.

e) **Listener Attach merge**: Merge of more multiple Listener Attach registrations propagated to a Bridge Port to result in a joint Listener Attach declaration on that Port.

f) **Associated Talker Announce declaration**: A Talker Announce declaration with which a given Listener Attach registration is associated is termed the associated Talker Announce declaration of that Listener Attach registration.

g) **Originating Listener Attach registration**: A Listener Attach registration from which a given Listener Attach declaration results is termed an originating Listener Attach registration of that Listener Attach declaration. A Listener Attach declaration can have more than one originating Listener Attach registration in the case of Listener Attach merge.

h) **Associated Talker Announce registration**: A Talker Announce registration with which a given Listener Attach declaration is associated is termed the associated Talker Announce registration of that Listener Attach declaration.

6.3.5.1 Association between Talker Announcement and Listener Attachment

A Listener Attachment is associated with a Talker Announcement that carries the same StreamId as that of the Listener Attachment. Depending on the type of the associated Taker Announcement, there are two types of the Listener Attachment, as follows:

a) **Single-Context Listener Attachment:** A Listener Attachment that is associated with a Single-Context Talker Announcement [item a) in 6.3.4.1.1].

b) **Multi-Context Listener Attachment:** A Listener Attachment that is associated with a Multi-Context Talker Announcement [item b) in 6.3.4.1.1].

A Single-Context Listener Attach registration is said to be associated with a Single-Context Talker Announce declaration, if all of the following conditions are met:

1) The Listener Attach registration is on the same Port as the Talker Announce declaration.
2) The StreamId (6.5.4.1) of the Listener Attach registration is identical to the StreamId (6.5.3.1) of the Talker Announce declaration.
3) The VID (6.5.4.2) of the Listener Attach registration is identical to the VID (6.5.3.5.3) of the Talker Announce declaration.

A Multi-Context Listener Attach registration is said to be associated with a Multi-Context Talker Announce declaration, if all of the following conditions are met:

4) The Listener Attach registration is on the same Port as the Talker Announce declaration.
5) The StreamId (6.5.4.1) of the Listener Attach registration is identical to the StreamId (6.5.3.1) of the Talker Announce declaration.
6) The set of VLAN Context(s) (6.5.4.4) indicated by the Listener Attach registration is identical to the set of VLAN Context(s) indicated by the Talker Announce declaration (6.5.3.9.2).

A Listener Attach declaration is said to be associated with a Talker Announce registration, if all of the following conditions are met:

7) The Listener Attach declaration is on the same Port as the Talker Announce registration.
8) The StreamId (6.5.4.1) of the Listener Attach declaration is identical to the StreamId (6.5.3.1) of the Talker Announce registration.
9) The Listener Attach declaration wholly or partially results from a Listener Attach registration whose associated Talker Announce declaration originates wholly or partially from the Taker Announce registration.

A Listener Attach registration with an associated Talker Announce declaration on a Bridge Port indicates a reservation request for transmission of a stream or a Compound or Member Stream through that Port and, upon successful resource allocation, is associated with a dedicated reservation in the Bridge Port. A

reservation maintained in a Bridge Port is identified by <StreamId, VID> with the corresponding values contained in the associated Listener Attach registration.

6.3.5.2 Listener Attach propagation

A Listener Attachment follows the path(s) traversed by its associated Talker Announcement in reversed direction.

A Listener Attach registration on a Bridge Port with an associated Talker Announce declaration is propagated to each other Port that contains an originating Talker Announce registration of the associated Talker Announce declaration. A Listener Attach registration on a Bridge Port without an associated Talker Announce declaration is not propagated by the Bridge.

6.3.5.3 Listener Attach merge

Listener Attach merge is applied, when more than one Listener Attach registration whose associated Talker Announce declaration has a common originating Talker Announce registration is propagated to the same Port, to result in a joint Listener Attach declaration on that Port.

In a Single-Context or Multi-Context Listener Attachment for a stream, Listener Attach merge can occur when multiple Listener Attachment processes originating from different Listeners of that stream converge on a Bridge Port, indicating a potential place for multicast forwarding the stream. Additionally, in a Multi-Context Listener Attachment for a Compound Stream, Listener Attach merge can also occur when multiple Listener Attachment processes running in different VLAN Contexts converge on a Bridge or Talker end station Port, indicating a potential place for applying stream splitting (6.3.9.4) to the stream.

6.3.5.4 Listener Attach status

The Listener Attach status of a Listener Attach declaration on a Port is associated with one or more paths, depending on the VLAN Context(s) in which the Listener Attach declaration is made, from the Port to each Listener whose participation in the Listener Attachment has been perceived on that Port, and indicates one of the following:

a) **Attach Ready:** In the case of the Single-Context Listener Attachment, indicating a reservation has been successfully made on each path to each Listener. In the case of the Multi-Context Listener Attachment, indicating a reservation has been successfully made on at least one path to one Listener, while the reservation status of each path is indicated by the Path status (6.3.5.5).

b) **Attach Fail:** A reservation has failed on each path to each Listener;

c) **Attach Partial Fail:** A reservation has been successfully made on the path to at least one of these Listeners, but has failed on the path to other Listeners. This status is only applicable to the Single-Context Listener Attachment.

NOTE—The Attach Partial Fail status is used only in the Single Context Listener Attachment for the same purpose for which the Listener Ready Failed status is used in MSRP as defined in [Q]:35.1.2.2. In the case of the Multi-Context Listener Attachment, there is no distinction made in the Listener Attach status between a full success and a partial success, while the latter case could arise, for example, when reservation has failed on one among all paths to a single Listener, or when reservation on all the paths has succeeded for some Listeners but failed for the others.

6.3.5.5 Path status

A Multi-Context Listener Attachment also signals the Path status on a per VLAN Context basis. The Path status of a Listener Attach declaration on a Port for a given VLAN Context is associated with a single path

within that VLAN Context from the Port to each Listener whose participation in the Listener Attachment has been perceived on that Port, and indicates one of the following:

a) **Faultless Path**: Resource allocation has succeeded without encountering any problems on the path.
b) **Faulty Path**: Resource allocation has encountered one or more problems on the path and failed wholly or partially on the path.
c) **Unresponsive Path**: No Listener Attachment with respect to the path has been received.

NOTE—In the Multi-Context Listener Attachment, resource allocation can succeed partially on a path as a result of the fact that the path goes through one or more places of stream merging in the network, as described in 6.3.9.5.

6.3.6 Resource allocation constraints

Resource allocation is constrained by limited capacity of end stations and Bridges to ensure bounded latency and zero congestion loss for reserved streams. The subsequent subclauses (6.3.6.1 through 6.3.6.3) describe three constraints taken into account in resource allocation.

6.3.6.1 Latency constraint

The latency constraint for a Bridge is specified in the per-RA class, per reception/transmission Port pair maxHopLatency parameter, as defined in item d) of 6.8.4.10. Such a parameter configured in a Bridge is used by the Bridge in resource allocation as a guaranteed latency upper bound provided to each stream reserved in a given RA class when transmitted through the hop from a neighboring station on a given reception Port to a given transmission Port.

The latency constraint for an end station is specified in the per-RA class parameter MaxLastHopLatency (6.5.2.2.5) received in an RA registration from a neighboring station. Such a parameter provided to an end station is used by the end station, when acting as a Listener, to impose a latency upper bound to streams when transmitted through the last-hop from that neighboring station to the end station.

A stream reservation request is said to meet the latency constraint of a Bridge or an end station (as a Listener), if allowing transmission of this stream would not cause any reserved stream to suffer from a latency exceeding the associated upper bound. A Talker Announcement failed due to not meeting the latency constraint is reported with the RAP Failure Code *LatencyExceeded* defined in Table 6-10.

The latency constraint configured on each hop along each path in the network, in conjunction with an accumulation of per-hop latency bounds in AccuMaxLatency (6.5.3.3), builds up the basis for bounded end-to-end latencies in resource allocation.

NOTE—The concept of per-hop latency bound requires the ability of each station participating in resource allocation to determine the worst-case maximum latency while checking the latency constraint. The method used for this purpose is specific to each RA class template.

6.3.6.2 Bandwidth constraint

The bandwidth constraint of a Bridge is specified in the per-RA class, per-transmission Port maxBandwidth parameter, as defined in item d) of 6.8.4.9. One such parameter configured in a Bridge places an upper bound on the amount of bandwidth that can be reserved for use by an RA class on a transmission Port.

The bandwidth constraint also applies to end stations. As in many circumstances the bandwidth availability for streams in an end station is controlled by the applications, no explicit bandwidth constraint parameters are specified by this standard for end stations, and it is a matter of each end station’s local decision.

A stream reservation request is said to meet the bandwidth constraint of a Bridge or end station, if the total bandwidth needed to transmit that stream and all reserved streams in the same RA class and on the same Port

would not exceed the associated upper bound. A Talker Announcement failed due to not meeting the bandwidth constraint is reported with the RAP Failure Code *BandwidthExceeded* defined in Table 6-10.

6.3.6.3 Resource constraint

The resource constraint is associated with the availability of configurable resources for streams in a Bridge or end station's local functions, such as filtering, buffering, shaping and FRER. A stream reservation request is said to meet the resource constraint of a station, if the station has sufficient resources available for in all the functions required for handling that stream. A Talker Announcement failed due to not meeting the resource constraint is reported with the RAP Failure Code *ResourceExceeded* defined in Table 6-10.

A measure of resource constraint and its determination method for streams are usually dependent on a particular implementation. No explicit parameters are specified by this standard for resource constraint.

6.3.7 Traffic specification usage rules and types

In resource allocation, the traffic specification (TSpec) carried in the Talker Announcement contains the information about the characteristics of the stream's data traffic.

6.3.7.1 Talker TSpec

A TSpec received via RESI (6.6.2) from higher layer entities in Talkers and specifying the amount of the stream data transmitted by the Talker into the network. The TSpec contained in this field remains unchanged during the propagation of the Talker Announcement across the network.

NOTE 1—Keeping a Talker TSpec unchanged during its propagation is to provide Bridges and Listener end stations with the original information about the stream traffic generated by Talkers. This information can be used, for example, by a specific intermediate Bridge in traffic shaping the stream to restore its original traffic characteristics.

There are also two types of TSpec supported by resource allocation, each using a different parameter set to describe the stream traffic characteristics:

a) **Talker Token Bucket TSpec**, a TSpec comprising the following set of parameters excluding any MAC framing and media-dependent overheads (e.g., preamble, 802.3 header, MAC header, VLAN tag, R-TAG, CRC, interframe gap):
   1) **MaxPayloadLength**: The maximum frame length of the traffic, in octets, associated with on any transmission Port.
   2) **MinPayloadLength**: The minimum frame length of the traffic, in octets, associated with on any transmission Port.
   3) **CommittedInformationRate**: the committed information rate, in bits per second.
   4) **CommittedBurstSize**: The committed burst size, in bits sent from the application.

b) **Interval-based TSpec**, a TSpec comprising the following set of parameters:
   1) **Interval**: The interval of time, in nanoseconds, taken as a fixed-length observation window to measure the maximum amount of the traffic.
   2) **MaximumFramePerInterval**: The maximum number of maximum sized frames can be observed from the traffic within one Interval [item 1) above].
   3) **MaximumFrameSize**: The maximum frame size of the traffic, in octets, excluding any MAC framing and media-dependent overheads (e.g., preamble, 802.3 header, MAC header, VLAN tag, R-TAG, CRC, interframe gap) associated with on any transmission Port.

NOTE 3—The Interval-based TSpec type is compatible with the TrafficSpecification defined in [Q]:46.2.3.5, except the TransmissionSelection parameter ([Q]:46.2.3.5.4).

6.3.7.2 Network TSpec

A TSpec received via RESI (6.6.2) from higher layer entities in Talkers and specifying the amount of the stream data transmitted by the Talker into the network. The TSpec contained in this field remains unchanged during the propagation of the Talker Announcement across the network.

NOTE 1—Keeping a Talker TSpec unchanged during its propagation is to provide Bridges and Listener end stations with the original information about the stream traffic generated by Talkers. This information can be used, for example, by a specific intermediate Bridge in traffic shaping the stream to restore its original traffic characteristics.

Token Bucket TSpec, a TSpec comprising the following set of parameters including the media-dependent overhead ([Q]:12.4.2.2) :

1) **MaxTransmittedFrameLength**: The maximum frame length of the traffic, in octets,associated with a transmission Port for the traffic.
2) **MinTransmittedFrameLength**: The minimum frame length of the traffic, in octets, associated with a transmission Port for the traffic.
3) **CommittedInformationRate**: the committed information rate, in bits per second.
4) **CommittedBurstSize**: The committed burst size, in bits.

6.3.7.3 Translation of Talker TSpec to Network TSpec

When initiating a Talker Announcement at a Talker, higher layer entities can supply either a Talker Token Bucket TSpec or an Interval-based TSpec as the Talker TSpec. The Network TSpec field in the Talker Announce attribute always contains a Token Bucket TSpec derived from the supplied Talker TSpec. When an Interval-based TSpec is supplied as the Talker TSpec, it is translated by the Taker to a Token Bucket TSpec as the Network TSpec as specified in Table 6-4. As the Talker Announcement propagates in the network, each Bridge on the path can adjust the Token Bucket TSpec values contained in the Network TSpec as necessary.

Table 6-4—Translation of an Interval-based TSpec to a Token Bucket TSpec

Interval-based TSpec	Token Bucket TSpec
MaximumFrameSize + ALL_OVERHEAD[a]	MaxTransmittedFrameLength
64 + MD_OVERHEAD[b]	MinTransmittedFrameLength
(MaximumFrameSize + ALL_OVERHEAD[a]) * MaximumFramePerInterval * 8 * 1e9 / Interval	CommittedInformationRate
(MaximumFrameSize + ALL_OVERHEAD[a]) * MaximumFramePerInterval * 8	CommittedBurstSize

[a] Including all MAC framing and media-dependent overheads (e.g., preamble, 802.3 header, MAC header, VLAN tag, R-TAG, CRC, interframe gap) associated with a transmission Port for the traffic.

[b] The media-dependent overhead [Q]:12.4.2.2 associated with a transmission Port for the traffic.

6.3.8 Stream Rank and reservation importance

RAP supports the concept of the Rank as specified in [Q]:46.2.3.2.1 and gives streams of Rank zero higher precedence than streams of Rank one in a manner that allows Bridges to make a reservation for a stream of

Rank zero in case of lacking resources by removing existing reservations made for one or more streams of Rank one. Each reservation removed under such circumstances is termed “the reservation preempted by Rank zero” and uses the RAP Failure Code *ReservationPreempted* as defined in Table 6-10.

NOTE—RAP allows reservations of Rank one streams to be preempted only by reservations of Rank zero streams. Reservations of streams with the same Rank, regardless Rank zero or one, are handled on a first-come first-served basis.

The order used by a Bridge in the selection of a reservation, in preference to others, to be preempted by Rank zero is determined based on the reservation importance, i.e., a least important reservation is to be removed first. Among a set of reservations on a Bridge Port, each identified by <StreamId, VID> and with a reservationAge value [item e) in 6.8.4.3], one reservation is considered more important than another, according to the following rules in the shown order:

a) A Reservation with a numerically larger reservationAge value is more important.
b) If reservationAge is the same, the one with a numerically smaller StreamId is more important.
c) If StreamId is the same, the one with a numerically smaller VID is more important.

6.3.9 Resource allocation for seamless redundancy

RAP supports resource allocation for transmitting streams via redundant paths using the seamless redundancy techniques specified by IEEE Std 802.1CB. The subsequent subclauses describe additional capabilities and operations required by resource allocation for seamless redundancy, while the definitive specification is contained in 6.5, 6.6 and 6.8. For illustrative purposes, some resource allocation examples for seamless redundancy are provided in Annex B.

6.3.9.1 FRER-capable stations

Resource allocation for seamless redundancy operates in a network where a set of FRER-capable stations, either or both FRER-capable end stations and FRER-capable Bridges, are placed at specific locations to perform FRER operations on Compound and Member Streams with corresponding FRER functions, as shown in Table 6-5.

RAP makes use of Multi-Context Talker Announcement and Multi-Context Listener Attachment in resource allocation for seamless redundancy. The potential locations in the network (including end stations) where each stream requesting seamless redundancy is subject to FRER operations along its paths are dependent on the placement of FRER-capable stations and the VLAN topologies configured for use with seamless redundancy, and are automatically determined on a per-stream basis during the stream’s Multi-Context Talker Announcement. Resource allocation, including configuration of the FRER functions needed for each stream, is executed during the stream’s Multi-Context Listener Attachment.

6.3.9.2 Redundancy Context

Resource allocation for seamless redundancy operates in a network in which one or more instances of redundant trees are established and maintained by other protocols such as ISIS-PCR ([Q]:45). Each Multi-Context Talker Announcement for a Compound Stream operates within a particular instance of redundant trees. The group of VLAN topologies associated with an instance of redundant trees comprising two or more trees forms the set of correlated VLAN Contexts, termed “the Redundancy Context”. A Redundancy Context is identified by the numerically smallest VLAN Context ID in the set, termed “the Redundancy Context ID”. Multiple Redundancy Contexts, each with distinct set of VLAN Context IDs, can exist in a network due to the existence of multiple instances of redundant trees.

NOTE—For example, there can be one instance of redundant trees for each Bridge located at the edge of the network and rooted at that Bridge, with the intention that each Compound Stream using the redundant trees rooted at a Bridge where it enters the network be established for transmission along the multiple paths that are disjoint from each other.

Table 6-5—FRER functions and operations in FRER-capable stations

<table>
<tr><th colspan="2" rowspan="2"></th><th colspan="3">FRER-capable stations</th></tr>
<tr><th>FRER-capable
RAP Talker</th><th>FRER-capable
RAP Bridge</th><th>FRER-capable
RAP Listener</th></tr>
<tr><td rowspan="5">FRER operations</td><td>redundancy tagging (6.3.9.3)</td><td colspan="2">sequence generation function ([CB]:7.4.1),
Stream encode/decode function ([CB]:7.6),
Redundancy tag ([CB]:7.8)</td><td rowspan="3">(none)</td></tr>
<tr><td rowspan="2">stream splitting (6.3.9.4)</td><td>Stream splitting function ([CB]:7.7)</td><td>(Multicast) [a]</td></tr>
<tr><td colspan="2">Active Destination MAC and VLAN Stream identification function ([CB]:6.6) [b]</td></tr>
<tr><td>stream merging (6.3.9.5)</td><td rowspan="2">(none)</td><td colspan="2">Active Destination MAC and VLAN Stream identification function ([CB]:6.6) [b],
sequence recovery function ([CB]:7.5),</td></tr>
<tr><td>redundancy untagging (6.3.9.3)</td><td colspan="2">Stream encode/decode function ([CB]:7.6),
Redundancy tag ([CB]:7.8)</td></tr>
</table>

[a] The stream splitting operation in a Bridge relies on the multicast ability of the Forwarding Process to replicate streams, without the need to use Stream splitting function.

[b] An end station can, but is not required to explicitly use this function; it is an implementation choice.

The per-Bridge parameter redundancyContext (6.8.4.11), configurable as a managed object defined in Table 7-8 and containing the configuration of each existing Redundancy Context in the network, provides the information needed by operations such as Talker Announce merge as described in 6.3.4.3. Proper operation of resource allocation for seamless redundancy requires a consistent Redundancy Context configuration among all the Bridges involved in resource allocation for seamless redundancy. An inconsistent Redundancy Context configuration, e.g. resulting from misconfiguration, can cause incorrect operation. To facilitate inconsistency detection, each Bridge advertises its redundancyContext value in the RA Advertisement (6.3.3) and maintains the per-Port array portRedundancyContext in 6.8.4.8. To prevent incorrect operation, a Multi-Context Talker Announcement is failed once it is registered on a Bridge Port whose portRedundancyContext table entry has a consistency value of FALSE, and reports the RAP Failure Code InconsistentRedundancyContext defined in Table 6-10.

6.3.9.3 Redundancy tagging and redundancy untagging

Redundancy tagging refers to a FRER operation in which a FRER-capable RAP Talker or a FRER-capable RAP Bridge uses the particular FRER functions, as shown in Table 6-5, to transform an ordinary stream into a Compound Stream by inserting a Redundancy tag (R-TAG) into the data frames of the stream. Redundancy untagging refers to a FRER operation in which a FRER-capable Bridge or a FRER-capable Listener uses the particular FRER function, as shown in Table 6-5, to transform a Compound Stream back into an ordinary stream by removing the R-TAG carried in the data frames of the Compound Stream.

The potential locations in the network (including end stations) where a stream requesting seamless redundancy is subject to redundancy tagging and redundancy untagging are determined automatically during the stream's Multi-Context Talker Announcement in accordance with the following rules:

a) Redundancy tagging is to be performed by the stream's Talker, if that Talker is FRER-capable, and by a FRER-capable Bridge where the stream enters the network otherwise. The Boolean flag RTagStatus (6.5.3.9.1) contained in the Talker Announcement is then set by the station responsible

for redundancy tagging. If neither of these conditions for redundancy tagging is fulfilled, the Talker Announcement is failed by the Bridge where the stream enters the network.

b) Redundancy untagging is to be performed by a FRER-capable Bridge on a transmission Port where all of the Member Streams previously split from the original stream are merged back into a single Compound Stream. In case that no such Bridge Port for stream merging (6.3.9.5) exists on the path towards a given Listener, that Listener is responsible for redundancy untagging, if it is FRER-capable, and fails the Talker Announcement otherwise. The Boolean flag RTagStatus (6.5.3.9.1) contained in the Talker Announcement is cleared by the station responsible for redundancy untagging.

The RAP Failure Code *RTagFailed* defined in Table 6-10 is reported in case of failure to meet the conditions of redundancy tagging or untagging as described above.

6.3.9.4 Stream splitting

Stream splitting refers to a FRER operation in which a FRER-capable Talker or a FRER-capable Bridge uses the particular FRER functions, as shown in Table 6-5, to split a Compound Stream or a Member Stream into multiple Member Streams.

The potential locations in a network (including end stations) where a stream requesting seamless redundancy is subject to stream splitting are determined automatically during the stream's Multi-Context Talker Announcement in accordance with the following rules:

a) If the stream's Talker is FRER-capable, it is responsible for stream splitting by making several separate Talker Announce declarations, each with a different VLAN Context from the Redundancy Context chosen by the stream. Otherwise, it makes a Talker Announce declaration that include all the VLAN Contexts in the Redundancy Context, which indicates that the stream is expected to be transformed into a Compound Stream (i.e., redundancy tagging) and then split into Member Streams by a FRER-capable Bridge proxying for this Talker.

b) Each FRER-capable Bridge participated in the stream's Talker Announcement is responsible for applying stream splitting to a stream indicated by a Talker Announce registration in the Bridge, if following condition is met:
The Talker Announce registration contains more than one VLAN Context and is propagated by the Bridge to at least two other Ports. And at least one of these Ports is not in all of the VLAN Contexts as contained in the Talker Announce registration. This also indicates the fact that the VLAN topologies in which the Talker Announce registration is propagated are disjoint somehow on these Ports.

NOTE—Stream splitting is distinguished from the normal multicast, as the former transforms replicated streams into separate Member Streams with different VIDs. If a FRER-capable Bridge propagates a Multi-Context Talker Announce registration to several Ports each with the same set of VLAN Contexts, it will apply just multicast instead of stream splitting to that Compound Stream.

Meeting the conditions described in item b) above requires a proper placement of FRER-capable stations and VLAN configuration in the network. To prevent incorrect stream splitting operation, a Multi-Context Talker Announcement is failed by a Bridge with the RAP Failure Code *StreamSplitFailed* defined in Table 6-10, in either of the following three circumstances:

c) A Talker Announce registration in a Bridge that is not FRER-capable meets b).

d) If the frames cannot be forwarded to the port of the given Talker Announce declaration element.

e) If the frames cannot be sent with the given VLAN ID configuration at the port of the given Talker Announce declaration element.

6.3.9.5 Stream merging

Stream merging refers to a FRER operation in which a FRER-capable RAP Bridge or a FRER-capable RAP Listener performs the particular FRER function, as shown in Table 6-5, to combine more than one Member Stream into a Compound Stream.

The potential locations in a network (including end stations) where a stream requesting seamless redundancy is subject to stream merging are determined automatically during the stream's Multi-Context Talker Announcement in accordance with the following rules:

a) Each FRER-capable RAP Bridge participated in propagation of the stream's Talker Announcement is responsible for stream merging, if the Bridge propagates more than one Talker Announce registration of the stream on different Ports to one or more common destination Ports. Such a common destination Port, also termed stream merging Port, is in fact located at a converging point of all or some of the VLAN topologies in the Redundancy Context used by the stream.

b) A FRER-capable RAP Listener of the stream is responsible for stream merging, if it receives more than one Talker Announce registration of the stream, each containing only a single VLAN Context or partial ones of the Redundancy Context used by the stream.

Note that a Bridge Port or a Listener where all of the Member Streams split from the original stream are subject to stream merging into a Compound Stream is also responsible for applying redundancy untagging (6.3.9.3) to the Compound Stream following the stream merging operation.

6.3.10 Relationship of RAP to MSRP

RAP has conceptual and functional similarities to MSRP. Table 6-6 summarizes the similarities and differences between RAP and MSRP.

Table 6-6—RAP/MSRP differences

	MSRP ([Q]:35)	**RAP**
Attribute registration framework	MRP ([Q]:10)	LRP (IEEE Std 802.1CS)
Operational modes	peer-to-peer (MSRPv0) MRP External Control (MSRPv1)	Native, Controlled and Proxy system
Attribute types	[Q]:Table 35-1	Table 6-9
Traffic classes for streams	SR class	RA class
Traffic specification	[Q]:35.2.2.8.4 (MSRPv0) [Q]:35.2.2.10.6 (MSRPv1)	6.5.3.6 6.5.3.7
Queuing and transmission functions for streams	FQTSS in [Q]:34	RA class template based
Support for multiple-path streams	only in MSRPv1 with CNC	6.3.9

6.4 Definition of LRP protocol elements

This subclause specifies the LRP protocol elements that are specific to the operation of RAP.

6.4.1 Value of AppId

The general format of AppIds is defined in [CS]:9.2. The AppId that identifies the RAP specified in this standard shall take the values shown in Table 6-7.

Table 6-7—The AppId value for RAP

Field	Value	Length (octets)	Offset (octets)
OUI or CID	00-80-C2	3	0
Application Sub-ID	1	1	3

6.4.2 Use of LLDP

When the IEEE Std 802.1AB Link Layer Discovery Protocol (LLDP) is deployed to advertise and discover target ports, the destination MAC address of the LLDP agent that is selected to carry the AppId (6.4.1) in an LRP ECP Discovery TLV ([CS]:C.2.1) and/or an LRP TCP Discovery TLV ([CS]:C.2.2), shall be the Nearest Bridge group address (01-80-C2-00-00-0E) as specified in [Q]:Table 8-1, [Q]:Table 8-2, and [Q]:Table 8-3.

6.4.3 Use of LRP ECP mechanism

When the LRP-DT ECP mechanism ([CS]:6.9.1) is deployed, the destination MAC address of the ECP instance operating on each target port of that system shall be the Nearest Bridge group address (01-80-C2-00-00-0E) as specified in [Q]:Table 8-1, [Q]:Table 8-2, and [Q]:Table 8-3.

6.4.4 Application Information TLV

An Application Information TLV, in the format specified in [CS]:9.4.2.10 and exchanged via LRP Hello LRPDUs ([CS]:9.4.2), is used by each RAP Participant to advertise the information about its capabilities and other operational parameters to its neighbor(s). Figure 6-7 shows the encoding of the Value field of the Application Information TLV:

	Octet	Length
ProtocolVersion	1	1

Figure 6-7—Value of Application Information TLV

a) **ProtocolVersion**: The ProtocolVersion for the version of RAP defined in this standard takes the hexadecimal value 0x00.

6.4.5 Use of LRP-DS service interface

For the purposes of the operation of a RAP Participant (6.7), a set of primitives and associated parameters chosen from the LRP-DS service interface defined in [CS]:10 are summarized in Table 6-8.

Table 6-8—LRP-DS service primitives used by RAP

Primitive	Subclause in IEEE Std 802.1CS-2020	Parameter	Item in IEEE Std 802.1CS-2020
LOCAL_TARGET_PORT.request	10.2.2	(see Table 6-13)	
NEIGHBOR_TARGET_PORT.request	10.2.3		
ASSOCIATE_PORTAL.request	10.2.4	rPortalId	10.2.4.1 a)
		rIsApproved	10.2.4.1 b)
FIRST_HELLO.indication [a]	10.2.5	iPortalId	10.2.5.1 a)
		iHelloData	10.2.5.1 b)
PORTAL_STATUS.indication	10.2.6	iPortalId	10.2.6.1 a)
		iAssociationStatus	10.2.6.1 b)
WRITE_RECORD.request	10.3.1.1	rPortalId	10.3.1.1.1 a)
		rRecordNumber	10.3.1.1.1 b)
		rRecordData	10.3.1.1.1 c)
DELETE_RECORD.request	10.3.2.1	rPortalId	10.3.2.1.1 a)
		rRecordNumber	10.3.2.1.1 b)
RECORD_WRITTEN.indication	10.3.2.3	iPortalId	10.3.2.3.1 a)
		iRecordNumber	10.3.2.3.1 b)
		iRecordData	10.3.2.3.1 c)

[a] It is assumed that a FIRST_HELLO.indication primitive invoked due to receiving a Hello message on a given Port by LRP be passed to a RAP Participant on that Port.

6.5 RAP attribute and TLV encoding definitions

This subclause defines the RAP attributes and associated TLV encoding.

6.5.1 General TLV definition

The information of RAP attributes is encoded in a Type-Length-Value (TLV) format, which consists of a 1-octet Type field that indicates the type of that TLV, a 2-octet Length field that indicates the number of octets in the Value field, and then a Value field. The Value field of a TLV encodes zero or more fixed-size parameters, followed by zero or more TLVs of specific types. Figure 6-8 illustrates the general TLV format.

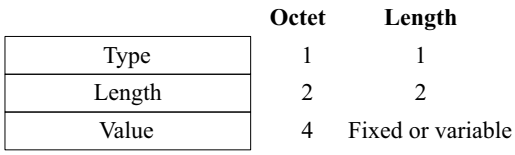


Figure 6-8—General TLV format

There are two categories of TLVs defined for RAP, attribute TLVs and sub-TLVs. An attribute TLV encodes a RAP attribute and can include one or more sub-TLVs in the Value field. An sub-TLV can further include one or more sub-TLVs. Attributes TLVs, as top-level TLVs, are never included in a sub-TLV or another attribute TLV. Table 6-9 lists the set of RAP attribute TLVs and sub-TLVs, and the values of the Type and Length field for each of them.

Table 6-9—RAP TLV and sub-TLV Type field and Length field values

TLV name	Type field (1 octet)	Length field (2 octets)	Reference
RA attribute TLV	0x00	variable	6.5.2
Talker Announce attribute TLV	0x01	variable	6.5.3
Listener Attach attribute TLV	0x02	10	6.5.4
RA Class Descriptor sub-TLV	0x20	variable	6.5.2.2
Redundancy Context sub-TLV	0x21	variable	6.5.2.3
Data Frame Parameters sub-TLV	0x22	8	6.5.3.5
Talker Token Bucket TSpec sub-TLV	0x23	16	6.5.3.6
Interval-based TSpec sub-TLV	0x24	8	6.5.3.7
Token Bucket TSpec sub-TLV	0x25	16	6.5.3.8
Redundancy Control sub-TLV	0x26	variable	6.5.3.9
VLAN Context Information sub-TLV	0x27	variable	6.5.3.9.2
Failure Information sub-TLV	0x28	9	6.5.3.10
Organizationally Defined sub-TLV	0x29	variable	6.5.3.11
VLAN Context Status sub-TLV	0x2A	variable	6.5.4.4
Reserved for future standardization	0x03-0x1F, 0x2B-0xFF	-	-

The subsequent subclauses specify the encoding of the Value field for each attribute TLV and sub-TLV, by using the "big-endian" convention, in which higher significance bytes and bits within each byte appear to the left of and above bytes and bits of lower significance. Any fields that are labeled as "Reserved" are transmitted as zero and ignored on receipt. In addition, the term "the stream" used in the specification of Talker Announce attribute TLV and the sub-TLVs thereof refers to a stream for which the Talker Announce attribute is declared.

6.5.2 RA attribute and TLV encoding

The RA attribute is used in RA Advertisements (6.3.3). Figure 6-9 shows the encoding of the Value field of the RA attribute TLV.

	Octet	Length
MaxInterferingFrameSize	1	2
1 or more RA Class Desriptor sub-TLVs	3	variable
zero or more Redundancy Context sub-TLVs	variable	variable

Figure 6-9—Value of RA attribute TLV

6.5.2.1 MaxInterferingFrameSize

A 2-octet unsigned integer, indicating the maximum frame size, in bytes, including media-dependent overhead ([Q]:12.4.2.2), that is allowed to be transmitted through the Port declaring the RA class attribute. The value of this parameter is determined by the operation of the underlying MAC Service on the Port.

6.5.2.2 RA Class Descriptor sub-TLV

One or more RA Class Descriptor sub-TLVs are included in the RA attribute TLV, each encoding in the Value field a set of parameters associated with a distinct RA class (6.3.2) as illustrated in Figure 6-10.

	Octet	Length
RaClassId	1	1
RaClassPriority	2	1
RTID	3	4
TrafficClass	7	1
MaxLastHopLatency	8	4
RaClassTemplateDefinedData	12	variable

Figure 6-10—Value of RA Class Descriptor sub-TLV

6.5.2.2.1 RaClassId

A 1-octet unsigned integer containing the RA class ID (6.3.2.1).

6.5.2.2.2 RaClassPriority

A 1-octet unsigned integer containing the RA class priority (6.3.2.2).

6.5.2.2.3 RTID

A 4-octet RTID (6.3.2.3) identifying the RA class template used by the RA class.

6.5.2.2.4 TrafficClass

A 1-octet unsigned integer, indicating the traffic class ([Q]:8.6.6) to which the RA class priority is mapped on the Port where the RA attribute is declared.

6.5.2.2.5 MaxLastHopLatency

A 4-octet unsigned integer, indicating a latency value, in nanoseconds, that specifies a latency constraint (6.3.6.1) for a neighboring end station on the Port where the RA attribute is declared.

The latency contained in this field is measured from a point located in this Port, to a point located in the neighboring end station, while the exact measurement points are specific to and defined by the RA class template being used by the RA class.

NOTE—This parameter is only intended for use by an end station when acting as a Listener and can be ignored at reception by a Bridge or a Talker-only end station. This also implies that an RA declaration made on a Port to which no Listener end station is expected to be connected can take an arbitrary value in this field.

6.5.2.2.6 RaClassTemplateDefinedData

The encoding of this field and the semantics associated with its values if any, is specific to the RA class template identified by the value contained in 6.5.2.2.3.

6.5.2.3 Redundancy Context sub-TLV

One or more Redundancy Context sub-TLVs are included in the RA attribute TLV, each describing a distinct Redundancy Context (6.3.9.2). The Value field contains one or more VLAN Context tuples, each encoding in the VlanContextId field a 12-bit VLAN Context ID, followed by a 4-bit Reserved field, as illustrated in Figure 6-11.

		Octet	Length
VLAN Context tuple 1	VlanContextId	1	12 bits
	Reserved	2	4 bits
	...		
VLAN Context tuple n	VlanContextId	2n-1	12 bits
	Reserved	2n	4 bits

Figure 6-11—Value of Redundancy Context sub-TLV

6.5.3 Talker Announce attribute and TLV encoding

The Talker Announce attribute is used in the Talker Announcement (6.3.4). Figure 6-12 shows the encoding of the Value Field of the Talker Announce attribute TLV.

	Octet	Length
StreamId	1	8
StreamRank	9	1
AccuMaxLatency	10	4
AccuMinLatency	14	4
Data Frame Parameters sub-TLV	18	11
Talker Token Bucket TSpec sub-TLV or Interval-based TSpec sub-TLV (Talker TSpec)	29	19 or 11
Token Bucket TSpec sub-TLV (Network TSpec)	variable	19
Redundancy Control sub-TLV *(only for Multi-Context Talker Announcement)*	variable	variable
0 or 1 Failure Information sub-TLV	variable	variable
0 or more Organizationally Defined sub-TLVs	variable	variable

Figure 6-12—Value of Talker Announce attribute TLV

For support of two types of the Talker Announcement, the encoding of the Talker Announce attribute follows the following rules:

a) The Talker Announce attribute used in the Single-Context Talker Announcement [item a) in 6.3.4.1.1] contain no Redundancy Control sub-TLV (6.5.3.9).

b) The Talker Announce attribute used in the Multi-Context Talker Announcement [item b) in 6.3.4.1.1] always contains a Redundancy Control sub-TLV (6.5.3.9).

6.5.3.1 StreamId

An 8-octet field encoding the StreamID element as specified in [Q]:46.2.3.1.

6.5.3.2 StreamRank

A 1-octet field encoding a Rank value as specified in [Q]:46.2.3.2.1.

6.5.3.3 AccuMaxLatency

A 4-octet field encoding the AccumulatedLatency element as specified in [Q]:46.2.5.2.

6.5.3.4 AccuMinLatency

A 4-octet unsigned integer, indicating the minimum latency, in nanoseconds, that a single frame of the stream can encounter when transmitted from the Talker along a given path to the Port declaring this Talker Announce attribute.

6.5.3.5 Data Frame Parameters sub-TLV

This sub-TLV contains a set of parameters whose values are carried in the data frames of the stream when transmitted through the declaring Port of this attribute. Figure 6-13 shows the encoding of the Value field of the Data Frame Parameters sub-TLV.

	Octet	Length
DestinationMacAddress	1	6
Priority	7	3 bits
Reserved	7	1 bit
VID	7	12 bits

Figure 6-13—Value of Data Frame Parameters sub-TLV

6.5.3.5.1 DestinationMacAddress

A 6-octet destination MAC address of the data frames of the stream.

6.5.3.5.2 Priority

A 3-bit unsigned integer, indicating the priority to be encoded in the PCP field of the VLAN tag ([Q]:6.9.3), with which the data frames of the stream are tagged. This priority value is used by each receiving Bridge to associate the stream to a local RA class of the same priority.

6.5.3.5.3 VID

A 12-bit VID. The semantics of this field is dependent on the type of the Talker Announcement in which the Talker Announce attribute is used, as follows:

a) In the case of the Single-Context Talker Announcement, this field indicates a VID to be encoded in the VLAN tag with which the data frames of the stream are tagged, and also a single VLAN Context used by the Talker Announce attribute.
b) In the case of the Multi-Context Talker Announcement, this field is set to the numerically smallest VlanContextId value contained in a VLAN Context Information sub-TLV (6.5.3.9.2).

6.5.3.6 Talker Token Bucket TSpec sub-TLV

This sub-TLV contains a Talker Token Bucket TSpec as defined in item a) of 6.3.7. Figure 6-14 shows the encoding of the Value field of the Talker Token Bucket TSpec sub-TLV.

	Octet	Length
MaxPayloadLength	1	2
MinPayloadLength	3	2
CommittedInformationRate	5	8
CommittedBurstSize	13	4

Figure 6-14—Value of Talker Token Bucket TSpec sub-TLV

6.5.3.6.1 MaxPayloadLength

A 2-octet unsigned integer, encoding the MaxPayloadLength parameter [item a).1) of 6.3.7.1].

6.5.3.6.2 MinPayloadLength

A 2-octet unsigned integer, encoding the MinPayloadLength parameter [item a).2) of 6.3.7.1].

6.5.3.6.3 CommittedInformationRate

A 8-octet unsigned integer, encoding the CommittedInformationRate parameter [item a).3) of 6.3.7.1].

6.5.3.6.4 CommittedBurstSize

A 4-octet unsigned integer, encoding the CommittedBurstSize parameter [item a).4) of 6.3.7.1].

6.5.3.7 Interval-based TSpec sub-TLV

This sub-TLV contains an Interval-based TSpec as defined in item b) of 6.3.7.1. Figure 6-15 shows the encoding of the Value field of the Interval-based TSpec sub-TLV.

	Octet	Length
Interval	1	4
MaximumFramesPerInterval	5	2
MaximumFrameSize	7	2

Figure 6-15—Value of Interval-based TSpec sub-TLV

6.5.3.7.1 Interval

A 4-octet unsigned integer, encoding the Interval parameter [item b).1) of 6.3.7.1].

6.5.3.7.2 MaximumFramesPerInterval

A 2-octet unsigned integer, encoding the MaximumFramesPerInterval parameter [item b).2) of 6.3.7.1].

6.5.3.7.3 MaximumFrameSize

A 2-octet unsigned integer, encoding the MaximumFrameSize parameter [item b).3) of 6.3.7.1].

6.5.3.8 Token Bucket TSpec sub-TLV

This sub-TLV contains a Token Bucket TSpec as defined in 6.3.7.2. Figure 6-16 shows the encoding of the Value field of the Token Bucket TSpec sub-TLV.

	Octet	Length
MaxTransmittedFrameLength	1	2
MinTransmittedFrameLength	3	2
CommittedInformationRate	5	8
CommittedBurstSize	13	4

Figure 6-16—Value of Token Bucket TSpec sub-TLV

6.5.3.8.1 MaxTransmittedFrameLength

A 2-octet unsigned integer, encoding the MaxTransmittedFramelength parameter [item 1) of 6.3.7.2].

6.5.3.8.2 MinTransmittedFrameLength

A 2-octet unsigned integer, encoding the MinTransmittedFramelength parameter [item 2) of 6.3.7.2].

6.5.3.8.3 CommittedInformationRate

A 8-octet unsigned integer, encoding the CommittedInformationRate parameter [item 3) of 6.3.7.2].

6.5.3.8.4 CommittedBurstSize

A 4-octet unsigned integer, encoding the CommittedBurstSize parameter [item 4) of 6.3.7.2].

6.5.3.9 Redundancy Control sub-TLV

The Redundancy Control sub-TLV contains the information exclusively used by the Multi-Context Talker Announcement. Figure 6-17 shows the encoding of the Value field of the Redundancy Control sub-TLV.

MaxTransmittedFrameLength	1	2
MinTransmittedFrameLength	3	2
CommittedInformationRate	5	8
CommittedBurstSize	13	4

Figure 6-17—Value of Redundancy Control sub-TLV

6.5.3.9.1 RTagStatus

A Boolean value indicating whether the data frames of the stream are (TRUE) or are not (FALSE) to contain a Redundancy tag (R-TAG, [CB]:7.8) when transmitted through the Port that declares this sub-TLV.

6.5.3.9.2 VLAN Context Information sub-TLV

One or more VLAN Context Information sub-TLVs are included in the Redundancy Control sub-TLV, each describing a distinct VLAN Context (6.3.4.1.1). The Value field encodes in the VlanContextId field a 12-bit VLAN Context ID, followed by a 4-bit Reserved field and then zero or one Failure Information sub-TLV (6.5.3.10), as illustrated in Figure 6-18.

	Octet	**Length**
RTagStatus	1	1 bit
Reserved	1	7 bits
1 or more VLAN Context Informtion sub-TLVs	2	variable

Figure 6-18—Value of VLAN Context Information sub-TLV

6.5.3.10 Failure Information sub-TLV

The Failure Information sub-TLV contains the information about the location and the type of a failure encountered in the Talker Announcement. Figure 6-19 shows the encoding of the Value field of the Failure Information sub-TLV.

VlanContextId	1	12 bits
Reserved	2	4 bits
0 or 1 Failure Information sub-TLV	3	variable

Figure 6-19—Value of Failure Information sub-TLV

6.5.3.10.1 SystemId

An 8-octet unsigned integer, indicating the identifier of a Bridge or end station at which the failure occurs. The SystemId value for an end station shall be the 6-octet MAC Address of the end station's port extended to 8 octets by prepending 2 octets of zero. The SystemId value for a Bridge shall be the 6-octet Bridge Identifier ([Q]:13.26.2) of the Bridge extended to 8 octets by prepending 2 octets of zero.

6.5.3.10.2 FailureCode

A 1-octet unsigned integer, containing the value of a RAP Failure Code defined in Table 6-10.

Table 6-10—RAP Failure Codes

Name	Value	Description	Reference
LatencyExceeded	0x01	Latency constraint not met	6.3.6.1
BandwidthExceeded	0x02	Bandwidth constraint not met	6.3.6.2
ResourceExceeded	0x03	Resource constraint not met	6.3.6.3
CrossingDomainBoundary	0x04	Talker Announcement across RA Class domain boundary	6.8.5.5
ResourceAllocationFailed	0x05	Resource allocation failed	6.8.5.42
ReservationPreempted	0x06	Reservation preempted by Rank zero streams	6.8.5.41
RTagFailed	0x07	Requirements for redundancy (un)tagging not fullfilled	6.3.9.3
StreamSplitFailed	0x08	Requirements for Stream splitting not fulfilled	6.3.9.4
MissingTalkerAnnounce	0x09	Missing Talker Announce at a Stream merging point	6.8.5.10
ListenerLackingFrer	0x0A	Listener lacking FRER capability	6.6.3.6
InconsistentRedundancyContext	0x0B	Inconsistent configuration of Redundancy Context	6.3.9.2

6.5.3.11 Organizationally Defined sub-TLV

The Organizationally Defined sub-TLV contains information defined by an organization that owns an OUI or a CID obtained from the IEEE registration authority. Figure 6-20 shows the encoding of the Value field of the Organizationally Defined sub-TLV.

	Octet	Length
SystemId	1	8
FailureCode	9	1

Figure 6-20—Value of Organizationally Defined sub-TLV

6.5.3.11.1 OUI/CID

A 3-octet OUI or an CID.

6.5.3.11.2 OrganizationallyDefinedData

The encoding of this field and the semantics associated with its values if any, is specific to and defined by the organization that owns the OUI/CID contained in 6.5.3.11.1.

6.5.4 Listener Attach attribute and TLV encoding

The Listener Attach attribute is used in the Listener Attachment (6.3.5). Figure 6-21 shows the encoding of the Value field of the Listener Attach attribute TLV.

	Octet	Length
OUI/CID	1	3
OrganizationallyDefinedData	4	variable

Figure 6-21—Value of Listener Attach attribute TLV

For support of two types of the Listener Attachment, the encoding of the Listener Attach attribute follows the following rules:

a) The Listener Attach attribute used in the Single-Context Listener Attachment [item a) in 6.3.5.1] contains no VLAN Context Status sub-TLV (6.5.4.4).

b) The Listener Attach attribute used in a Multi-Context Listener Attachment [item b) in 6.3.5.1] contains one VLAN Context Status sub-TLV (6.5.4.4).

6.5.4.1 StreamId

An 8-octet field encoding the StreamID element as specified in [Q]:46.2.3.1.

6.5.4.2 VID

A 12-bit integer, indicating a VID to be encoded in the VLAN tag with which the data frames of the stream are tagged.

6.5.4.3 ListenerAttachStatus

A 4-bit field, taking one of the following enumerated values to indicate the Listener Attach status (6.3.5.4):

a) **1: Attach Ready** [item a) in 6.3.5.4];
b) **2: Attach Fail** [item b) in 6.3.5.4];
c) **3: Attach Partial Fail** [item c) in 6.3.5.4].

6.5.4.4 VLAN Context Status sub-TLV

The VLAN Context Status sub-TLV contains the information exclusively used by the Multi-Context Listener Attachment. The Value field contains one or more VLAN Context tuples, each encoding in VlanContextId field a 12-bit VLAN Context ID, followed by a 4-bit PathStatus field, as illustrated in Figure 6-22.

	Octet	Length
StreamId	1	8
VID	9	12 bits
ListenerAttachStatus	10	4 bits
VLAN Context Status sub-TLV *(only for Multi-Context Listener Attachment)*	11	variable

Figure 6-22—Value of VLAN Context Status sub-TLV

6.5.4.4.1 PathStatus

A 4-bit field, taking one of the following enumerated values to indicate the Path status (6.3.5.5):

a) **0: Faultless Path** [item a) in 6.3.5.5];
b) **1: Faulty Path** [item b) in 6.3.5.5];
c) **2: Unresponsive Path** [item c) in 6.3.5.5].

6.6 RAP Endpoint

6.6.1 RAP Endpoint overview

A RAP Endpoint associated with an end station is responsible for the following:

— Provision of service interface to higher layer entities (termed “RAP Endpoint user”) for controlling of resource allocation by end stations.
— Handling of resource allocation for an end station.
— Generation of attributes for declarations and processing of received attribute registrations.

The operation of a RAP Endpoint is specified by the following:

— RAP Endpoint Service Interface (RESI) in 6.6.2.
— RAP Endpoint procedures in 6.6.3.

6.6.2 RAP Endpoint Service Interface (RESI)

The RESI provided to RAP Endpoint user comprises a set of primitives and associated parameters, which are divided into three groups, as summarized in Table 6-11.

Table 6-11—RAP Endpoint Service Interface primitives

	primitive	reference
RA primitives (6.6.2.1)	DECLARE_RA.request	6.6.2.1.1
	REGISTER_RA.indication	6.6.2.1.2
TA primitives (6.6.2.2)	ANNOUNCE_STREAM.request	6.6.2.2.1
	ANNOUNCE_STREAM.indication	6.6.2.2.2
	TERMINATE_STREAM.request	6.6.2.2.3
	TERMINATE_STREAM.indication	6.6.2.2.4
LA primitives (6.6.2.3)	ATTACH_STREAM.request	6.6.2.3.1
	ATTACH_STREAM.indication	6.6.2.3.2
	DETACH_STREAM.request	6.6.2.3.3
	DETACH_STREAM.indication	6.6.2.3.4

6.6.2.1 RA primitives

The RA primitives specified in 6.6.2.1.1 and 6.6.2.1.2 are used for controlling and monitoring RA Advertisement (6.3.3) in an end station.

The method of using the RA primitives is specific to each end station. For example, if the RAP Endpoint user in an end station is made aware of the RA class and Redundancy Context configurations in the network through a higher layer protocol, it can make an RA declaration without having to wait for an RA registration to be received from a neighboring station. Otherwise, the end station relies on the information received in an RA registration from a neighboring station to make an RA declaration.

6.6.2.1.1 DECLARE_RA.request()

This primitive is used by the RAP Endpoint user to request the RAP Endpoint to make an RA declaration. It has the following parameters:

a) **MaxInterferingFrameSize** (6.5.2.1)
b) One or more sets of the following parameters, each associated with a distinct RA class (6.5.2.2):
    1) **RAClassId** (6.5.2.2.1)
    2) **RaClassPriority** (6.5.2.2.2)
    3) **RTID** (6.5.2.2.3)
    4) **TrafficClass** (6.5.2.2.4)
    5) **MaxLastHopLatency** (6.5.2.2.5)
    6) **RaClassTemplateDefinedData** (6.5.2.2.6)
c) Zero or more sets of **VLAN Context ID**s, each associated with an Redundancy Context (6.5.2.3).

6.6.2.1.2 REGISTER_RA.indication()

The REGISTER_RA.indication primitive is used by the RAP Endpoint to inform the RAP Endpoint user about an RA registration. This primitive has the same set of parameters as that defined in 6.6.2.1.1.

6.6.2.2 TA primitives

The TA primitives specified in the subsequent clauses (6.6.2.2.1 through 6.6.2.3.4) are used for controlling and monitoring Talker Announcement (6.3.4).

6.6.2.2.1 ANNOUNCE_STREAM.request()

This primitive is used by the RAP Endpoint user in a Talker to request the RAP Endpoint to initiate a Talker Announcement. It has the following parameters:

a) **StreamId** (6.5.3.1)
b) **StreamRank** (6.5.3.2)
c) **AccuMaxLatency** (6.5.3.3)
d) **AccuMinLatency** (6.5.3.4)
e) A 3-tuple of {**DestinationMacAddress** (6.5.3.5.1), **Priority** (6.5.3.5.2) and **VID** (6.5.3.5.3)}
f) Either of the following as TalkerTSpec (6.5.3.6):
    1) A 4-tuple of {**MaxTransmittedFrameLength** (6.5.3.6.1), **MinTransmittedFrameLength** (6.5.3.6.2), **CommittedInformationRate** (6.5.3.6.3) and **CommittedBurstSize** (6.5.3.6.4)}, indicating a Token Bucket TSpec (6.5.3.6)
    2) A 3-tuple of {**Interval** (6.5.3.7.1), **MaximumFramesPerInterval** (6.5.3.7.2) and **MaximumFrameSize** (6.5.3.7.3)}, indicating an Interval-based TSpec (6.5.3.6)
g) **Redundancy Context ID** (6.3.9.2), if supplied, indicating a Multi-Context Talker Announcement and otherwise, indicating a Single-Context Talker Announcement
h) A 2-tuple of **{OUI/CID** (6.5.3.11.1) and **OrganizationallyDefinedData** (6.5.3.11.2)}, if supplied, indicating the use of the Organizationally Defined sub-TLV (6.5.3.11)
i) **Failure Information**, set to either zero or a non-zero RAP Failure Code defined in Table 6-10.

The RAP Endpoint shall not accept an ANNOUNCE_STREAM.request with a StreamId value for which it has initiated a Talker Announcement since a previous ANNOUNCE_STREAM.request.

6.6.2.2.2 ANNOUNCE_STREAM.indication()

This primitive is used by the RAP Endpoint to inform the RAP Endpoint user in a Listener about the result of processing a Talker Announcement received. In addition to all the parameters defined in 6.6.2.2.1, this primitive has the following additional parameter for use only in the case of a Multi-Context Talker Announcement:

a) A set of 3-tuples, each with {**VlanContextId**, **SystemId, FailureCode**} and corresponding to a VLAN Context Information sub-TLV (6.5.3.9.2), where SystemId and FailureCode are provided only when that VlanContextId is reported with a RAP Failure Code in the Talker Announcement.

NOTE—As a response to a received ANNOUNCE_STREAM.indication, the RAP Endpoint user of a Listener can, if it is interested in receiving that stream, initiates a Listener Attachment by issuing an ATTACH_STREAM.request (6.6.2.3.1) with the StreamId contained in the received primitive.

6.6.2.2.3 TERMINATE_STREAM.request()

This primitive is used by the RAP Endpoint user in a Talker to request the RAP Endpoint to terminate a Talker Announcement. It has a single parameter StreamId (6.5.3.1).

If a Talker Announcement to be terminated is associated with a stream being transmitted by the Talker, it is required that the Talker stops transmitting the stream before issuing a TERMINATE_STREAM.request.

NOTE—This requirement can be viewed as part of the fundamental principle of resource allocation, i.e., reservation prior to stream transmission.

6.6.2.2.4 TERMINATE_STREAM.indication()

This primitive is used by the RAP Endpoint to inform the RAP Endpoint user in a Listener that a Talker Announcement previously received has been removed. It has a single parameter StreamId (6.5.3.1).

NOTE—As a response to a received TERMINATE_STREAM.indication, the RAP Endpoint user of a Listener can, if it has initiated a corresponding Listener Attachment, terminate that Listener Attachment by issuing a DETACH_STREAM.request (6.6.2.3.3) with the StreamId contained in the received primitive.

6.6.2.3 LA primitives

The Listener primitives specified in the following subclauses (6.6.2.3.1 through 6.6.2.2.4) are used for controlling and monitoring Listener Attachment (6.3.5).

6.6.2.3.1 ATTACH_STREAM.request()

This primitive is used by the RAP Endpoint user in a Listener to request the RAP Endpoint to initiate a Listener Attachment. It has a single parameter StreamId (6.5.4.1).

NOTE—An ATTACH_STREAM.request issued with a given StreamId won’t actually cause initiation of a Listener Attachment, if a Talker Announce registration with that StreamId is not present in the RAP Endpoint.

6.6.2.3.2 ATTACH_STREAM.indication()

This primitive is used by the RAP Endpoint to inform the RAP Endpoint user in a Talker about the result of processing a Listener Attachment received. It has the following parameters:

a) **StreamId** (6.5.4.1)
b) **VID** (6.5.4.2)
c) **ListenerAttachStatus** (6.5.4.3)

d) A set of 2-tuples, each with {**VlanContextId**, **PathStatus**} (6.5.4.4)

NOTE—The ListenerAttachStatus value and in the case of a Multi-Context Listener Attachment the set of PathStatus values contained in a received ATTACH_STREAM.indication can be used by the RAP Endpoint user to instruct the Talker whether and when to start transmission of the stream concerned, according to the semantics associated with these status parameters.

6.6.2.3.3 DETACH_STREAM.request()

This primitive is used by the RAP Endpoint user in a Listener to request the RAP Endpoint to terminate a Listener Attachment previously initiated. It has a single parameter StreamId (6.5.4.1).

NOTE—The RAP Endpoint user of a Listener can issue a DETACH_STREAM.request to terminate a specific Listener Attachment it has initiated previously at any time as desired, including but not limited to the case when receiving a TERMINATE_STREAM.indication (6.6.2.2.4) that indicates the associated Talker Announcement has been terminated.

6.6.2.3.4 DETACH_STREAM.indication()

This primitive is used by the RAP Endpoint to inform the RAP Endpoint user in a Talker that a Listener Attachment previously received for a Talker Announcement initiated by the Talker. It has a single parameter StreamId (6.5.3.1).

NOTE—A notification of a removed Listener Attachment occurs only when the RAP Endpoint of a Talker has removed any Listener Attach registration previously received as belonging to that Listener Attachment. In this case, it is up to the RAP Endpoint user to determine whether to retain the Talker Announcement concerned.

6.6.3 RAP Endpoint procedures

The subsequent subclauses (6.6.3.1 through 6.6.3.12) specify a set of procedures for the RAP Endpoint to process request primitives received at RESI (6.6.2) and indication primitives received at RPSI (6.7.2).

The procedures specified in these subclauses assume that the information about each attribute being declared and each attribute being registered is maintained in a database and accessible for the RAP Endpoint. A description about the data structure and operation methods of such a database is not provided; it is an implementation choice.

6.6.3.1 handleDeclareRaRequest()

This procedure is invoked when the RAP Endpoint receives at RESI a DECLARE_RA.request (6.6.2.1.1). It constructs the RA attribute (6.5.2) with the parameter values supplied in the primitive and issues a DECLARE_ATTR.request (6.7.2.1) at RPSI with the constructed attribute.

6.6.3.2 handleRaRegIndication()

This procedure is invoked when the RAP Endpoint receives at RPSI an ATTR_REG.indication (6.7.2.3) that indicates an RA registration. It issues at RESI a REGISTER_RA.indication (6.6.2.1.2) with the parameter values abstracted from the RA registration supplied in the primitive.

6.6.3.3 handleAnnounceStreamRequest()

This procedure is invoked when the RAP Endpoint receives at RESI an ANNOUNCE_STREAM.request (6.6.2.2.1). It performs the following actions:

a) If there exist one or more Talker Announce declarations associated with the same StreamId (6.5.3.1) as indicated in the primitive, terminate the procedure.
b) Otherwise, construct a Talker Announce attribute TLV (6.5.3) in tTaAttr with the values supplied in items a), b), c), d), e), f) and h) of 6.6.2.2.1.
c) If a non-zero RAP Failure Code value is supplied in i) of 6.6.2.2.1, construct a Failure Information sub-TLV (6.5.3.10) with that RAP Failure Code and the SystemId value of this end station, and then append it to tTaAttr.
d) Construct in tTaAttr.NetworkTSpec a Token Bucket TSpec sub-TLV (6.5.3.6) with the values either copied from those supplied in item f).1) of 6.6.2.2.1 for a Token Bucket TSpec, or translated from those supplied in item f).2) of 6.6.2.2.1 for an Interval-based TSpec according to Table 6-4.
e) If no value is supplied in g) of 6.6.2.2.1, issue a DECLARE_ATTR.request (6.7.2.1) at RPSI with tTaAttr and then terminate. Otherwise, set tRedundancyContextId to the value supplied.
f) If the end station is not FRER-capable, perform the following:
    1) Construct a Redundancy Control sub-TLV (6.5.3.9) with RTagStatus (6.5.3.9.1) set FALSE and with a set of VLAN Context Information sub-TLVs (6.5.3.9.2) constructed for each VLAN Context ID belonging to the Redundancy Context identified by tRedundancyContextId, and then append it to tTaAttr.
    2) Set tAttr.VID to tRedundancyContextId.
    3) Issue a DECLARE_ATTR.request (6.7.2.1) at RPSI with tTaAttr.
g) Otherwise, i.e., the end station is FRER-capable, for each VLAN Context ID (tVlanContextId) belonging to the Redundancy Context identified by tRedundancyContextId, perform the following:
    1) Copy tTaAttr to tTaAttrSplit.
    2) Construct a Redundancy Control sub-TLV (6.5.3.9) with RTagStatus (6.5.3.9.1) set TRUE and with a single VLAN Context Information sub-TLV (6.5.3.9.2) constructed for tVlanContextId, and then append it to tTaAttrSplit.
    3) Set tTaAttrSplit.VID to tVlanContextId.
    4) Issue a DECLARE_ATTR.request (6.7.2.1) at RPSI with tTaAttrSplit.

6.6.3.4 handleTerminateStreamRequest(pStreamId)

This procedure is invoked when the RAP Endpoint receives a TERMINATE_STREAM.request (6.6.2.2.3) at RESI with a StreamId value (pStreamId). It performs the following actions:

a) If there is no Talker Announce declaration associated with pStreamId, terminate.
b) Otherwise, for each Talker Announce declaration associated with pStreamId, issue at RPSI a WITHDRAW_ATTR.request (6.7.2.2) with the corresponding Talker Announce attribute and then remove the Talker Announce declaration.

6.6.3.5 handleTaReg(pTaAttr)

This procedure is invoked when the RAP Endpoint receives at RPSI an ATTR_REG.indication (6.7.2.3) that indicates a Talker Announce registration (pTaAttr). It performs the following actions:

a) If pAttr indicates a Multi-Context Talker Announcement and pTaAttr.RedundancyControl contains only part of the VLAN Contexts of a Redundancy Context declared by this end station, call **handleTaMerge**(pTaAttr.StreamId) and terminate.
b) Copy pTaAttr to tTaAttr.

c) If tTaAttr indicates **Announce Success**, perform checks of resource allocation constraints (6.3.6) on tTaAttr and then the following:
   1) If all the resource allocation constraints are met, update both tTaAttr.AccuMaxLatency and tTaAttr.AccuMinLatency.
   2) Otherwise, fail tTaAttr with the RAP Failure Code associated with a failure encountered during the constraint checking.

d) Issue at RESI an ANNOUCE_STREAM.indication (6.6.2.2.2) with its parameters set to the corresponding values contained in tTaAttr.

6.6.3.6 handleTaMerge(pStreamId)

This procedure is invoked by the **handleTaReg**() procedure (6.6.3.5) to process a Multi-Context Talker Announcement identified by a StreamId (pStreamId) and subject to Talker Announce merge (6.3.4.3) in the end station. It performs the following actions:

a) Search for each existing Talker Announce registration (tTaAttr) associated with pStreamId. If no such tTaAttr is found, issue a TERMINATE_STREAM.indication (6.6.2.2.4) with pStreamId and then terminate. Otherwise, proceed.

b) Construct a Talker Announce attribute in tTaAttrMerge and fill each of the following fields with the common value contained in the corresponding field of each tTaAttr found in a): tTaAttrMerge.StreamId, tTaAttrMerge.StreamRank, tTaAttrMerge.DestinationMacAddress and tTaAttrMerge.Priority.

c) If one of the following conditions is met, fail tTaAttrMerge with the indicated RAP Failure Code and then proceed to g) below:
   1) *ListenerLackingFrer*, if this end station is not FRER-capable.
   2) *RTagFailed,* if there is at least one tTaAttr whose RTagStatus contains FALSE.

d) Copy each VLAN Context Information sub-TLV (6.5.3.9.2) contained in each tTaAttr found in a) to tTaAttrMerge.RedundancyControl, and set tTaAttrMerge.VID to the numerically smallest VLAN Context ID copied.

e) If each tTaAttr found in a) indicates **Announce Fail**, fail tTaAttrMerge with the numerically smallest RAP Failure Code contained in tTaAttrMerge.RedundancyControl and then proceed to g).

f) If tTaAttrMerge indicates **Announce Success**, perform checks of resource allocation constraints (6.3.6) on tTaAttrMerge and then the following:
   1) If all the resource allocation constraints are met, update both tTaAttrMerge.AccuMaxLatency and tTaAttrMerge.AccuMinLatency.
   2) Otherwise, fail tTaAttrMerge with the RAP Failure Code associated with a failure encountered during the constraint checking.

g) Issue at RESI an ANNOUCE_STREAM.indication (6.6.2.2.2) with its parameters set to the corresponding values contained in tTaAttrMerge.

6.6.3.7 handleTaDereg(pTaAttr)

This procedure is invoked when the RAP Endpoint receives at RPSI an ATTR_DEREG.indication (6.7.2.4) that indicates a Talker Announce deregistration (pTaAttr). It performs the following actions:

a) Set tStreamId = pTaAttr.StreamId and then remove the Talker Announce registration in pTaAttr.

b) If there is no remaining Talker Announce registration associated with tStreamId, issue at RESI a TERMINATE_STREAM.indication (6.6.2.2.4) with tStreamId.

c) Otherwise, call **handleTaMerge**(tStreamId).

6.6.3.8 handleAttachStreamRequest(pStreamId)

This procedure is invoked when the RAP Endpoint receives at RESI an ATTACH_STREAM.request (6.6.2.3.1) with a StreamId value (pStreamId). It performs the following actions:

a) Search for each Talker Announce registration (tTaAttr) associated with pStreamId. If no such tTaAttr is found, terminate; otherwise proceed.
b) If there is no reservation associated with pStreamId in the end station, and the Talker Announcement identified by pStreamId is not failed, make a reservation for pStreamId.
c) For each tTaAttr found in a), issue at RPSI a DECLARE_ATTR.request (6.7.2.1) with a Listener Attach attribute (tLaAttr) constructed as follows:
    1) Set tLaAttr.StreamId to tTaAttr.StreamId.
    2) Set tLaAttr.VID to tTaAttr.VID.
    3) Set tLaAttr.ListenerAttachStatus to **Attach Ready**, if there is a reservation associated with pStreamId, and **Attach Fail**, otherwise.
    4) If tTaAttr indicates a Multi-Context Talker Announcement, construct a VLAN Context Status sub-TLV (6.5.4.4) with each VLAN Context Id contained in tTaAttr, and the associated PathStatus set to **Faultless Path** if tLaAttr indicates **Attach Ready**, and **Faulty Path** if tLaAttr indicates **Attach Fail**, and then append it to tLaAttr.

6.6.3.9 handleDetachStreamRequest(pStreamId)

This procedure is invoked when the RAP Endpoint receives at RESI a DETACH_STREAM.request (6.6.2.3.3) with a StreamId value (pStreamId). It performs the following actions:

a) If there is no Listener Attach declaration associated with pStreamId, terminate.
b) If there is a reservation associated with pStreamId, remove the reservation.
c) For each Listener Attach declaration associated with pStreamId, issue at RPSI a WITHDRAW_ATTR.request (6.7.2.2) with the corresponding Listener Attach attribute and then remove the Listener Attach declaration.

6.6.3.10 handleLaReg(pLaAttr)

This procedure is invoked when the RAP Endpoint receives at RPSI an ATTR_REG.indication (6.7.2.3) that indicates a Listener Attach registration (pLaAttr). It performs the following actions:

a) Search for a Talker Announce declaration (tTaAttr) with which pLaAttr is associated, in accordance with 6.3.5.1. If no such tTaAttr is found, terminate; otherwise, proceed.
b) If tLaAttr indicates either **Attach Ready** or **Attach Partial Fail**, tTaAttr indicates **Announce Success**, and there is no reservation associated with pLaAttr, then make a reservation for pLaAttr.
c) Else if tLaAttr indicates **Attach Fail** and there is a reservation associated with pLaAttr, then remove the reservation.
d) If there is more than one Talker Announce declaration whose StreamId is the same as pLaAttr.StreamId, call **handleLaMerge**(pLaAttr.StreamId); otherwise, issue at RESI an ATTACH_STREAM.indication (6.6.2.3.2) with its parameters set to the corresponding values contained in pLaAttr.

6.6.3.11 handleLaMerge(pStreamId)

This procedure is invoked by the **handleLaReg**() procedure (6.6.3.10) to process a Multi-Context Listener Attachment identified by a StreamId (pStreamId) and subject to Listener Attach merge (6.3.5.3) in the end station. It performs the following actions:

a) Search for each Listener Attach registration (tLaAttr) associated with pStreamId.
b) Construct a Listener Attach attribute (tLaAttrMerge), as follows:
    1) Set tLaAttrMerge.StreamId to pStreamId.
    2) Set tLaAttrMerge.VID to the Redundancy Context Id used by this Listener Attachment.
    3) Set tLaAttrMerge.ListenerAttachStatus to **Attach Ready**, if at least one tLaAttr found in a) indicates **Attach Ready**, and **Attach Fail**, otherwise.
    4) Construct a VLAN Context Status sub-TLV (6.5.4.4) with each VlanContextId and PathStatus pair contained in each tLaAttr found in a), and append it to tLaAttrMerge.
c) Issue at RESI an ATTACH_STREAM.indication (6.6.2.3.2) with its parameters set to the corresponding values contained in tLaAttrMerge.

6.6.3.12 handleLaDeReg()

This procedure is invoked when the RAP Endpoint receives an ATTR_DEREG.indication (6.7.2.4) at RPSI that indicates a Listener Attach deregistration (pLaAttr). It performs the following actions:

a) Set tStreamId to pLaAttr.StreamId, remove the reservation associated with pLaAttr, if any, and then remove the Listener Attach registration in pLaAttr.
b) If there is no remaining Listener Attach registration associated with tStreamId, issue at RESI a DETACH_STREAM.indication (6.6.2.3.4) with tStreamId.
c) Otherwise, call **handleLaMerge**(tStreamId).

6.7 RAP Participant

6.7.1 RAP Participant overview

A RAP Participant associated with a Port of an end station or Bridge is responsible for the following:

— Cooperation with the underlying LRP in managing Portal creation and association on that Port.
— Provision of attribute declaration and registration services to a RAP Endpoint or RAP Propagator (termed “RAP Participant user”).
— Maintenance of attribute declaration and registration databases.

The operation of a RAP Participant is described in terms of a RAP Participant Service Interface (RPSI) as specified in 6.7.2, and a per-Port RAP Participant state machine as specified in the following subclauses:

— RAP Participant state machine diagrams in 6.7.3.
— RAP Participant state machine variables in 6.7.4.
— RAP Participant state machine procedures in 6.7.5.

6.7.2 RAP Participant Service Interface (RPSI)

The RPSI provides a set of primitives (Table 6-12) for interaction between each RAP Participant and its RAP Participant user. The rPortRef (or iPortRef) parameter in a request (or indication) primitive indicates a RAP Participant to (or from) which an RPSI primitive is issued.

Table 6-12—RAP Participant Service Interface primitives

primitive name	reference
DECLARE_ATTR.request(rPortRef, rAttr)	6.7.2.1
WITHDRAW_ATTR.request(rPortRef, rAttr)	6.7.2.2
ATTR_REG.indication(iPortRef, iAttr)	6.7.2.3
ATTR_DEREG.indication(iPortRef, iAttr)	6.7.2.4
DEC_OPER_STATUS.indication(iPortRef, iStatus)	6.7.2.5

6.7.2.1 DECLARE_ATTR.request(rPortRef, rAttr)

This primitive is used by the RAP Participant user to request the RAP Participant on a Port (rPortRef) to make a declaration for an attribute (rAttr).

6.7.2.2 WITHDRAW_ATTR.request(rPortRef, rAttr)

This primitive is invoked by the RAP Participant user to request the RAP Participant on a Port (rPortRef) to withdraw an existing declaration for an attribute (rAttr).

6.7.2.3 ATTR_REG.indication(iPortRef, iAttr)

This primitive is used by a RAP Participant on a Port (iPortRef) to inform the RAP Participant user about registration of an attribute (iAttr).

6.7.2.4 ATTR_DEREG.indication(iPortRef, iAttr)

This primitive is used by a RAP Participant on a Port (iPortRef) to inform the RAP Participant user about deregistration of an attribute (iAttr).

6.7.2.5 DEC_OPER_STATUS.indication(iPortRef, iStatus)

This primitive is used by a RAP Participant on a Port (iPortRef) to notify the RAP Participant user of whether its attribute declaration function is operational (iStatus == TRUE) or not (iStatus == FALSE).

6.7.3 RAP Participant state machine diagrams

A RAP Participant associated with a local target Port of an end station or Bridge maintains a single RAP Participant state machine. The RAP Participant state machine diagram is shown in Figure 6.23.

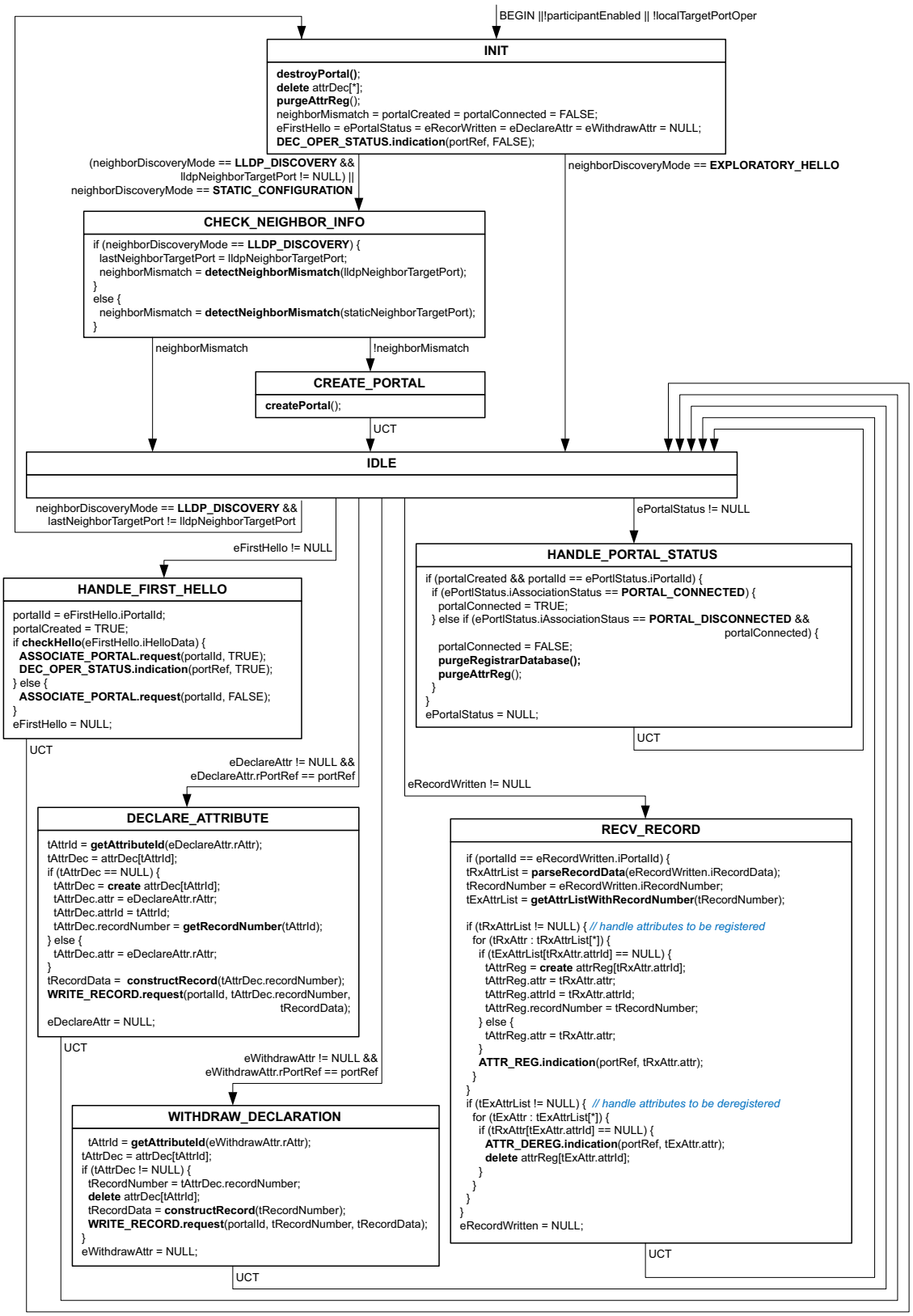


Figure 6-23—RAP Participant state machine diagram

6.7.4 RAP Participant state machine variables

There is one instance of the RAP Participant state machine variables per RAP Participant state machine.

6.7.4.1 portRef

A reference to the local target Port with which the RAP Participant state machine is associated.

NOTE—The portRef variable is only used inside a RAP Instance for interaction between each RAP Participant and the RAP Participant user. There is no specified relationship between portRef and localTargetPort.portId [item b) in 6.7.4.7.2].

6.7.4.2 participantEnabled

A Boolean variable indicating whether the operation of the RAP Participant state machine is administratively enabled (TRUE) or not (FALSE).

6.7.4.3 neighborDiscoveryMode

An administratively assigned value, indicating the operation mode in which neighbor discovery is performed on the local target Port, and taking one of the following enumerated values:

1) **LLDP_DISCOVERY**: The information about a neighbor target port to be passed to the underlying LRP is obtained through the exchange of LRP Discovery TLVs ([CS]:C) by use of LLDP, and contained in lldpNeighborTargetPort (6.7.4.7.4).
2) **STATIC_CONFIGURATION**: The information about a neighbor target port to be passed to the underlying LRP is statically configured by the management and contained in staticNeighborTargetPort (6.7.4.7.3).
3) **EXPLORATORY_HELLO**: No neighbor target port information needs to be passed to the underlying LRP.

6.7.4.4 helloTime

An administratively assigned integer value, in the range 30 through 65535, for the Hello Time parameter in a Local Target Port request (item c):1) in [CS]:10.2.2.1) issued by the RAP Participant state machine to the underlying LRP. The default value is 30.

6.7.4.5 completeListTimerReset

An administratively assigned integer value, in the range x through y, for the cplCompleteListTimerReset parameter in a Local Target Port request (item c):3) in [CS]:10.2.2.1] issued by the RAP Participant state machine to the underlying LRP. The default value is z.

<< Editor's note: The editor is unsure what values for x, y, z to take. Contribution is required. >>

6.7.4.6 exploreHelloRecvEnabled

An administratively assigned Boolean value for the imPplExploreRecv parameter in a Neighbor Target Port request (item b):3):iii) in [CS]:10.2.3.1) issued by the RAP Participant state machine to the underlying LRP.

6.7.4.7 Variables of TargetPort data type

6.7.4.7.1 TargetPort data type

A structure that consists of a collection of configuration parameters for a target port, as follows:

a) **chassisId**: The Chassis ID ([AB]:8.5.2).
b) **portId**: The Port ID ([AB]:8.5.3).
c) **ecpCapable**: A Boolean value indicating whether the target port supports the LRP-DT ECP mechanism (TRUE) or not (FALSE).
d) **tcpCapable**: A Boolean value indicating whether that the target port supports the LRP-DT TCP mechanism (TRUE) or not (FALSE).
e) **tcpPort:** A 2-byte TCP port number for the target port ([CS]:C.2.2.6.1).
f) **addrIPv4**: A 4-byte IPv4 address for the target port (item 1) in [CS]:C.2.2.6.2) or NULL.
g) **addrIPv6:** A 16-byte IPv6 address for the target port (item 2) in [CS]:C.2.2.6.2) or NULL.

6.7.4.7.2 localTargetPort

A structure of the TargetPort data type (6.7.4.7.1). It contains the administratively configured parameters of the local target port.

6.7.4.7.3 staticNeighorTargetPort

A structure of the TargetPort data type (6.7.4.7.1). It contains the administratively configured parameters of a neighbor target port to which the local target port is to be connected.

6.7.4.7.4 lldpNeighborTargetPort

A structure of the TargetPort data type (6.7.4.7.1). It contains the configuration parameters of a neighbor target port discovered by LLDP.

The values contained in this variable are derived from a matching LRP ECP Discovery TLV and/or a matching LRP TCP Discovery TLV currently stored in the LLDP remote systems MIB on the local target port and meeting the following two conditions:

— received by an LLDP agent using a MAC address as specified in 6.4.2.
— containing an Application descriptor ([CS]:C.2.1.6 and [CS]:C.2.2.6) with an AppId value as specified in Table 6-7.

In the presence of the matching TLV(s), each member variable of lldpNeighborTargetPort is set as follows:

a) **chassisId**: Set to the Chassis ID value ([AB]:8.5.2) of the received LLDPDUs that carry the matching LRP Discovery TLV(s).
b) **portId**: Set to the Port ID value ([AB]:8.5.3) of the received LLDPDUs that carry the LRP Discovery TLV(s).
c) **ecpCapable**: Set TRUE if a matching LRP ECP Discovery TLV is present, indicating that the neighbor Port supports the LRP-DT ECP mechanism, and set FALSE otherwise.
d) **tcpCapable**: Set TRUE if a matching LRP TCP Discovery TLV is present, indicating that the neighbor Port supports the LRP-DT TCP mechanism, and set FALSE otherwise.
e) **tcpPort**: If tcpCapable in item d) is TRUE, set to the value contained in the TCP port number field ([CS]:C.2.2.6.1) of the matching Application descriptor in the received LRP TCP Discovery TLV.

f) **addrIPv4**: If tcpCapable in item d) is TRUE and the matching Application descriptor in the received LRP TCP Discovery TLV indicates an IPv4 address, set to the corresponding value encoded in the Address info field ([CS]:C.2.2.6.2); otherwise, set to NULL.

g) **addrIPv6:** If tcpCapable in item d) is TRUE and the matching Application descriptor in the received LRP TCP Discovery TLV indicates an IPv6 address, set to the corresponding value encoded in the Address info field ([CS]:C.2.2.6.2); otherwise, set to NULL.

If no matching LRP Discovery TLV(s) is present, the lldpNeighborTargetPort variable set to NULL, indicating that there is currently no neighbor discovered by LLDP.

NOTE—This standard does not specify the means for keeping lldpNeighborTargetPort up to date with the information maintained by LLDP. A Native system could run a local procedure to access the LLDP MIBs residing within the same system. A Proxy system with a RAP Participant that has LRP discovery TLVs exchanged on a Controlled system's target port could update this variable by observing the contents of the lrpLldpEcpTlvRecvInfo and lrpLldpTcpTlvRecvInfo managed objects ([CS]:11.6.2.3 and [CS]:11.6.2.6, respectively) of that Controlled system via management interfaces.

6.7.4.7.5 lastNeighborTargetPort

A structure of the TargetPort data type (6.7.4.7.1) used to contain the values copied from lldpNeighborTargetPort (6.7.4.7.4) since the latter was last updated.

6.7.4.8 neighborMismatch

A Boolean variable, set TRUE when detecting a mismatch between the local target port and the neighbor target port to be connected.

NOTE—The purpose of targetPortMismatch is to supply diagnostic information, through the associated managed object, for the management to take appropriate actions, if necessary, when its value is TRUE.

6.7.4.9 localTargetPortOper

A Boolean variable indicating the status of the Internal Sublayer Service MAC_Operational parameter ([AC]:11.2) associated with the local target port.

NOTE—If the Continuity Check protocol ([Q]:20.1) is deployed to detect physical link connectivity on the local target port, the MEP Continuity Check Receiver ([Q]:19.2.8) will cause that port's MAC_Operational parameter to be false upon detecting a connectivity fault.

6.7.4.10 portalCreated

A Boolean value indicating whether a Portal for the operation of the RAP Participant on the local Port has been created by the underlying LRP (TRUE) or not (FALSE).

6.7.4.11 portalId

The identifier of a Portal created by the underlying LRP for the local target port, set to the corresponding parameter value contained in a FIRST_HELLO.indication (Table 6-8) received from the underlying LRP.

6.7.4.12 portalConnected

A Boolean value indicating whether a Portal association for the Portal as indicated in portalId (6.7.4.11) has been established by the underlying LRP (TRUE) or not (FALSE).

6.7.4.13 attrDec

An array in which each element represents an attribute declaration maintained by the RAP Participant and consists of the following member variables:

a) **attr**: An attribute instance as specified in 6.5.
b) **attrId**: An integer identifier generated by the getAttributeId procedure (6.7.5.5) for the attribute instance contained in item a).
c) **recordNumber**: The record number of a record in which the attribute instance contained in item a) is carried.

The attrDec variable is associated with a key <attrId>.

6.7.4.14 attrReg

An array in which each element represents an attribute registration maintained by the RAP Participant and consists of the following member variables:

a) **attr**: An attribute instance as specified in 6.5.
b) **attrId**: An integer identifier generated by the getAttributeId procedure (6.7.5.5) for the attribute instance contained in item a).
c) **recordNumber**: The record number of a record in which the attribute instance contained in item a) is carried.

The attrReg variable is associated with a key <attrId>.

6.7.4.15 eFirstHello

An event pointer (6.2.4) associated with the FIRST_HELLO.indication primitive (Table 6-8).

6.7.4.16 ePortalStatus

An event pointer (6.2.4) associated with the PORTAL_STATUS.indication primitive (Table 6-8).

6.7.4.17 eRecordWritten

An event pointer (6.2.4) associated with the RECORD_WRITTEN.indication primitive (Table 6-8).

6.7.4.18 eDeclareAttr

An event pointer (6.2.4) associated with the DECLARE_ATTR.request primitive (6.7.2.1).

6.7.4.19 eWithdrawAttr

An event pointer (6.2.4) associated with the WITHDRAW_ATTR.request primitive (6.7.2.2).

6.7.5 RAP Participant state machine procedures

6.7.5.1 detectNeighborMismatch()

This procedure determines whether there is a mismatch between the local target port and the neighbor target port to be connected, as follows:

```
detectNeighborMismatch(pNeighborTargetPort) {
  if ((pNeighborTargetPort.ecpCapable && localTargetPort.ecpCapable) ||
```

```
    (pNeighborTargetPort.tcpCapable && localTargetPort.tcpCapable &&
     ((pNeighborTargetPort.addrIPv4 != NULL &&
       localTargetPort.addrIPv4!= NULL) ||
      (pNeighborTargetPort.addrIPv6 != NULL &&
       localTargetPort.addrIPv6 != NULL)
     )
    ){
   return FALSE;
 } else {
   return TRUE;
 }
```

6.7.5.2 createPortal()

This procedure requests the underlying LRP to create a Portal for the RAP Participant, by issuing a LOCAL_TARGET_PORT.request primitive and a NEIGHBOR_TARGET_PORT.request primitive, with the parameters values shown in Table 6-13.

Table 6-13—Input parameter values for Portal Creation

		neighborDiscoveryMode		
		LLDP_DISCOVERY	STATIC_CONFIGURATION	EXPLORATORY_HELLO
Local Target Port request input parameters (item in [CS]:10.2.2.1)	a).1)	the AppId value specified in Table 6-7		
	b).1)	localTargetPort.chassisId		
	b).2)	localTargetPort.portId		
	b).3)	localTargetPort.ecpCapable		
	b).4).i)	localTargetPort.addrIPv4		
	b).4).ii)	localTargetPort.addrIPv6		
	b).4).iii)	localTargetPort.tcpPort		
	c).1)	helloTime		
	c).2)	the value specified in 6.4.4		
	c).3)	completeListTimerReset		
Neighbor Target Port request input parameters (item in [CS]:10.2.3.1)	a).1)	a 3-tuple of {AppId, localTargetPort.chassisId, localTargetPort.portId}		
	b).1)	lldpNeighborTargetPort.chassisId	staticNeighborTargetPort.chassisId	none
	b).2)	lldpNeighborTargetPort.portId	staticNeighborTargetPort.portId	none
	b).3).i)	if lldpNeighborTargetPort.ecpCapable == TRUE, set to the ECP MAC address as specified in 6.4.3, otherwise NULL	if staticNeighborTargetPort.ecpCapable == TRUE, set to the ECP MAC address as specified in 6.4.3, otherwise NULL	the ECP MAC address as specified in 6.4.3
	b).3).ii)	FALSE		TRUE
	b).3).iii)	exploreHelloRecvEnabled		TRUE
	b).4).i)	lldpNeighborTargetPort.addrIPv4	staticNeighborTargetPort.addrIPv4	none
	b).4).ii)	lldpNeighborTargetPort.addrIPv6	staticNeighborTargetPort.addrIPv6	none
	b).4).iii)	lldpNeighborTargetPort.tcpPort	staticNeighborTargetPort.tcpPort	none

6.7.5.3 checkHello(pHelloData)

This procedure checks the contents of a Hello LRPDU (pHelloData) passed from the underlying LRP to the RAP Participant state machine, to determine whether to approve a Portal association, as follows:

a) If pHelloData contains an Application Information TLV whose value field is encoded as specified in 6.4.4, approve the association by returning a value of TRUE.

b) Otherwise, deny the association by returning a value of FALSE.

6.7.5.4 destroyPortal()

This procedure is used by the RAP Participant state machine to request the underlying LRP to destroy the Portal previously created (if any), by issuing a LOCAL_TARGET_PORT.request (Table 6-8) with the input parameters (items defined in [CS]:10.2.2.1) set as follows:

a) item a).1) set to the AppId value defined in Table 6-7.
b) item b).1) set to localTargetPort.chassisId.
c) item b).2) set to localTargetPort.portId.
d) item b).3) set FALSE.
e) other items left empty.

6.7.5.5 getAttributeId(pAttr)

This procedure generates and returns an integer identifier for an attribute instance (pAttr) to be declared or registered on the local target port. Each attribute declaration or registration maintained by the RAP Participant state machine is identified by a subset of the parameters encoded in the attribute TLV, as specified in Table 6-14.

Table 6-14—Attribute identification parameters in a RAP Participant

RAP attribute	Identification Parameters in the attribute TLV
RA attribute	Type == 0x00
Talker Announce attribute	Type == 0x01, StreamId (6.5.3.1), VID (6.5.3.5.3)
Listener Attach attribute	Type == 0x02, StreamId (6.5.4.1), VID (6.5.4.2)

NOTE—An attribute identifier generated by this procedure is only of local significance to and used within a particular RAP Participant state machine.

6.7.5.6 getRecordNumber(pAttrId)

This procedure allocates and returns a 4-octet record number as defined in [CS]:Table 9-7, for a new attribute declaration identified by pAttrId.

The assigned record number could be a new one that is not used by any of the other entries in attrDec, or an existing one that is being used by one or more of the other entries in attrDec. Assignment of a record number to multiple declarations, which causes the attribute instances of these declarations to be serialized into a single record, shall not cause the record to exceed the maximum record size determined by the underlying LRP.

The strategies deployed by this procedure for assignment of record numbers are not specified by this standard.

NOTE—According to the operation of LRP, record number assignment is performed by an LRP application on the Applicant size of a Portal connection. Within a Portal, there is no relationship between a record in an applicant database and a record with the same record number in the registrar database. Thus, there is no requirement for an LRP application to use the same record number assignment strategies on both sides of a Portal connection.

6.7.5.7 constructRecord(pRecordNumber)

This procedure searches in attrDec for all the elements whose recordNumber value matches the pRecordNumber value, and returns an octet string in one of the following cases:

a) No matching element found: a zero length octet string.
b) Only one matching element: an octet string holding the attr value of the matching attrDec element.
c) More than one matching element: an octet string holding the concatenation of the attr values of all the matching attrDec elements in any order.

6.7.5.8 parseRecordData(pRecordData)

This procedure parses the data (pRecordData) received in a record from the underlying LRP into RAP attributes in accordance with the attribute TLV formats defined in 6.5.

If one or more attribute instances are parsed from the record, the procedures returns an array in which each element corresponds to an attribute instance and consists of the following member variables:

a) **attr**: An attribute instance parsed from the given record.
b) **attrId**: The return value of invoking the getAttributeId procedure (6.7.5.5) with the attribute contained in a).

Otherwise, the procedure returns a NULL value.

6.7.5.9 getAttrListWithRecordNumber(pRecordNumber)

This procedure searches in attrReg for each element whose recordNumber value matches the pRecordNumber value. If one or more matching elements are found, it returns an array in which each element corresponds to a matching element in attrReg and consists of the following member variables:

a) **attr**: copied from the attr value of a matching element in attrReg.
b) **attrId**: copied from the attrId value of a matching element in attrReg.

Otherwise, the procedure returns a NULL value.

6.7.5.10 purgeRegistrarDatabase()

This procedure is used by the RAP Participant state machine to request the underlying to purges all records in the Portal's registrar database, as follows:

```
purgeRegistrarDatabase() {
  for (tAttrReg : attrReg[*]) {
    DELETE_RECORD.request(portalId, tAttrReg.recordNumber);
  }
}
```

6.7.5.11 purgeAttrReg()

This procedure purges all attribute registrations contained in attrReg, as follows:

```
purgeAttrReg() {
  for (tAttrReg : attrReg[*]) {
    ATTR_DEREG.indication(portRef, tAttrReg.attr);
    delete attrReg[tAttrReg.attrId];
  }
}
```

6.8 RAP Propagator

6.8.1 RAP Propagator overview

Editor's note: Add a paragraph that describes that the RAP Propagator does both attribute propagation and attribute processing.

The operation of a RAP Propagator is described in terms of a per-Bridge RAP Propagator state machine as specified in the following subclauses:

- RAP Propagator state machine diagrams in 6.8.2.
- RAP Propagator state machine events in 6.8.3.
- RAP Propagator state machine variables in 6.8.4.
- RAP Propagator state machine procedures in 6.8.5.

6.8.2 RAP Propagator state machine diagrams

The operation of the RAP Propagator state machine is specified by the state machine diagram in Figure 6-24. The actions executed on entry to each state of the state machine (except for the IDLE state in which no

actions are defined) are shown in Figure 6-25, Figure 6-26, Figure 6-27, Figure 6-28, Figure 6-29 and Figure 6-30.

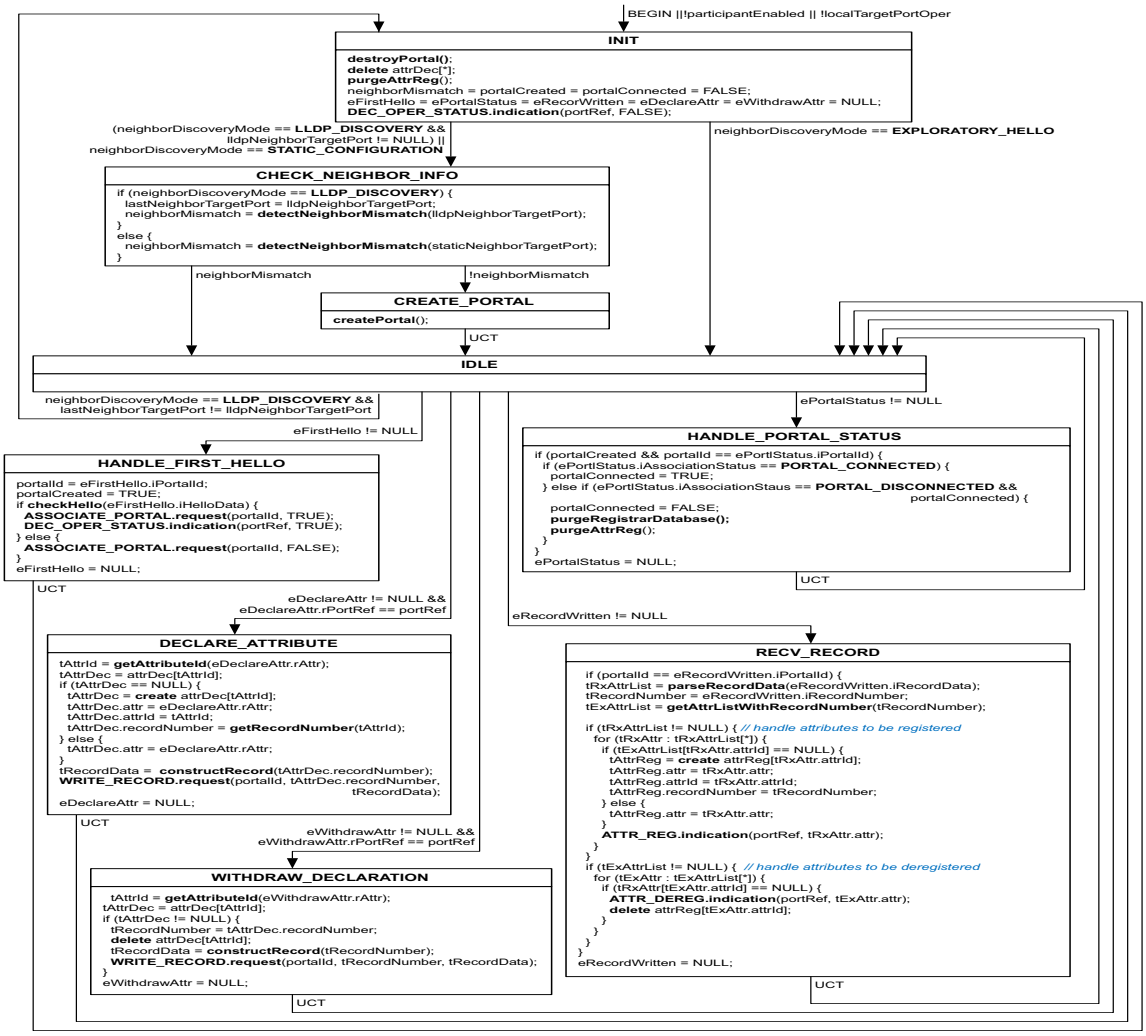


Figure 6-24—RAP Propagator state machine diagram

Figure 6-25 defines the actions in the INIT state for initialization of the RAP Propagator state machine, and the actions in the HANDLE_RA_REG state for handling of an ATTR_REG.indication (6.7.2.3) that indicates an RA registration.

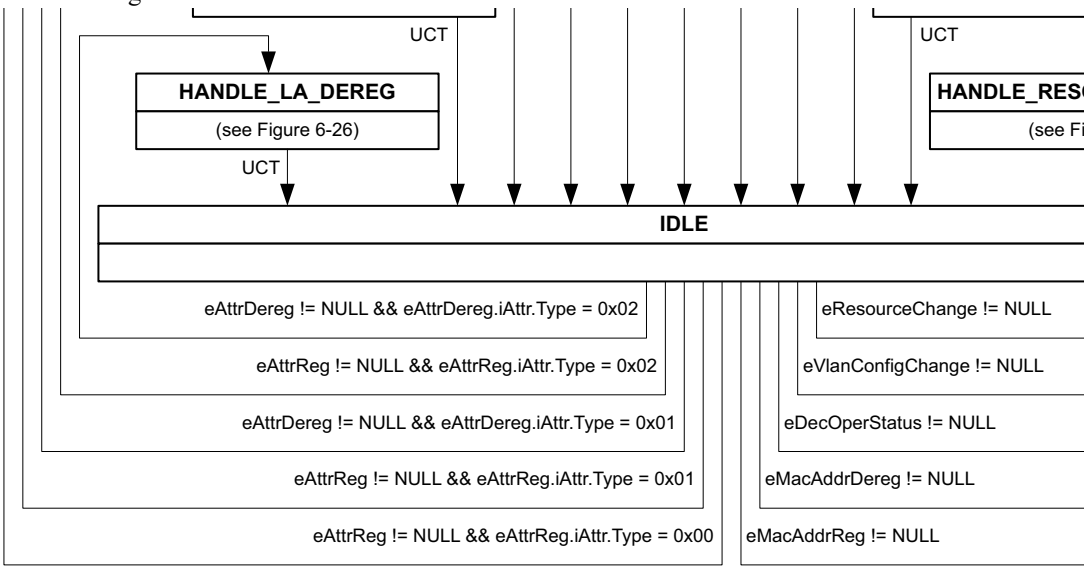


Figure 6-25—Initialization and handling of RA registrations in a RAP Propagator

Figure 6-26 defines the actions in the HANDLE_TA_REG state for handling of an ATTR_REG.indication (6.7.2.3) that indicates a Talker Announce registration, and the actions in the HANDLE_TA_DEREG state for handling of an ATTR_DEREG.indication (6.7.2.4) that indicates a Taker Announce deregistration.

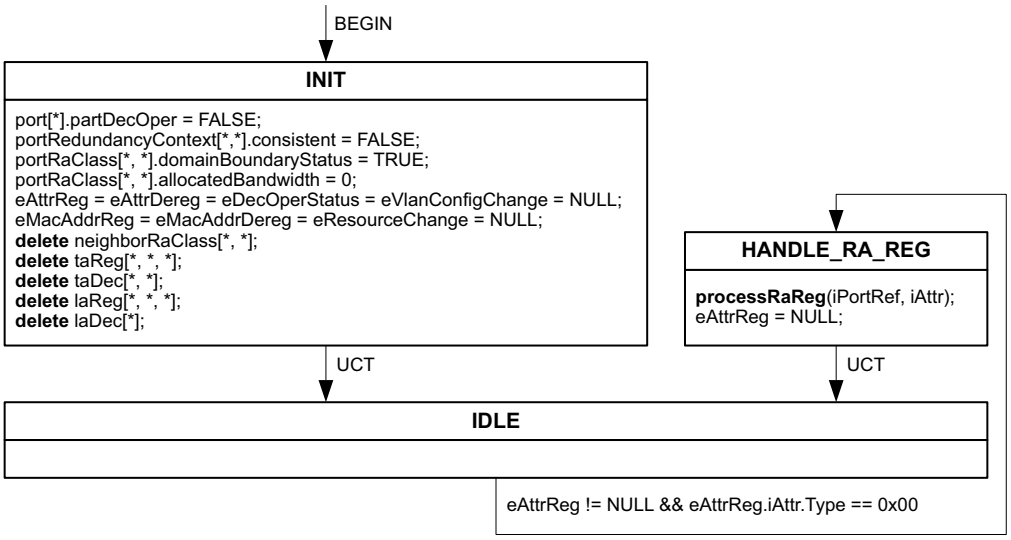


Figure 6-26—Handling of Talker Announce registrations and deregistrations in a RAP Propagator

Figure 6-27 defines the actions in the HANDLE_LA_REG state for handling of an ATTR_REG.indication (6.7.2.3) that indicates a Listener Attach registration, and the actions in the HANDLE_LA_DEREG state for handling of an ATTR_DEREG.indication (6.7.2.4) that indicates a Listener Attach deregistration.

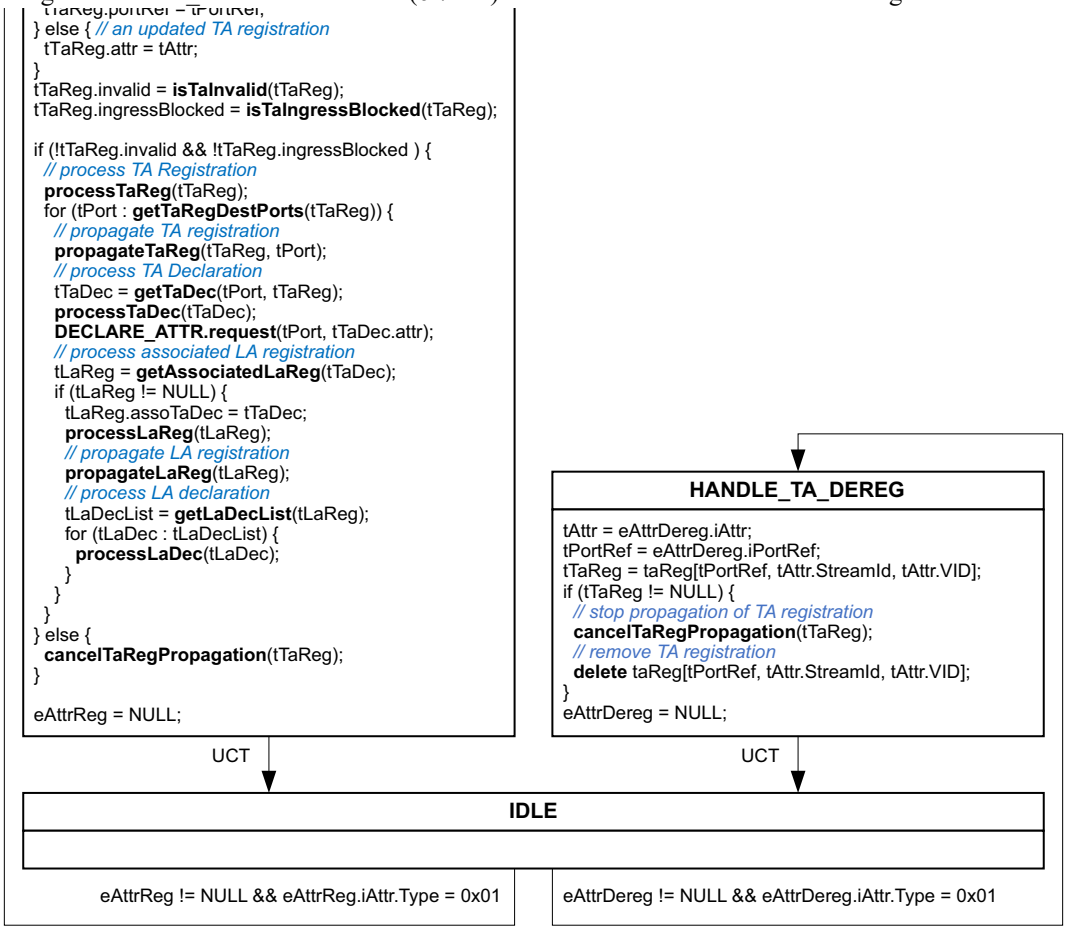


Figure 6-27—Handling of Listener Attach registrations and deregistrations in a RAP Propagator

Figure 6-28 defines the actions in the HANDLE_MAC_ADDR_REG state for processing of a MAC_ADDR_REG.indication (6.8.3.2), and the actions in the HANDLE_MAC_ADDR_DEREG state for processing of a MAC_ADDR_DEREG.indication (6.8.3.3).

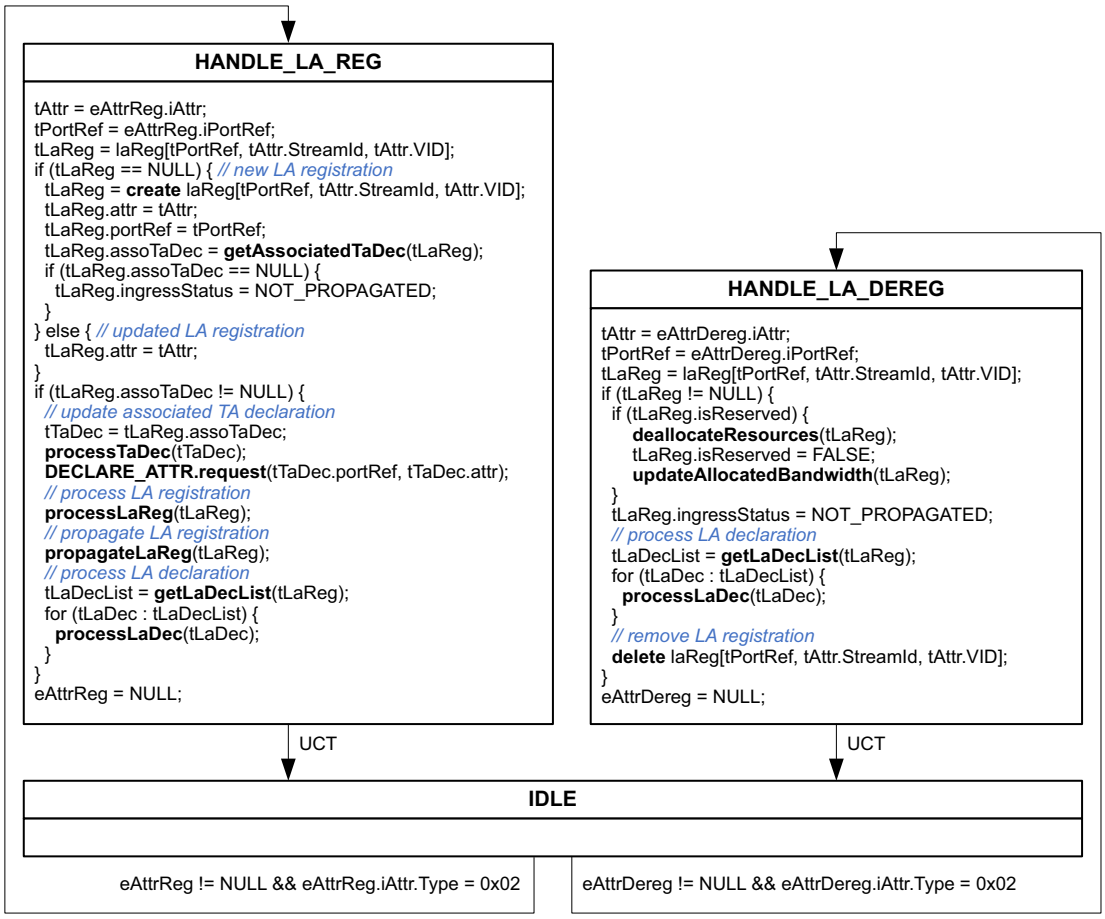


Figure 6-28—Handling of MAC address registrations and deregistrations in a RAP Propagator

Figure 6-29 defines the actions in the HANDLE_DEC_OPER_STATUS state for handling of a DEC_OPER_STATUS.indication (6.7.2.5), and the actions in the HANDLE_VLAN_CONFIG_CHANGE state for handling of a VLAN_CONFIG_CHANGE.indication (6.8.3.1).

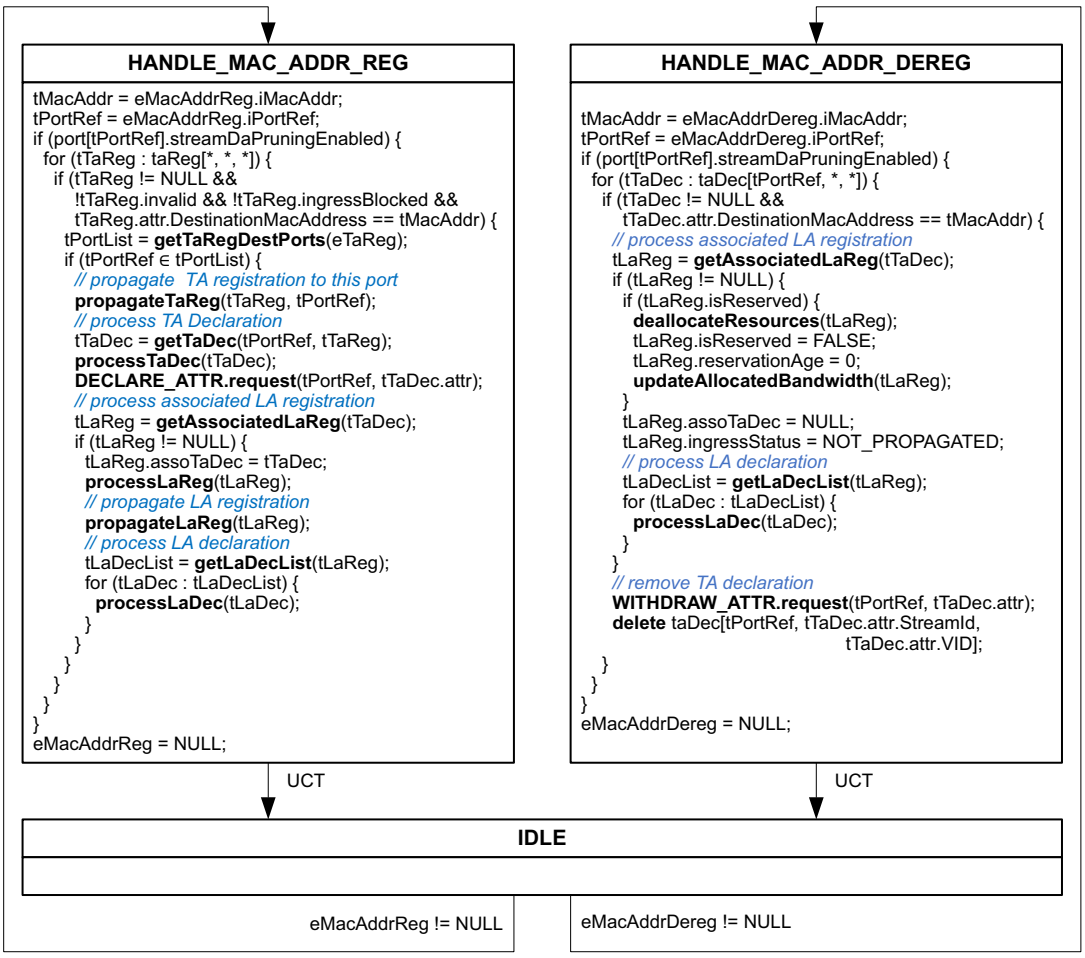


Figure 6-29—Handling of Port status changes and VLAN topology changes in a RAP Propagator

Figure 6-30 defines the actions in the HANDLE_RESOURCE_CHANGE state for handling of a RESOURCE_CHANGE.indication (6.8.3.4).

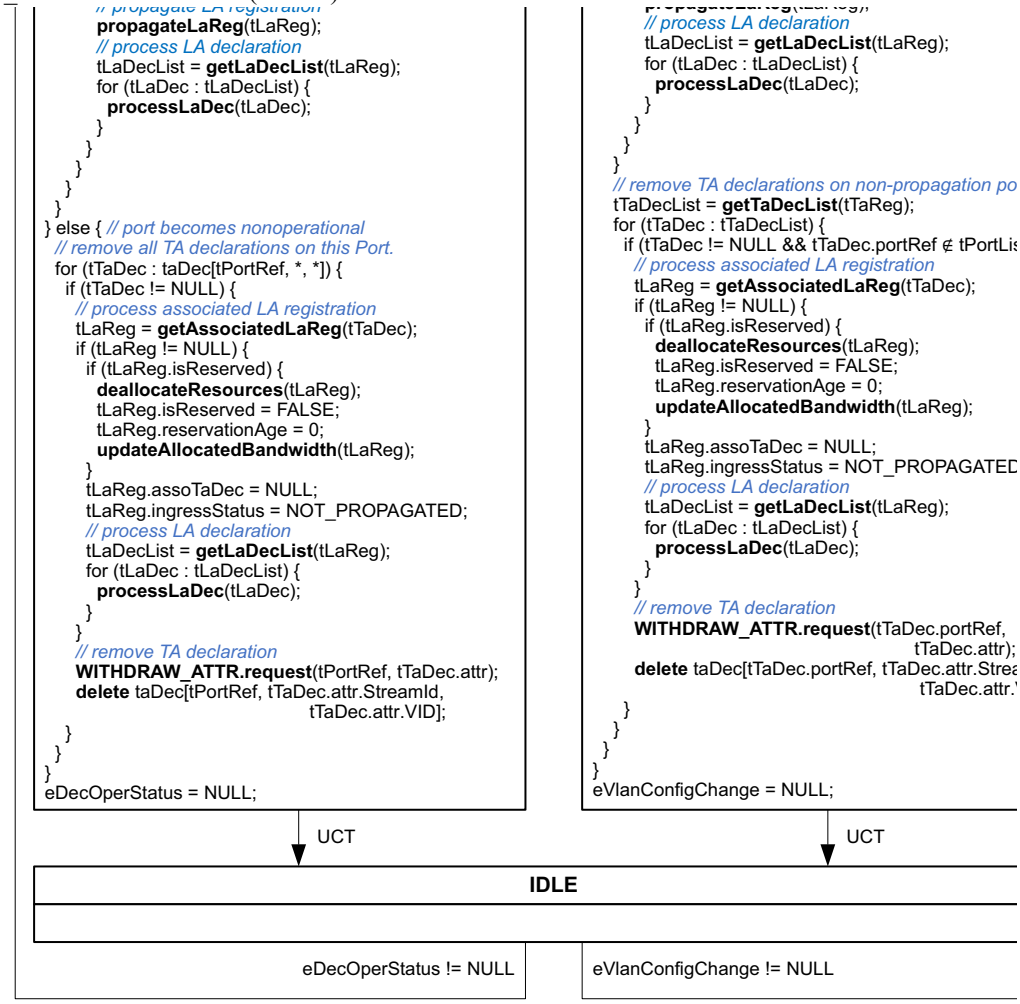


Figure 6-30—Handling of resource availability changes in a RAP Propagator

6.8.3 RAP Propagator state machine events

Transitions from the IDLE state in the RAP Propagator state machine are driven by the events generated by the per-Port RAP Participants through the indication primitives as defined in 6.7.2, and by the underlying mechanisms in the Bridge through the set of indication primitives as defined in 6.8.3.1, 6.8.3.2 and 6.8.3.3.

6.8.3.1 VLAN_CONFIG_CHANGE.indication()

This primitive is invoked to inform the RAP Propagator state machine about changes in the VLAN configuration. Such a change can result from a spanning tree ([Q]:7.3) or VLAN topology ([Q]:7.4) reconfiguration.

6.8.3.2 MAC_ADDR_REG.indication(iPortRef, iMacAddr)

This primitive is invoked to inform the RAP Propagator state machine about registration of a MAC address (iMacAddr) on a Port (iPortRef).

6.8.3.3 MAC_ADDR_DEREG.indication(iPortRef, iMacAddr)

This primitive is invoked to inform the RAP Propagator state machine about deregistration of a MAC address (iMacAddr) on a Port (iPortRef).

6.8.3.4 RESOURCE_CHANGE.indication(iIncreased)

This primitive is invoked to inform the RAP Propagator state machine about an increase (iIncreased = TRUE) or a decrease (iIncreased = FALSE) in the remaining resources of this Bridge available for allocation to streams, as a consequence of one or more reservations being recently removed or created, respectively.

NOTE—As specified in Figure 6-30, this primitive is used to trigger reprocessing of those Talker Announce declarations that could be impacted by the indicated event to keep their status up-to-date. Proper usage of these primitives can help reduce repetitive computing tasks and avoid excessive processing burden. For example, issuing only a single primitive for a batch of reservations removed by the deallocateResources procedure (6.8.5.44) could be more efficient than issuing one for each of them. The method used to decide when and how to issue this primitive is an implementation choice.

6.8.4 RAP Propagator state machine variables

6.8.4.1 taReg

The taReg variable is an array in which each element represents a Talker Announce registration maintained by the RAP Propagator state machine and consists of the following member variables:

a) **portRef**: The portRef (6.7.4.1) value of a Bridge Port holding the Talker Announce registration.
b) **attr**: The Talker Announce attribute (6.5.3) of the Talker Announce registration.
c) **invalid**: A Boolean indicating whether the Talker Announce registration is invalid (TRUE) or not (FALSE), as determined by the isTaInvalid() procedure (6.8.5.3).
d) **ingressBlocked**: A Boolean indicating whether the Talker Announce registration is ingress blocked (TRUE) or not (FALSE), as determined by the isTaIngressBlocked() procedure (6.8.5.4).
e) **ingressStatus**: The ingress status of the Talker Announce registration, taking one of the following enumerated values:
    1) **TA_RECV_FAIL (0)**: The Talker Announce registration is failed by an upstream station.
    2) **TA_INGRESS_SUCCESS (1)**: The Talker Announce registration is neither failed by an upstream station nor failed on ingress of this Bridge.
    3) **TA_INGRESS_FAIL (2)**: The Talker Announce registration is failed on ingress of this Bridge with a RAP Failure Code indicated in ingressFailureCode [item e), below].
f) **ingressFailureCode**: A value of zero, indicating the Talker Announce registration is not failed on ingress of this Bridge, or the value of a RAP Failure Code defined in Table 6-10.
g) **rTagging**: A Boolean indicating whether the stream of this Talker Announce registration is subject to redundancy tagging (6.3.9.3) by this Bridge (TRUE) or not (FALSE). This variable is used only by a FRER-capable Bridge in processing a Multi-Context Talker Announcement.

The taRag variable is associated with a key <portRef, attr.StreamId, attr.VID>.

6.8.4.2 taDec

The taDec variable is an array in which each element represents a Talker Announce declaration being made by the RAP Propagator state machine and consists of the following member variables:

a) **portRef**: The portRef (6.7.4.1) value of a Bridge Port holding the Talker Announce declaration.
b) **attr**: The Talker Announce attribute (6.5.3) of the Talker Announce declaration.
c) **origTaReg**: A reference to an element in taReg (6.8.4.1) that is the single originating Talker Announce registration of the Talker Announce declaration, or NULL indicating the use of origTaRegList in item d) below.

d) **origTaRegList**: An array of references to one or more elements in taReg (6.8.4.1) that are originating Talker Announce registrations of the Talker Announce declaration. Each element in origTaRegList has a single variable origTaRegRef, which contains a reference to an originating Talker Announce registration. This variable is used only by a FRER-capable Bridge in handling a Multi-Context Talker Announce declaration that is subject to Talker Announce merge (6.3.4.3).

e) **rUntagging**: A Boolean indicating whether the stream of this Talker Announce declaration is subject to redundancy untagging (6.3.9.3) on the Port as indicated in portRef (TRUE) or not (FALSE). This variable is used only by a FRER-capable Bridge in handling a Multi-Context Talker Announce declaration that is subject to Talker Announce merge (6.3.4.3).

The taDec variable is associated with a key <portRef, attr.StreamId, attr.VID>.

6.8.4.3 laReg

The laReg variable is an array in which each element represents a Listener Attach registration maintained by the RAP Propagator state machine and consists of the following member variables:

a) **portRef**: The portRef (6.7.4.1) value of a Bridge Port holding the Listener Attach registration.

b) **attr**: The Listener Attach attribute (6.5.4) of the Listener Attach registration.

c) **assoTaDec**: A reference to an element in taDec (6.8.4.2) that is the associated Talker Announce declaration of the Listener Attach registration, or NULL indicating no associated Talker Announce declaration exists.

d) **isReserved**: A Boolean value indicating whether the Listener Attach registration is associated with a reservation in the Bridge (TRUE) or not (FALSE).

e) **reservationAge**: A 32-bit unsigned integer, indicating the time, in seconds, since a reservation associated with the Listener Attach registration was successfully made, and set to zero when the reservation is removed.

f) **ingressStatus**: The ingress status of the Listener Attach registration, taking one of the following enumerated values:

1) **NOT_PROPAGATED (0)**: The Listener Attach registration has no associated Talker Announce declaration and is not propagated.
2) **ATTACH_READY (1)**: The Listener Attach registration is propagated with **Attach Ready** [item a) in 6.5.4.3].
3) **ATTACH_FAIL (2)**: The Listener Attach registration is propagated with **Attach Fail** [item b) in 6.5.4.3].
4) **ATTACH_PARTIAL_FAIL (3)**: This Listener Attach registration is propagated with **Attach Partial Fail** [item c) in 6.5.4.3].

The laReg variable is associated with a key <portRef, attr.StreamId, attr.VID>.

6.8.4.4 laDec

The laDec variable is an array in which each element represents a Listener Attach declaration maintained by the RAP Propagator state machine and consists of the following member variables:

a) **assoTaReg**: A reference to an element in taReg (6.8.4.1) that is the associated Talker Announce registration of the Listener Attach declaration.

b) **attr**: The Listener Attach attribute (6.5.4) of the Listener Attach declaration.

c) **origLaRegList**: An array of references to one or more elements in laReg (6.8.4.3) that are originating Listener Attach registrations of the Listener Attach declaration and subject to Listener Attach merge (6.3.5.3). Each element in origLaRegList has a single variable origLaRegRef, which contains a reference to an originating Listener Attach registration.

The laDec variable is associated with a key <assoTaReg>.

6.8.4.5 localRaClass

The localRaClass variable is an array in which each element indicates an RA class supported by the Bridge and consists of the following member variables:

a) **id**: An RA class ID (6.3.2.1).
b) **priority**: An RA class priority (6.3.2.2)
c) **rtid**: An RTID (6.3.2.3)
d) **templateDefinedData:** The value to be contained in the RaClassTemplateDefinedData field (6.5.2.2.6) of the RA Class Descriptor sub- TLV for this RA class and carried in the RA attribute declared by this Bridge.

The localRaClass variable is associated with a key <id>.

6.8.4.6 neighborRaClass

The neighborRaClass variable is an array in which each element indicates an RA class declared by a neighboring station on a Port of this Bridge and consists of a set of parameters copied from an RA Class Descriptor sub-TLV (6.5.2.2) contained in an RA registration on that Port, as follows:

a) **portRef**: The portRef (6.7.4.1) value of the RA registration.
b) **id**: The value of RaClassId (6.5.2.2.1).
c) **priority**: The value of RaClassPriority (6.5.2.2.2).
d) **rtid**: The value of RTID (6.5.2.2.3).
e) **trafficClass**: The value of TrafficClass (6.5.2.2.4).
f) **maxLastHopLatency:** The value of MaxLastHopLatency (6.5.2.2.5).
g) **templateDefinedData:** The value in the RaClassTemplateDefinedData (6.5.2.2.6).

The neighborRaClass variable is associated with a key <portRef, id>.

6.8.4.7 port

The port variable is an array in which each element corresponds to a Bridge Port and consists of the following member variables:

a) **portRef**: The portRef (6.7.4.1) value of the associated Port.
b) **streamDaPruningEnabled:** A Boolean indicating whether Stream DA Pruning (6.3.4.1.2) is administratively enabled (TRUE) or disabled (FALSE) on the Port.
c) **partDecOper:** A Boolean indicating whether the attribute declaration function of the RAP Participant associated with the Port is operational (TRUE) or not (FALSE).
d) **portTransmitRate:** The transmission rate, in bits per second, of the underlying MAC service on the Port.
e) **maxInterferingFrameSize:** The value of the MaxInterferingFrameSize parameter (6.5.2.1) associated with the Port.
f) **neighborMaxInterferingFrameSize**: The value of the MaxInterferingFrameSize parameter (6.5.2.1) contained in an RA registration on the Port and associated with a neighboring station's Port making the corresponding RA declaration.
g) **maxPropagationDelay**: An unsigned integer, indicating the maximum latency, in nanoseconds, a frame can experience when transmitted from the underlying physical medium on the Port to a reception port connected via a LAN to the Port.

h) **minPropagationDelay**: An unsigned integer, indicating the minimum latency, in nanoseconds, a frame can experience when transmitted from the underlying physical medium on the Port to a reception port connected via a LAN to the Port.

The port variable is associated with a key <portRef>.

6.8.4.8 portRedundancyContext

The portRedundancyContext variable is an array in which each element is associated with a redundancy Context supported by the Bridge and a Bridge Port, and consists of the following member elements:

a) **portRef**: The portRef (6.7.4.1) value of the associated Port.
b) **redundancyContextId**: The redundancy Context ID (6.8.4.10) of the associated redundancy Context.
c) **consistent**: A Boolean indicating whether the Port is consistent for the redundancy Context (TRUE) or not (FALSE).

6.8.4.9 portRaClass

The portRaClass variable is an array in which each element is associated with an RA class supported by the Bridge and a Bridge Port, and consists of the following member elements:

a) **portRef**: The portRef (6.7.4.1) value of the associated Port.
b) **raClassId**: The RA class ID (6.3.2.1) of the associated local RA class.
c) **domainBoundaryPort:** A Boolean indicating whether the Port is a domain boundary port for the RA class (TRUE) or not (FALSE).
d) **maxBandwidth**: An unsigned integer, indicating the maximum amount of bandwidth that can be allocated to the streams reserved in the RA class on the Port. The bandwidth value is represented as a percentage of the portTransmitRate [item d) in 6.8.4.7] value on that Port and expressed as a fixed-point number scaled by a factor of 1,000,000; i.e., 100,000,000 (the maximum value) represents 100%.
e) **allocatedBandwidth**: An unsigned integer, indicating the amount of bandwidth that has been allocated to the streams reserved in the RA class on the Port. The bandwidth value is represented as a percentage of the portTransmitRate [item d) in 6.8.4.7] value on that Port and expressed as a fixed-point number scaled by a factor of 1,000,000; i.e., 100,000,000 (the maximum value) represents 100%.
f) **maxLastHopLatency:** The value to be contained in the MaxLastHopLatency field (6.5.2.2.5) of an RA Class Descriptor sub-TLV associated with the RA class in the RA attribute declared on the Port.

The portRaClass variable is associated with a key <portRef, raClassId>.

6.8.4.10 portPairRaClass

The portPairRaClass variable is an array in which each element is associated with an RA class supported by the Bridge and a reception Port transmission Port pair, and consists of the following member variables:

a) **receptionPortRef**: The portRef (6.7.4.1) value of the associated reception Port.
b) **transmissionPortRef**: The portRef (6.7.4.1) value of the associated transmission Port.
c) **raClassId**: The RA class ID (6.3.2.1) of the associated local RA class.
d) **maxHopLatency**: An unsigned integer, indicating a latency value, in nanoseconds, that specifies a latency constraint as described in 6.3.6.1. The latency is measured from a point located in an upstream station connected via a LAN to the reception Port, to a point located in the transmission Port, while the exact measurement points are specific to and defined by the RA class template being used by the RA class.

The portPairRaClass variable is associated with a key <receptionPortRef, transmissionPortRef, raClassId>.

6.8.4.11 redundancyContext

The redundancyContext variable is an array in which each element represents a Redundancy Context supported by the Bridge and consists of the following member variables:

a) **id**: The Redundancy Context ID of the Redundancy Context, as defined in 6.3.9.2.
b) **vlanContextList**: The set of VLAN Context IDs associated with the Redundancy Context.

The redundancyContext variable is associated with a key <id>.

6.8.4.12 frerCapable

A Boolean value, indicating whether the Bridge is a FRER-capable RAP Bridge (TRUE) or not (FALSE).

6.8.4.13 maxProcessingDelay

An unsigned integer, indicating the maximum delay, in nanoseconds, that a frame can experience during the forwarding process of the Bridge until it is placed into an outbound queue.

6.8.4.14 minProcessingDelay

An unsigned integer, indicating the minimum delay, in nanoseconds, that a frame can experience during the forwarding process of the Bridge until it is placed into an outbound queue.

6.8.4.15 eAttrReg

An event pointer (6.2.4) associated with the ATTR_REG.indication primitive (6.7.2.3).

6.8.4.16 eAttrDereg

An event pointer (6.2.4) associated with the ATTR_DEREG.indication primitive (6.7.2.4).

6.8.4.17 eDecOperStatus

An event pointer (6.2.4) associated with the DEC_OPER_STATUS.indication primitive (6.7.2.5).

6.8.4.18 eVlanConfigChange

An event pointer (6.2.4) associated with the VLAN_CONFIG_CHANGE.indication primitive (6.8.3.1).

6.8.4.19 eMacAddrReg

An event pointer (6.2.4) associated with the MAC_ADDR_REG.indication primitive (6.8.3.2).

6.8.4.20 eMacAddrDereg

An event pointer (6.2.4) associated with the MAC_ADDR_DEREG.indication primitive (6.8.3.3).

6.8.4.21 eResourceChange

An event pointer (6.2.4) associated with the RESOURCE_CHANGE.indication primitive (6.8.3.4).

6.8.5 RAP Propagator state machine procedures

6.8.5.1 processRaReg(pPortRef, pRaAttr)

This procedure processes an RA registration (pRaAttr) on a Port (pPortRef), as follows:

```
processRaReg(pPortRef, pRaAttr) {
  port[pPortRef].neighborMaxInterferingFrameSize =
                                                pRaAttr.MaxInterferingFrameSize;
  // store neighbor's RA class settings received on this Port
  delete neighborRaClass[pPortRef, *];
  for (tDescriptor : pRaAttr.RAClassDescriptor[*]) {
    tNeighborRaClass = create neighborRaClass[pPortRef,tDescriptor.RaClassId];
    tNeighborRaClass.portRef = pPortRef;
    tNeighborRaClass.id = tDescriptor.RaClassId;
    tNeighborRaClass.priority = tDescriptor.RaClassPriority;
    tNeighborRaClass.rtid = tDescriptor.RTID;
    tNeighborRaClass.trafficClass = tDescriptor.TrafficClass;
    tNeighborRaClass.maxLastHopLatency = tDescriptor.MaxLastHopLatency;
    tNeighborRaClass.templateDefinedData =
                                       tDescriptor.RaClassTemplateDefinedData;
  }

  // determine RA class domain boundary status on this Port
  for (tLocalRaClass : localRaClass[*]) {
    tNeighborRaClass = neighborRaClass[pPortRef, tLocalRaClass.id];
    if (tNeighborRaClass != NULL &&
        tNeighborRaClass.priority == tLocalRaClass.priority) {
      portRaClass[pPortRef, tLocalRaClass.id].domainBoundaryPort = FALSE;
    } else {
      portRaClass[pPortRef, tLocalRaClass.id].domainBoundaryPort= TRUE;
    }
  }

  // update redundancy text consistency on this Port
  UpdateRedundancyContextConsistency( pPortRef, pRaAttr );
}
```

6.8.5.2 UpdateRedundancyContextConsistency(pPortRef, pRaAttr)

This procedure updates portRedundancyContext array by comparing the Redundancy Context configuration in this Bridge, as indicated by the value of redundancyContext (6.8.4.10), and in the neighbor Bridge, as indicated by the Redundancy Context sub-TLV(s).

For each item in the portRedundancyContext array:

1) The consistent element is set to TRUE if both this Bridge's and the neighbor's Redundancy Contexts have the same vlanContextList for the same redundancyContextId.
2) The consistent element is set to FALSE if the redundancyContextId exists in both this Bridge and the neighbor, but the vlanContextList does not match.
3) The consistent element is set to FALSE if this Bridge has a vlanContextList for a redundancyContextId, but the neighbor does not.

6.8.5.3 isTaInvalid(pTaReg)

This procedure determines whether a Talker Announce registration referenced by a taReg element (pTaReg) is invalid (TRUE) or not (FALSE). It searches in taReg for an existing Talker Announce registration element (tTaReg) with the same StreamId as pTaAttr. The procedure returns TRUE, if one or more matching tTaReg elements are found in taReg and at least one tTaReg meets one or more of the following conditions:

a) **isSingleContext**(pTaReg.attr) != **isSingleContext**(tTaReg.attr);
b) pTaReg.attr.StreamRank != tTaReg.attr.StreamRank;
c) pTaReg.attr.DestinationMacAddress != tTaReg.attr.DestinationMacAddress;
d) pTaReg.attr.Priority != tTaReg.attr.Priority;
e) **isSingleContext**(pTaReg.attr) && (pTaReg.portRef == tTaReg.portRef) && (pTaReg.attr.VID != tTaReg.attr.VID);
f) **isSingleContext**(pTaReg.attr) && (pTaReg.portRef != tTaReg.portRef) && (pTaReg.attr.VID == tTaReg.attr.VID);
g) If **isSingleContext**(pTaReg.attr) == FALSE, the set of VLAN Context IDs contained in the Redundancy Control sub-TLV in pTaReg.attr is neither the same nor a subset of any Redundancy Context configured in redundancyContext (6.8.4.11).

Otherwise, the procedure returns FALSE.

6.8.5.4 isTaIngressBlocked(pTaReg)

This procedure returns TRUE if a Talker Announce registration referenced by a taReg element (pTaReg) meets all the conditions for Ingress Blocking (6.3.4.1.3) and FALSE otherwise.

6.8.5.5 processTaReg(pTaReg)

This procedure processes a Talker Announce registration referenced by a taReg element (pTaReg), as follows:

```
processTaReg(pTaReg) {
  pTaReg.ingressFailureCode = 0;
  if (getTaStatus(pTaReg.attr) == Announce Fail) {
    // TA failed at an upstream station
    pTaReg.ingressStatus = TA_RECV_FAIL;
    return;
  }

  // check of RA class domain boundary conditions
  tLocalRaClass = getLocalRaClass(pTaReg.attr.Priority);
  tNeighborRaClass = getNeighborRaClass(pTaReg.portRef, pTaReg.attr.Priority);
  if (tLocalRaClass == NULL || tNeighborRaClass == NULL ||
                               tLocalRaClass.id != tNeighborRaClass.id){
    pTaReg.ingressStatus = TA_INGRESS_FAIL;
    pTaReg.ingressFailureCode = getFailureCodeValue(CrossingDomainBoundary);
    return;
  }
  // a Multi-Context Talker Announce registration
  if (!isSingleContext(pTaReg.attr) {
    tRedundancyContexId = getRedundancyContextId(pTaReg.attr.VID);
    if (!portRedundancyContext [pTaReg.portRef,
                               tRedundancyContexId].consistent){
    // check of Redundancy Context consistency
```

```
      pTaReg.ingressStatus = TA_INGRESS_FAIL;
      pTaReg.ingressFailureCode =
                              getFailureCodeValue(InconsistentRedundancyContext);
      return;
    } else if (!pTaReg.attr.RedundancyControl.RTagStatus && !frerCapable) {
    // check of redundancy tagging conditions
      pTaReg.ingressStatus = TA_INGRESS_FAIL;
      pTaReg.ingressFailureCode = getFailureCodeValue(RTagFailed);
      return;
    }
  }
  // set redundancy tagging
  if (isRedundancyTagging(pTaReg)) {
    pTaReg.rTagging = TRUE;
  } else {
    pTaReg.rTagging = FALSE;
  }
  pTaReg.ingressStatus = TA_INGRESS_SUCCESS;
}
```

6.8.5.6 propagateTaReg(pTaReg, pPortRef)

This procedure propagates a Talker Announce registration referenced by a taReg element (pTaReg) to a Port (pPortRef), by associating it with a new or existing Talker Announce declaration in taDec, as follows:

```
propagateTaReg(pTaReg, pPortRef) {
  tTaDec = getTaDec(pPortRef, pTaReg);
  if (tTaDec == NULL) { // a new propagated TA registration

    if (!isStreamMerging(pTaReg, pPortRef) { // not subject to TA merge
      tPropagatedVlanContextList = getPropagatedVlanContextList(pTaReg,
                                                                pPortRef);
      tVid = min(tPropagatedVlanContextList);
      tTaDec = create taDec[pPortRef, pTaReg.attr.StreamId, tVid];
      tTaDec.portRef = pPortRef;
      tTaDec.origTaReg = pTaReg;
    } else { // subject to TA merge
      // search for a TA declaration to merge
      tRedundancyContexId = getRedundancyContextId(pTaReg.attr.VID);
      tMergedVlanContexList = getMergedVlanContextList(tRedundancyContexId,
                                                       pPortRef);
      // use the numerically smallest VLAN Context ID in the merged VLAN Contexts
      tVid = min(tMergedVlanContexList);
      tTaDec = taDec[pPortRef, pTaReg.attr.StreamId, tVid];
      if (tTaDec == NULL) { // 1st TA to be merged
        tTaDec = create taDec[pPortRef, pTaReg.attr.StreamId, tVid];
        tTaDec.portRef = pPortRef;
        tTaDec.origTaReg = NULL;
      }
      // insert it to TA merge list
      tOrigTaReg = create tTaDec.origTaRegList[pTaReg];
      tOrigTaReg.origTaRegRef = pTaReg;
    }
  }
```

```
}
```

6.8.5.7 cancelTaRegPropagation(pTaReg)

This procedure stops propagation of a Talker Announce registration referenced by a taReg element (pTaReg), as follows:

```
cancelTaRegPropagation(pTaReg) {
  for (tTaDec : getTaDecList(pTaReg)) {
    tRemoveTaDec = TRUE;
    if (tTaDec.origTaReg == NULL) { // a merged TA declaration
      delete tTaDec.origTaRegList[pTaReg];
      if (tTaDec.origTaRegList != NULL) { // having other TA registrations
        // update taDec for remaining TA registration(s)
        tRemoveTaDec = FALSE;
        processTaDec(tTaDec);
        DECLARE_ATTR.request(tTaDec.portRef, tTaDec.attr);
      }
    }
    // process LA registration
    tLaReg = getAssociatedLaReg(tTaDec);
    if (tLaReg != NULL) {
      if (tRemoveTaDec) {
        if (tLaReg.isReserved) {
          deallocateResources(tLaReg);
          tLaReg.isReserved = FALSE;
          tLaReg.reservationAge = 0;
          updateAllocatedBandwidth(tLaReg);
        }
        tLaReg.assoTaDec = NULL;
        tLaReg.ingressStatus = NOT_PROPAGATED;
      } else {
        processLaReg(tLaReg);
      }
    }

    if (tRemoveTaDec) { // remove TA declaration
      WITHDRAW_ATTR.request(tTaDec.portRef, tTaDec.attr);
      delete taDec[tTaDec.portRef, tTaDec.attr.StreamId, tTaDec.attr.VID];
    }
  }
  // remove associated LA declaration
  tLaDec = laDec[pTaReg];
  WITHDRAW_ATTR.request(tLaDec.portRef, tLaDec.attr);
  delete laDec[pTaReg];
}
```

6.8.5.8 processTaDec(pTaDec)

This procedure processes a Talker Announce declaration referenced by a taDec element (pTaDec), as follows:

```
processTaDec(pTaDec) {
  if (pTaDec.origTaReg != NULL) { // not subject to TA merge
```

```
    processTaDecNonMerge(pTaDec);
  } else { // subject to TA merge
    processTaDecMerge(pTaDec);
  }
}
```

6.8.5.9 processTaDecNonMerge(pTaDec)

This procedure processes a Talker Announce declaration referenced by a taDec element (pTaDec) that is not subject to Talker Announce merge (6.3.4.3), as follows:

```
processTaDecNonMerge(pTaDec) {
  tTaReg = pTaDec.origTaReg;
  pTaDec.attr = tTaReg.attr;

  if(!isSingleContext(pTaDec.attr)) { // a Multi-Context TA
    tPropagatedVlanContextList = getPropagatedVlanContextList(tTaReg,
                                                              pTaDec.portRef);
    pTaDec.attr.VID = min(tPropagatedVlanContextList);
    // remove VLAN Context Information sub-TLV(s) not propagated
    for (tVlanContextInfoSubTlv : pTaDec.attr.VLANContextInformation[*]) {
      tVlanContextId = tVlanContextInfoSubTlv.VlanContextId;
      if (tVlanContextId NOT IN tPropagatedVlanContextList) {
        delete pTaDec.attr.VLANContextInformation[tVlanContextId];
      }
    }
  }

  if (tTaReg.ingressStatus == TA_RECV_FAIL) {
    // TA failed at an upstream Bridge, propagate it as is
    return;
  } else if (tTaReg.ingressStatus == TA_INGRESS_FAIL) {
    // TA failed at ingress of this Bridge
    failTaDec(pTaDec, tTaReg.ingressFailureCode);
  } else if (isStreamSplitFailed) {
    failTaDec(pTaDec, getFailureCodeValue(StreamSplitFailed));
    return;
  }else if (tTaReg.ingressStatus == TA_INGRESS_SUCCESS){ // TA succeeded so far
    setAccuLatencies(pTaDec); // set accumulated latencies
    setNetworkTSpec(pTaDec); // set TSpec
    if (tTaReg.rTagging) {
      pTaDec.attr.RedundancyControl.RTagStatus = TRUE;
    }
    tLaReg = getAssociatedLaReg(pTaDec);
    if (tLaReg == NULL || !tLaReg.isReserved) { // the stream not yet reserved
      // check resource allocation constraints
      tEgressFailureCode = checkReservationConstraints(pTaDec);
      if (tEgressFailureCode != 0) { // fail TA at egress
        failTaDec(pTaDec, tEgressFailureCode);
        return;
      }
    }
  }
}
```

6.8.5.10 processTaDecMerge(pTaDec)

This procedure processes a Talker Announce declaration referenced by a taDec element (pTaDec) that is subject to Talker Announce merge (6.3.4.3), and determines the Talker Announce attribute in pTaDec.attr, as follows:

a) Set each of the following fields in pTaDec.attr to the value of the corresponding field contained in one of the taReg element listed in pTaDec.origTaRegList: StreamId (6.5.3.1), StreamRank (6.5.3.2), DestinationMacAddress (6.5.3.5.1) and Priority (6.5.3.5.2).

b) Set pTaDec.attr.VID to the VID value used in the key of pTaDec in taDec, as determined when creating pTaDec by the propagateTaReg() procedure (6.8.5.6).

c) Set the Redundancy Control sub-TLV (6.5.3.9) in pTaDec.attr.RedundancyControl as follows:

   1) tRedundancyContextId = **getRedundancyContextId**(pTaDec.attr.VID);
   2) tMergedVlanContextList = **getMergedVlanContextList**(tRedundancyContextId; pTaDec.portRef
   3) If **isRedundancyUntagging**(pTaDec) == TRUE, set pTaDec.RedundancyControl.RTagStatus to FALSE and set pTaDec.rUntagging to TRUE; otherwise, set pTaDec.rUntagging to FALSE.
   4) For each element tTaReg in pTaDec.origTaRegList, copy each VLAN Context Information sub-TLV (6.5.3.9.2) in tTaReg.attr whose VlanContextId value is listed in tMergedVlanContextList to pTaDec.RedundancyControl. If a copied VLAN Context Information sub-TLV does not contain a Failure Information sub-TLV (6.5.3.10) and the tTaReg from which the sub-TLV is copied indicates in ingressStatus TA_INGRESS_FAIL, construct a Failure Information sub-TLV using the RAP Failure Code value contained in tTaReg.ingressFailureCode and the SystemId value of this Bridge, and then append it to the copied sub-TLV.
   5) For each VLAN Context listed in tMergedVlanContextList but not indicated in any VLAN Context Information sub-TLV copied to pTaDec.RedundancyControl in b).4) above, construct a VLAN Context Information sub-TLV for that VLAN Context with a Failure Information sub-TLV that contains the value of the RAP Failure Code *MissingTalkerAnnounce* defined in Table 6-10 and the SystemId value of this Bridge, and then append the constructed VLAN Context Information sub-TLV to pTaDec.attr.RedundancyControl.

d) If pTaDec.origTaRegList contains no element tTaReg whose ingressStatus indicates TA_INGRESS_SUCCESS, perform the following:

   1) Set tEgressFailureCode to the numerically smallest RAP Failure Code contained in the VLAN Context Information sub-TLV(s) in pTaDec.attr.RedundancyControl.
   2) Call **failTaDec**(pTaDec, tEgressFailureCode).

e) Otherwise, i.e., pTaDec.origTaRegList contains at least one element tTaReg whose ingressStatus indicates TA_INGRESS_SUCCESS, perform the following:

   1) Call **setAccuLatencies**(pTaDec) and then **setNetworkTSpec**(pTaDec).
   2) tEgressFailureCode = **checkReservationConstraints**(pTaDec). If tEgressFailureCode != 0, call **failTaDec**(pTaDec, tEgressFailureCode).

6.8.5.11 processLaReg(pLaReg)

This procedure processes a Listener Attach registration referenced by a laReg element (pLaReg), as follows:

```
processLaReg(pLaReg) {
  if (pLaReg.assoTaDec != NULL) { // has an associated TA declaration
    tTaDec = pLaReg.assoTaDec;
    if (pLaReg.attr.ListenerAttachStatus == Attach Fail ||
                getTaStatus(tTaDec.attr) == Announce Fail) {
      // reservation conditions not satisfied
      if (pLaReg.isReserved) { // remove the existing reservation
```

```
        deallocateResources(pLaReg);
        pLaReg.isReserved = FALSE;
        pLaReg.reservationAge = 0;
        updateAllocatedBandwidth(pLaReg);
      }
      pLaReg.ingressStatus = ATTACH_FAIL; // propagate LA as Attach Fail
    } else { // reservation conditions satisfied
      tFailureCode = 0;
      if (!pLaReg.isReserved) { // make a new reservation
        if (tTaDec.attr.StreamRank == 0) {
          preemptReservationsForRank0(pLaReg);
        }
        if (tTaDec.origTaReg != NULL) { // for a non-merged stream
          tFailureCode = allocateResourcesNonMerged(pLaReg);
        } else { // for a merged stream
          tFailureCode = allocateResourcesMerged(pLaReg);
        }
        if (tFailureCode == 0) { // resource allocation succeeded
          pLaReg.isReserved = TRUE;
          if (pLaReg.attr.ListenerAttachStatus == Attach Ready) {
            pLaReg.ingressStatus = ATTACH_READY;
          } else {
            pLaReg.ingressStatus = ATTACH_PARTIAL_FAIL;
          }
        }
      } else if (pLaReg.isReserved && tTaDec.origTaReg == NULL) {
        // update the existing reservation for a merged stream
        tFailureCode = allocateResourcesMerged(pLaReg);
        if (tFailureCode == 0) { // updating reservation succeeded
          pLaReg.isReserved = TRUE;
        }
      }
      // handling reservation failure
      if (tFailureCode != 0) {
        pLaReg.isReserved = FALSE;
        pLaReg.ingressStatus = ATTACH_FAIL;
        // fail the TA declaration
        failTaDec(tTaDec, tFailureCode);
        DECLARE_ATTR.request(tTaDec.portRef, tTaDec.attr);
      }
      updateAllocatedBandwidth(pLaReg);
    }
  } else { // no associated TA declaration
    pLaReg.ingressStatus = NOT_PROPAGATED; // not propagate LA registration
  }
}
```

6.8.5.12 propagateLaReg(pLaReg)

This procedure propagates a given Listener Attach registration referenced by a laReg element (pLaReg), by associating it with one or more new or existing Listener Attach declaration elements in laDec, as follows:

```
propagateLaReg(pLaReg) {
  tTaDec = pLaReg.assoTaDec;
```

```
if (tTaDec != NULL) {
  if (tTaDec.origTaReg != NULL) { // associated TA declaration is not merged
    tLaDec = laDec[tTaDec.origTaReg];
    if (tLaDec = NULL) { // propagated to a new LA declaration
      tLaDec = create laDec[tTaDec.origTaReg];
      tLaDec.assoTaReg = tTaDec.origTaReg;
      tOrigLaReg = create tLaDec.origLaRegList[pLaReg];
      tOrigLaReg.origLaRegRef = pLaReg;
    } else if (tLaDec.origLaRegList(pLaReg) == NULL) {
      // propagated to an existing LA declaration
      tOrigLaReg = create tLaDec.origLaRegList[pLaReg];
      tOrigLaReg.origLaRegRef = pLaReg;
    }
  } else { // associated TA declaration is merged
    for (tTaReg : tTaDec.origTaRegList[*]) {
      tLaDec = laDec[tTaReg];
      if (tLaDec = NULL) { // propagated to a new LA declaration
        tLaDec = create laDec[tTaReg];
        tLaDec.assoTaReg = tTaReg;
        tOrigLaReg = create tLaDec.origLaRegList[pLaReg];
        tOrigLaReg.origLaRegRef = pLaReg;
      } else if (tLaDec.origLaRegList(pLaReg) == NULL) {
        // propagated to an existing LA declaration
        tOrigLaReg = create tLaDec.origLaRegList[pLaReg];
        tOrigLaReg.origLaRegRef = pLaReg;
      }
    }
  }
}
```

6.8.5.13 processLaDec(pLaDec)

This procedure processes a Listener Attach declaration referenced by a laDec element (pLaDec), as follows:

```
processLaDec(pLaDec) {
  tNumPropagated = tNumReady = tNumFailed = 0;
  for (tLaReg : pLaDec.origLaRegList[*] {
    if (tLaReg.ingressStatus == NOT_PROPAGATED) {
      delete pLaDec.origLaRegList[tLaReg];
      continue;
    } else if (tLaReg.ingressStatus == ATTACH_READY) {
      tNumReady++;
    } else if (tLaReg.ingressStatus == ATTACH_FAIL) {
      tNumFailed++;
    }
    tNumPropagated++;
  }
  if (tNumPropagated == 0) { // remove this LA declaration
    WITHDRAW_ATTR.request(pLaDec.assoTaReg.portRef, pLaDec.attr);
    delete laDec[pLaDec.assoTaReg];
    return;
  }
  // construct the Listener Attach attribute
```

```
pLaDec.attr.StreamId = pLaDec.assoTaReg.attr.StreamId;
pLaDec.attr.VID = pLaDec.assoTaReg.attr.VID;
if (isSingleContext(pLaDec.assoTaReg) { // a single context LA declaration
  if (tNumPropagated == tNumReady) { // all propagated as Attach Ready
    pLaDec.attr.ListenerAttachStatus = Attach Ready;
  } else if (tNumPropagated == tNumFailed) { // all propagated as Attach Fail
    pLaDec.attr.ListenerAttachStatus = Attach Fail;
  } else { // Otherwise: declare Attach Partial Fail
    pLaDec.attr.ListenerAttachStatus = Attach Partial Fail;
  }
} else { // a Multi-Context LA declaration
  pLaDec.VLANContextStatus = constructVlanContextStatus(pLaDec);
  if (tNumReady != 0) { // at least one propagated as Attach Ready
    pLaDec.attr.ListenerAttachStatus = Attach Ready;
  } else {
    pLaDec.attr.ListenerAttachStatus = Attach Fail;
  }
  setVidInMultiContextLaDec(pLaDec);
}
DECLARE_ATTR.request(pLaDec.assoTaReg.portRef, pLaDec.attr);
}
```

6.8.5.14 getTaRegDestPorts(pTaReg)

This procedure determines the Port(s) to which a Talker Announce registration referenced by a taReg element (pTaReg) is to be propagated. It returns a list, possibly empty, that includes each possible portRef (6.7.4.1) value for which all of the following conditions are met:

a) portRef != pTaReg.portRef;

b) port[portRef].partDecOper == TRUE;

c) If **isSingleContext**(pTagReg.attr) == TRUE, portRef is in the VLAN Context as indicated by the pTaReg.attr.VID value according to 6.5.3.5.3;

d) If **isSingleContext**(pTagReg.attr) == FALSE, portRef is in at least one of the VLAN Contexts as indicated in the VLAN Context Information sub-TLV(s) contained in pTaReg.attr, according to 6.5.3.9.2;

e) If port[portRef].streamDaPruningEnabled == TRUE, pTaReg.attr.DestinationMacAddress is registered on the Port referenced by portRef, according to 6.3.4.1.2.

6.8.5.15 isRedundancyTagging(pTaReg)

This procedure determines whether a stream as indicated by a Talker Announce registration element (pTaReg) in taReg is subject to redundancy tagging (6.3.9.3). It returns TRUE, if all of the following conditions are met:

a) **isSingleContext**(pTaReg.attr) != TRUE;

b) frerCapable == TRUE;

c) pTaReg.attr.RedundancyControl.RTagStatus == FALSE;

Otherwise, it returns FALSE.

6.8.5.16 isRedundancyUntagging(pTaDec)

This procedure determines whether a stream as indicated by a Talker Announce declaration element (pTaDec) in taDec is subject to redundancy untagging (6.3.9.3). It returns TRUE, if the following condition is met:

a) Let tRedundancyContextId = **getRedundancyContextId**(pTaDec.attr.VID) and tMergedVlanContextList = **getMergedVlanContextList**(tRedundancyContextId, pTaDec.portRef). The tMergedVlanContextList contains all the VLAN Contexts in redundancyContext[tRedundancyContextId].

Otherwise, it returns FALSE.

6.8.5.17 isStreamSplitFailed(pTaReg)

This procedure determines whether a Talker Announce declaration element (pTaDec) in taDec is to be failed. It returns true if either of the conditions described in c), d) and e) of 6.3.9.4 are met.

For a Bridge, doing 802.1Q forwarding an example pseudo code is given below.

```
isStreamSplitFailed (pTaDec) {
    If (isStreamSplitPoint(pTaDec.origTaReg && !frerCapable) {
        return true;
    }

    tLaDec = laDec[pTaDec.origTaReg]
    if (laDec.attr.VID NOT IN
taDec.attr.VLANContextInformation[*].VlanContextId) {
        return true;
    }
    return false;
}
```

6.8.5.18 isStreamSplittingPoint(pTaReg)

This function checks if the stream is to be split at this Bridge.

6.8.5.19 isStreamMerging(pTaReg, pPortRef)

This procedure determines whether a stream as indicated by a Talker Announce registration element (pTaReg) in taReg is subject to stream merging (6.3.9.5) on the Port (pPortRef). It returns TRUE, if all of the following conditions are met:

a) **isSingleContext**(pTaReg.attr) != TRUE;

b) frerCapable == TRUE;

c) Let tPropagatedVlanContextList = **getPropagatedVlanContextList**(pTaReg, pPortRef), tRedundancyContextId = **getRedundancyContextId**(pTaReg.attr.VID), and tMergedVlanContextList = **getMergedVlanContextList**(tRedundancyContextId, pPortRef). The list of VID(s) contained in tPropagatedVlanContextList is a subset (less than) of the list of VID(s) contained in tMergedVlanContextList.

Otherwise, it returns FALSE.

6.8.5.20 getPropagatedVlanContextList(pTaReg, pPortRef)

This procedures returns a list containing each VLAN Context ID used in propagation of a given Talker Announce registration element (pTaReg) in taReg to a given Port (pPortRef), according to the VLAN Context(s) indicated in pTaReg.attr and the VLAN membership of the Port, as described in 6.3.4.1.1.

6.8.5.21 getMergedVlanContextList(pRedundancyContextId, pPortRef)

This procedures returns a list containing each VLAN Context ID associated with the Redundancy Context identified by pRedundancyContextId and including the Port pPortRef in its member set.

6.8.5.22 getRedundancyContextId(pVlanContextId)

This procedures returns the Redundancy Context ID of a Redundancy Context in redundancyContext (6.8.4.11) that includes a VLAN Context ID contained in pVlanContextId.

6.8.5.23 getTaDec(pPortRef, pTaReg)

This procedure searches in taDec for a Talker Announce declaration element that contains in either origTaReg or origTaRegList a reference to a given Talker Announce registration element (pTaReg) in taReg and contains in portRef a value matching pPortRef. If such an element is found in taDec, it returns a reference to the matching element in taDec. Otherwise, it returns NULL.

6.8.5.24 getTaDecList(pTaReg)

This procedure searches in taDec for one or more Talker Announce declaration elements, each containing in either origTaReg or origTaRegList a reference to a given Talker Announce registration element (pTaReg) in taReg. If such elements are found in taDec, it returns a list of references to the matching element(s) in taDec. Otherwise, it returns NULL.

6.8.5.25 getAssociatedTaDec(pLaReg)

This procedure returns a reference to a taDec element that contains a Talker Announce declaration with which a given Listener Attach registration referenced by a laReg element (pLaReg) is associated, in accordance with 6.3.5.1, or NULL if such a taDec element does not exist.

6.8.5.26 getAssociatedLaReg(pTaDec)

This procedure returns a reference to a laReg element that contains a Listener Attach registration associated with a given Talker Announce declaration referenced by a taDec element (pTaDec), in accordance with 6.3.5.1, or NULL if such a laReg element does not exist.

6.8.5.27 getLaDecList(pLaReg)

This procedure searches in laDec for one or more Listener Attach declaration elements, each containing in origLaRegList a reference to a given Listener Attach registration element (pLaReg) in laReg. If such elements are found in laDec, it returns a list of references to the matching element(s) in laDec. Otherwise, it returns NULL.

6.8.5.28 isSingleContextTa(pTaAttr)

This procedure returns TRUE if the Talker Announce attribute supplied in pTaAttr indicates a Single-Context Talker Announcement according to [item a) in 6.5.3], and returns FALSE if pTaAttr indicates a Multi-Context Talker Announcement according to the [item b) in 6.5.3].

6.8.5.29 getTaStatus(pTaAttr)

This procedure returns the Talker Announce status of the Talker Announce attribute supplied in pTaAttr, according to 6.3.4.2.

6.8.5.30 getFailureCodeValue(pFailureCodeName)

This procedure returns the value of a RAP Failure Code (pFailureCodeName) defined in Table 6-10.

6.8.5.31 setAccuLatencies(pTaDec)

This procedure determines the value of the maximum and minimum accumulated latency values for a Talker Announce declaration element (pTaDec), as follows:

```
setAccuLatencies(pTaDec) {
  tTxPort = pTaDec.portRef;
  tRaClass = getLocalRaClass(pTaDec.attr.Priority);
  if (pTaDec.origTaRegList == NULL) { // a stream not subject to stream merging
    tTaReg = pTaDec.origTaReg;
    tRxPort = tTaReg.portRef;
    pTaDec.attr.AccuMaxLatency +=
           portPairRaClass[tRxPort, tTxPort, tRaClass.id].maxHopLatency;
    pTaDec.attr.AccuMinLatency += minProcessingDelay +
                              port[tRxPort].minPropagationDelay +
       tTaReg.attr.NetworkTSpec.MinTransmittedFrameLength*8*1e9 /
                                 port[tRxPort].portTransmitRate;
  } else { // a stream subject to stream merging
    tMaxAccuMaxLatency = tMinAccuMinLatency = 0;
    for (tTaReg : pTaDec.origTaRegList[*]) {
      tRxPort = tTaReg.portRef;
      if (tTaReg.ingressStatus == TA_INGRESS_SUCCESS) {
        tAccuMaxLatency =
           portPairRaClass[tRxPort, tTxPort, tRaClass.id].maxHopLatency;
        tAccuMinLatency = minProcessingDelay +
                                          port[tRxPort].minPropagationDelay +
                  tTaReg.attr.NetworkTSpec.MinTransmittedFrameLength * 8 * 1e9 /
                                             port[tRxPort].portTransmitRate;
      if (tAccuMaxLatency > tMaxAccuMaxLatency) {
        tMaxAccuMaxLatency = tAccuMaxLatency;
      }
      if (tMinAccuMinLatency == 0 || tAccuMinLatency < tMinAccuMinLatency) {
        tMinAccuMinLatency = tAccuMinLatency;
      }
    }
    pTaDec.attr.AccuMaxLatency = tMaxAccuMaxLatency;
    pTaDec.attr.AccuMinLatency = tMinAccuMinLatency;
  }
}
```

6.8.5.32 setNetworkTSpec(pTaDec)

This procedures determines the value of the NetworkTSpec (6.5.3.9) for a Talker Announce declaration element (pTaDec) in taDec, as follows:

a) If pTaDec.origTaReg == NULL, determine the pTaDec.attr.NetworkTSpec value for a stream that is to be merged from each Member Stream listed in pTaDec.origTaRegList and whose ingressStatus indicates TA_INGRESS_SUCCESS. The algorithm for determining the TSpec in this case depends on the mechanism used by a Bridge for stream merging and is out of the scope of this standard.

b) Adjust the pTaDec.attr.NetworkTSpec value as necessary to take into account the factors such as the ones described in [Q]:V.7 and [Q]:V.8 for ATS.

c) Adjust the frame size parameters in pTaDec.attr.NetworkTSpec if redundancy tagging or untagging is applied to this stream, as follows:

```
if ((pTaDec.origTaReg != NULL) && pTaDec.origTaReg.rTagging) {
  // increase frame sizes by a R-TAG length of 6 octets
  pTaDec.attr.NetworkTSpec.MaxTransmittedFrameLength += 6;
  pTaDec.attr.NetworkTSpec.MinTransmittedFrameLength += 6;
} else if (pTaDec.rUntagging) {
  // decrease frame sizes by a R-TAG length of 6 octets
  pTaDec.attr.NetworkTSpec.MaxTransmittedFrameLength -= 6;
  pTaDec.attr.NetworkTSpec.MinTransmittedFrameLength -= 6;
}
```

6.8.5.33 failTaDec(pTaDec, pFailureCode)

This procedure fails a Talker Announce declaration referenced by a taDec element (pTaDec), as follows:

a) Construct a Failure Information sub-TLV (6.5.3.10) using the SystemId value of this Bridge and the RAP Failure Code indicated in pFailureCode.

b) Append the Failure Information sub-TLV constructed in item a) to the Talker Announce attribute contained in pTaDec.attr, in accordance with 6.5.3.

c) If **isSingleContext**(pTaDec.attr) == FALSE, append the Failure Information sub-TLV constructed in item a) to each VLAN Context Information sub-TLV (6.5.3.9.2) in pTaDec.attr that currently contains no Failure Information sub-TLV.

6.8.5.34 constructVlanContextStatus(pLaDec)

This procedure constructs and returns in tVlanContextStatus a VLAN Context Status sub-TLV (6.5.4.4) for a Multi-Context Listener Attach declaration element (pLaDec) in laDec, as follows:

a) For each originating Listener Attach registration (tLaReg) contained in pLaDec.origLaRegList, perform the following actions:
   1) tPropagatedVlanContextStatus = tLaReg.attr.VLANContextStatus;
   2) Delete in tPropagatedVlanContextStatus each VLAN Context tuple for a VLAN Context that is not indicated in pLaDec.assoTaReg.attr.RedundancyControl.
   3) If tLaReg.ingressStatus ==ATTACH FAIL, set the PathStatus (6.5.4.4.1) of each VLAN Context tuple contained in tPropagatedVlanContextStatus to **Faulty Path**.

b) Construct a VLAN Context Status sub-TLV in tVlanContextStatus, as follows:
   1) Copy each VLAN Context tuple contained in each tPropagatedVlanContextStatus obtained in a) to tVlanContextStatus.
   2) For each VLAN Context indicated in pLaDec.assoTaReg.attr.RedundancyControl but not contained in tVlanContextStatus, create a VLAN Context tuple for that VLAN Context with its PathStatus set to **Unresponsive Path** and then append it to tVlanContextStatus.

c) If there is more than one VLAN Context tuple in tVlanContextStatus indicating the same VLAN Context, retain only one of them and delete the others, with the following rules:
   1) If there is one whose PathStatus is **Faulty Path**, retain this one;
   2) Else if there is one whose PathStatus is **Faultless Path**, retain this one;
   3) Otherwise, retain any one.

d) Return tVlanContextStatus.

6.8.5.35 setVidInMultiContextLaDec(pLaDec)

This procedure determines the VID (6.5.4.2) value for a Multi-Context Listener Attach declaration element (pLaDec) in laDec, as follows:

a) If there is only one VLAN Context indicated in pLaDec.attr.VLANContextStatus, set pLaDec.attr.VID to the VLAN Context ID of that VLAN Context and the procedure terminates.

b) Let tPortList = **getTaRegDestPorts**(pLaDec.assoTaReg); For each Port (tPortRef) in tPortList, let tPropagatedVlanContextList = **getPropagatedVlanContextList**(pLaDec.assoTaReg, tPortRef);

c) If there is one or more VLAN Contexts commonly present in each tPropagatedVlanContextList obtained in b) above, set pLaDec.attr.VID to the numerically smallest VLAN Context ID of the common VLAN Context(s). Otherwise, set pLaDec.attr.VID to the numerically smallest VLAN Context ID of the VLAN Contexts in pLaDec.attr.VLANContextStatus.

6.8.5.36 constructRaAttr(pPortRef)

This procedure constructs and returns an RA attribute (tRaAttr) in preparation for an RA declaration on a Port (pPortRef), as follows:

a) For each localRaClass (6.8.4.5) element (tLocalRaClass), construct an RA Class Descriptor sub-TLV (raDescTlv) in accordance with 6.5.2.2 and fill its Value field, as follows:

```
raDescTlv.RaClassID            = tLocalRaClass.id;
raDescTlv.RaClassPriority      = tLocalRaClass.priority;
raDescTlv.RTID                 = tLocalRaClass.rtid;
raDescTlv.TrafficClass         = getTrafficClass(pPortRef,
                                                 tLocalRaClass.priority);
raDescTlv.MaxLastHopLatency    =
              portRaClass[pPortRef, tLocalRaClass.id].maxLastHopLatency;
raDescTlv.RaClassTemplateDefinedData =
                                          tLocalRaClass.templateDefinedData;
```

b) For each redundancyContext (6.8.4.11) element (tRedundancyContext), if any, construct a Redundancy Context sub-TLV and fill its Value field with each VLAN Context ID contained in tRedundancyContext.vlanContextList, in accordance with 6.5.2.3.

c) Construct an RA attribute TLV (tRaAttr) and fills its Value field with the value of port[pPortRef].maxInterferingFrameSize and the sub-TLV(s) constructed in item a) and b) above, in accordance with 6.5.2.

d) Return tRaAttr.

6.8.5.37 getLocalRaClass(pPriority)

This procedure returns a reference to a localRaClass (6.8.4.5) element whose priority value matches the pPriority value.

6.8.5.38 getNeighborRaClass(pPortRef, pPriority)

This procedure returns a reference to a neighborRaClass (6.8.4.6) element whose portRef value matched the pPortRef value and whose priority value matches the pPriority value.

6.8.5.39 getTrafficClass(pPortRef, pPriority)

This procedure returns the traffic class to which a priority (pPriority) is assigned on a given Bridge Port (pPortRef).

6.8.5.40 checkReservationConstraints(pTaDec)

This procedure performs checks to determine whether a stream as indicated by a Talker Announce declaration referenced by a taDec element (pTaDec) meets the resource allocation constraints (6.3.6). This subclause specifies only the input and output parameters, along with the rules of handling the stream Rank, but not the detailed algorithms that are specific to the RA class template used by the stream specific or dependent on the particular Bridge implementation.

In computing the current available resources, the existing reservations in this Bridge are treated according to the pTaDec.attr.StreamRank value, as follows:

a) If pTaDec.attr.StreamRank contains the value zero, only those existing reservations being made for the streams of Rank zero are treated as “reserved”, and all other existing reservations for the streams of Rank one are treated as “not reserved” as they can be preempted as described in 6.3.8.
b) If pTaDec.attr.StreamRank contains the value one, all of the existing reservations regardless of the stream Rank are treated as “reserved”.

The procedure checks each reservation constraint defined in 6.3.6, for which there is no requirement for the order in which one constraint is examined after another. Note that when the stream is subject to stream merging (6.3.9.5) on a FRER-capable RAP Bridge’s Port, as indicated by pTaDec.origTaReg == NULL, the procedure takes into account only each Member Stream listed in pTaDec.origTaRegList and whose ingressStatus indicates TA_INGRESS_SUCCESS.

In case that one or more of these constraints are not met, the procedure returns the value of the RAP Failure Code associated with one constraint not met, as described in 6.3.6. It returns the value zero, if all of the constraints are met.

The procedures described in 6.8.5.46, 6.8.5.47, and 6.8.5.48 provide example of latency computation algorithms for the RA class templates defined in Table 6-3.

6.8.5.41 preemptReservationsForRank0(pLaReg)

This procedure is invoked prior to making a reservation for a stream of Rank zero and indicated by a Listener Attach registration referenced by a laReg element (pLaReg) to perform the following actions:

a) Determine whether it is necessary to remove one or more reservations made for the streams of Rank one, if necessary, determine which of those reservations are to be removed according to the rules of reservation importance as defined in 6.3.8.

   NOTE—For making the decisions required by a), this procedure can use the same algorithms as used by the checkResevationConstraints() procedure (6.8.5.40) with different assumptions about which of the existing reservations are treated “reserved” or “not reserved”.

b) For each reservation to be moved as determined in a), if any, perform the following actions with regard to the Listener Attach registration (tLaReg) with which the reservation is associated:

```
deallocateResources(tLaReg);
```

```
        tLaReg.isReserved = FALSE;
        tLaReg.reservationAge = 0;
        updateAllocatedBandwidth(tLaReg);
        tTaDec = tLaReg.assoTaDec;
        failTaDec(tTaDec, getFailureCodeValue(ReservationPreempted));
        DECLARE_ATTR.request(tTaDec.portRef, tTaDec.attr);
        tLaReg.ingressStatus = ATTACH_FAIL;
        for (tLaDec : getLaDecList(tLaReg)) {
          processLaDec(tLaDec);
        }
```

6.8.5.42 allocateResourcesNonMerged(pLaReg)

This procedure is invoked to create a reservation for a Listener Attach registration referenced by a laReg element (pLaReg) and indicating a stream that is not subject to stream merging (6.3.9.5).

The procedure takes all necessary actions, such as configuration of shaper, buffer size and FRER functions, to ensure the QoS required by the stream. The information about the characteristics of the stream is contained in both the Listener Attach registration (pLaReg) and its associated Talker Announce declaration (pLaReg.assoTaDec).

If all the required actions have been successfully carried out, the time measurement for pLaReg.reservationAge is started and the procedure returns a value of zero. If some of the required actions have failed, the procedure cancels the actions already taken, if any, and returns the value of the RAP Failure Code *ResourceAllocationFailed* defined in Table 6-10.

6.8.5.43 allocateResourcesMerged(pLaReg)

This procedure is invoked to create a new reservation or update an existing reservation for a Listener Attach registration referenced by a laReg element (pLaReg) and indicating a stream subject to stream merging (6.3.9.5) on a FRER-capable RAP Bridge's Port.

NOTE—The need for updating an existing reservation arises from dynamic changes in the presence or status of the propagated Talker Announce registration of each Member Stream at a stream merging port. For example, assuming a total of three Member Streams are expected to be merged, a reservation was initially made only for two of them propagated with the Announce Success status. The reservation needs to be adjusted later on when the third one is also propagated as Announce Success, or when one previously with Announce Success is changed to Announce Fail or not any more propagated to the stream merging port.

The procedure takes all necessary actions, such as configuration of shaper, buffer size and FRER functions, to ensure the QoS required by the stream. The information about the stream characteristics is contained in both the Listener Attach registration (pLaReg) and its associated Talker Announce declaration (pLaReg.assoTaDec). Note that a reservation created or updated by this procedure only allows forwarding and merging of each Member Stream listed in pLaReg.assoTaDec.origTaRegList and whose ingressStatus indicates TA_INGRESS_SUCCESS.

If all the required actions have been successfully carried out, the time measurement for pLaReg.reservationAge is started (only when creating a new reservation) and the procedure returns a value of zero. If some of the required actions have failed, the procedure cancels the actions already taken, if any, and returns the value of the RAP Failure Code *ResourceAllocationFailed* defined in Table 6-10.

6.8.5.44 deallocateResources(pLaReg)

This procedure is invoked to remove an existing reservation associated with a Listener Attach registration referenced by a laReg element (pLaReg), by reversing all the changes to the underlying Bridge mechanisms previously made for this reservation.

6.8.5.45 updateAllocatedBandwidth(pLaReg)

This procedure is invoked, when creating or removing a reservation associated with a Listener Attach registration referenced by a laReg element (pLaReg), to update the corresponding allocatedBandwidth value [item e) in 6.8.4.9], as follows:

```
updateAllocatedBandwidth(pLaReg) {
  tRaClass = getLocalRaClass(pLaReg.assoTaDec.attr.priority);
  tPort = pLaReg.portRef;
  portRaClass[tPort, tRaClass.id].allocatedBandwidth = 0;
  for (tLaReg: laReg[tPort, *, *] {
    tTaDec = getAssociatedTaDec(tLaReg);
    if (tTaDec != NULL && tTaDec.attr.priority == tRaClass.priority &&
                                                    tLaReg.isReserved) {
      portRaClass[tPort, tRaClass.id].allocatedBandwidth +=
                  ceil(1e8 * tTaDec.attr.NetworkTSpec.CommittedInformationRate /
                                                 port[tPort].portTransmitRate);
    }
  }
}
```

<< Editor's note: The following subclauses are only intended as example algorithms specific to SP and ATS. Consider to move them to an informative annex, and to provide necessary explanatory text/figures. >>

6.8.5.46 checkLatencyConstraintSP(pTaDec)

This subclause provides an example of checking the latency constraint (6.3.6.1) for resource allocation in RA classes using the RA class template 00-80-C2-00 for Strict Priority as specified in Table 6-3. The procedure is described as part of the **checkReservationConstraints**() procedure (6.8.5.40) invoked during processing of a Talker Announce declaration element (pTaDec) in taDec and is intended for use only in resource allocation without seamless redundancy.

```
checkLatencyConstraintSP(pTaDec) {
  tTaReg = pTaDec.origTaReg;
  tRxPort = tTaReg.portRef; // reception port of this stream
  tTxPort = pTaDec.portRef; // transmission port of this stream
  tMaxLowerPrioFrameSize = port[tRxPort].neighborMaxInterferingFrameSize;
  for (tObsvRaClass: localRaClass[*]) {
    if (!portRaClass[tRxPort, tObsvRaClass.id].domainBoundaryPort) {
      tSumBursts = getInterferingBurstSizeSP(tObsvRaClass.id, tTaReg, tTxPort);
      // collect info from all existing reservations made for streams received
         on that Port.
      for (tLaDec: laDec[*]) {
        if (tLaDec.assoTaReg.portRef == tRxPort &&
            tLaDec.attr.ListenerAttachStatus != Attach Fail) {
          tSumBursts += getInterferingBurstSizeSP(tObsvRaClass.id,
                                                  tLaDec.assoTaReg, tTxPort);
        }
      }
```

```
      tSumBursts += tMaxLowerPrioFrameSize * 8;
      tMaxQueuingDelay = ceil(1e9 * tSumBursts /
                                                port[tRxPort].portTransmitRate);
      tMaxFrameSize = tTaReg.attr.NetworkTSpec.MaxTransmittedFrameLength;
      tMaxSFDelay = tMaxFrameSize * 8 * 1e9 / port[tRxPort].portTransmitRate;
      tCurrMaxHopLatency = tMaxQueuingDelay + port[tRxPort].maxPropagationDelay
                                            + tMaxSFDelay + maxProcessingDelay;
      if (tCurrMaxHopLatency > portPairRaClass[tRxPort, tTxPort,
                                               tObsvRaClass.id].maxHopLatency) {
        tFailureCode = getFailureCodeValue(LatencyExceeded);
        return tFailureCode; // latency constraint not met
      }
    }
  }
  return 0; // latency constraint met
}
```

6.8.5.47 getInterferingBurstSizeSP(pObservedRaClassId, pTaReg, pTxPort)

As part of the example in 6.8.5.46 for Strict Priority, this procedure computes the amount of interference generated by transmission of a stream referenced by a taReg element (pTaReg) on a Port in an upstream station, to an RA class identified by a given RA class ID (pObservedRaClassId) on that Port, as follows:

```
getInterferingBurstSizeSP(pObservedRaClassId, pTaReg, pTxPort) {
  tAttr = pTaReg.attr;
  tMinFrameSize = tAttr.NetworkTSpec.MinTransmittedFrameLength; // bytes
  tRxPort = pTaReg.portRef;
  tObsvTrafficClass = neighborRaClass[tRxPort, pObservedRaClassId].trafficClass;
  tIntfRaClassId = getLocalRaClass(tAttr.priority).id;
  tIntfTrafficClass = neighborRaClass[tRxPort, tIntfRaClassId].trafficClass;
  if (tObsvTrafficClass == tIntfTrafficClass) {
    tTimeFrame = tAttr.AccuMaxLatency - tAttr.AccuMinLatency;
    tIntfBurstSize = tAttr.NetworkTSpec.CommittedBurstSize +
                     tAttr.NetworkTSpec.CommittedInformationRate * tTimeFrame;
  } else if (tObsvTrafficClass < tIntfTrafficClass) {
    tTimeFrame = tAttr.AccuMaxLatency – tAttr.AccuMinLatency +
            portPairRaClass[tRxPort, pTxPort, pObservedRaClassId].maxHopLatency;
    tIntfBurstSize = tAttr.NetworkTSpec.CommittedBurstSize +
                     tAttr.NetworkTSpec.CommittedInformationRate * tTimeFrame;
  } else {
    tIntfBurstSize = 0;
}
return tIntfBurstSize;
}
```

6.8.5.48 checkLatencyConstraintATS(pTaDec)

This procedure provides an example of checking the latency constraint (6.3.6.1) for resource allocation in RA classes using the RA class template 00-80-C2-01 for Asynchronous Traffic Shaping as specified in Table 6-3. The procedure is described as part of the **checkReservationConstraints**() procedure (6.8.5.40) invoked during processing of a Talker Announce declaration element (pTaDec) in taDec and is intended for use only in resource allocation without seamless redundancy.

```
checkLatencyConstraintATS(pTaDec) {
  tTaReg = pTaDec.origTaReg;
```

```
tRxPort = tTaReg.portRef; // reception port of this stream
tTxPort = pTaDec.portRef; // transmission port of this stream
tAttr = tTaReg.attr;
tMaxLowerPrioFrameSize = port[tRxPort].neighborMaxInterferingFrameSize;
for (tObsvRaClass: localRaClass[*]) {
  if (!portRaClass[tRxPort, tObsvRaClass.id].domainBoundaryPort) {
    tObsvTrafficClass =
                      neighborRaClass[tRxPort, tObsvRaClass.id].trafficClass;
    tSumRates = 0; // bits/s
    tSumBursts = tAttr.NetworkTSpec.CommittedBurstSize; // bits
    tMinEqualPrioFrameSize = tAttr.NetworkTSpec.MinTransmittedFrameLength;
    // collect info from all existing reservations made for streams received
       on that Port.
    for (tLaDec: laDec[*]) {
      if (tLaDec.assoTaReg.portRef == tRxPort &&
          tLaDec.attr.ListenerAttachStatus != Attach Fail) {
        tIntfAttr = tLaDec.assoTaReg.attr;
        tIntfRaClassId = getLocalRaClass(tIntfAttr.priority).id;
        tIntfTrafficClass =
                       neighborRaClass[tRxPort, tIntfRaClassId].trafficClass;
        if (tIntfTrafficClass > tObsvTrafficClass) {
          tSumBursts += tIntfAttr.NetworkTSpec.CommittedBurstSize;
          tSumeRates += tIntfAttr.NetworkTSpec.CommittedInformationRate;
        } else if (tIntfTrafficClass == tObsvTrafficClass) {
          tSumBursts += tIntfAttr.NetworkTSpec.CommittedBurstSize;
          if (tIntfAttr.NetworkTSpec.MinTransmittedFrameLength <
                                          tMinEqualPrioFrameSize) {
            tMinEqualPrioFrameSize =
                              tIntfAttr.NetworkTSpec.MinTransmittedFrameLength;
          }
        }
      }
    }
    tSumBursts += (tMaxLowerPrioFrameSize – tMinEqualPrioFrameSize) * 8;
    tRemainingDateRate = port[tRxPort].portTransmitRate – tSumRates;
    tMaxQueuingDelay = ceil(tSumBursts / tRemainingDateRate +
           tMinEqualPrioFrameSize * 8 / port[tRxPort].portTransmitRate) * 1e9;
    tMaxFrameSize = tAttr.NetworkTSpec.MaxTransmittedFrameLength;
    tMaxSFDelay = tMaxFrameSize * 8 * 1e9 / port[tRxPort].portTransmitRate;
    tCurrMaxHopLatency = tMaxQueuingDelay + port[tRxPort].maxPropagationDelay
                                          + tMaxSFDelay + maxProcessingDelay;
    if (tCurrMaxHopLatency > portPairRaClass[tRxPort, tTxPort,
                                             tObsvRaClass.id].maxHopLatency) {
      tFailureCode = getFailureCodeValue(LatencyExceeded);
      return tFailureCode; // latency constraint not met
    }
  }
}
return 0; // latency constraint met
}
```

6.8.5.49 checkLatencyConstraintCBS(pTaDec)

This subclause provides an example of checking the latency constraint (6.3.6.1) for resource allocation in RA classes using the RA class template 99-99-99-99 for Credit Based Shaper as specified in Table 6-3. The procedure is described as part of the checkReservationConstraints() procedure (6.8.5.39) invoked during processing of a Talker Announce declaration element (pTaDec) in taDec and is intended for use only in resource allocation without seamless redundancy.

The calculations used by this function shall be as described in Clause 6.6. in IEEE Std 802.1BA-2021.

6.9 RAP/MSRP Converter

<< Editor's note: The is a placeholder for specifying a RAP/MSRP Converter that provides translation functions required to support interoperability of RAP with MSRP in a RAP/MSRP Gateway as shown in Figure 6-3.>>

6.9.1 MSRP/RAP Converter overview

A MSRP/RAP Converter associated with a RAP/MSRP Gateway (5.6) is responsible for the following:

a) Link between MSRP Participants' Stream Registration Service Interface (SRSI) ([Q]:35.2.3) and RAP Participant users' RAP Participant Service Interface (RPSI) (6.7.2)
b) Provision of mapping mechanism between SRSI and RPSI primitives
c) Translation of MSRP Attributes and corresponding Failure Codes to RAP Attributes and Error Codes and vice versa

The operation of a MSRP/RAP Converter is specified in the following subclauses:

d) MSRP/RAP Converter Service Interface Primitive Translation in 6.9.2
e) MSRP/RAP Converter Attribute Translation in 6.9.3
f) MSRP/RAP Converter Failure Code Translation in 6.9.4

6.9.2 MSRP/RAP Converter Service Interface Primitive Translation

In an MSRP/RAP Gateway architecture, the MSRP Participant represents an equivalent component of the RAP Participant. So that the RAP Participant user (RAP Propagator) and MSRP Participant can continue to use their existing sets of Service Interface Primitives (see [Q]: 35.2.3, 6.7.2), the primitives occurring at the converter SRSI and RPSI interfaces must be translated into each other as shown in Table INTERFACES.

<Editor's note: Table INTERFACES will be added>

6.9.3 MSRP/RAP Converter Attribute Translation

6.9.3.1 Attribute Value Translation

Table ATTRIBUTES provides a translation between MSRP attributes (see [Q]: 35.2.2) and RAP attributes (see 6.5) that are exchanged through the primitives as specified in clause 6.9.2.

<Editor's note: Table ATTRIBUTES will be added>

6.9.3.2 Failure Code Translation

Table FAILURE_CODES provides a translation between MSRP failure codes (see [Q], Table 46-15) and RAP failure codes (see Table 6-10).

<Editor's note: Table ERROR_CODES will be added>

7. Resource Allocation Protocol (RAP) management

Editor's note: If RAP/MSRP Gateway requires management this has to be added to this clause.

This subclause defines the following managed objects that allows administrative configuration of RAP:

a) RAP Participant Table (7.1)
b) RAP Propagator Bridge Table (7.2)
c) RAP Propagator Port Table (7.3)
d) RA Class Bridge Table (7.4)
e) RA Class Port Table (7.5)
f) RA Class Port Pair Table (7.6)
g) Priority Regeneration Override Table (7.7)
h) Redundancy Context Table (7.8)
i) Talker Announce Registration Port Table (7.9)
j) Talker Announce Declaration Port Table (7.10)
k) Listener Attach Registration Port Table (7.11)
l) Listener Attach Declaration Port Table (7.12)

7.1 RAP Participant Table

There is one RAP Participant Table per RAP Participant (6.7). The table contains a set of parameters that supports the operation of a RAP Participant for an end station or Bridge Port, as detailed in Table 7-1.

Table 7-1—RAP Participant Table

Name	Data type	Operations supported[a]	References
participantEnabled	Boolean	RW	6.7.4.2
neighborDiscoveryMode	Enumerated	RW	6.7.4.3
helloTime	integer	RW	6.7.4.4
completeListTimerReset	integer	RW	6.7.4.5
exploreHelloRecvEnabled	Boolean	RW	6.7.4.6
localTargetPort	TargetPort	RW	6.7.4.7.1 6.7.4.7.2
staticNeighorTargetPort	TargetPort	RW	6.7.4.7.1 6.7.4.7.3
lldpNeighborTargetPort	TargetPort	R	6.7.4.7.1 6.7.4.7.4
neighborMismatch	Boolean	R	6.7.4.8
portalConnected	Boolean	R	6.7.4.12

[a] R = read only access; RW = Read/Write access.

7.2 RAP Propagator Bridge Table

There is one RAP Propagator Bridge Table per RAP Propagator (6.8). The table contains a set of parameters that supports the operation of the RAP Propagator for a Bridge as a whole, as detailed in Table 7-2.

Table 7-2—RAP Propagator Bridge Table

Name	Data type	Operations supported[a]	References
frerCapable	Boolean	R	6.8.4.12
maxProcessingDelay	unsigned integer	R	6.8.4.13
minProcessingDelay	unsigned integer	R	6.8.4.14

[a] R = read only access; RW = Read/Write access.

7.3 RAP Propagator Port Table

There is one RAP Propagator Port Table per Bridge Port for a RAP Propagator (6.8). The table contains a set of parameters that supports the operation of the RAP Propagator for a Bridge Port, as detailed in Table 7-3.

Table 7-3—RAP Propagator Port Table

Name	Data type	Operations supported[a]	References
streamDaPruningEnabled	Boolean	RW	6.8.4.7 b)
maxInterferingFrameSize	unsigned integer	R	6.8.4.7 e)
maxPropagationDelay	unsigned integer	R	6.8.4.7 g)
minPropagationDelay	unsigned integer	R	6.8.4.7 h)

[a] R = read only access; RW = Read/Write access.

7.4 RA Class Bridge Table

There is one RA Class Bridge Table per RAP Propagator (6.8). Each table row contains a set of parameters that defines an RA class (6.3.2) for a Bridge as a whole, as detailed in Table 7-4.

Table 7-4—RA Class Bridge Table row elements

Name	Data type	Operations supported[a]	References
raClassId	unsigned integer [0..255]	RW	6.3.2.1
raClassPriority	unsigned integer [0..7]	RW	6.3.2.2
rtid	integer	RW	6.3.2.3
templatedDefinedData	octet string	RW	6.5.2.2.6

[a] R = read only access; RW = Read/Write access.

7.5 RA Class Port Table

There is one RA Class Port Table per Bridge Port for a RAP Propagator (6.8). Each table row contains a set of parameters associated with an RA class configured in the RA Class Bridge Table (7.4) and a Bridge Port, as detailed in Table 7-5.

Table 7-5—RA Class Port Table row elements

Name	Data type	Operations supported[a]	References
raClassId	unsigned integer [0..255]	RW	6.8.4.9 b)
domainBoundaryPort	Boolean	R	6.8.4.9 c)
maxBandwidth	unsigned integer	RW	6.8.4.9 d)
allocatedBandwidth	unsigned integer	R	6.8.4.9 e)
maxLastHopLatency	unsigned integer	RW	6.8.4.9 f)

[a] R = read only access; RW = Read/Write access.

7.6 RA Class Port Pair Table

There is one RA Class Port Pair Table per reception transmission Port pair of a Bridge for a RAP Propagator (6.8). Each table row is associated with an RA class present in the RA Class Bridge Table (7.4) and contains a set of parameters for a pair of a reception Port and a transmission Port, as detailed in Table 7-6.

Table 7-6—RA Class Port Pair Table row elements

Name	Data type	Operations supported[a]	References
receptionPort	Port Number	RW	6.8.4.10 a)
transmissionPort	Port Number	RW	6.8.4.10 b)
raClassId	unsigned integer [0..255]	RW	6.8.4.10 c)
maxHopLatency	unsigned integer	RW	6.8.4.10 d)

[a] R = read only access; RW = Read/Write access.

7.7 RA Class Domain Boundary Priority Regeneration Table

There is one RA Class Domain Boundary Priority Regeneration Table per Bridge Port for a RAP Propagator (6.8). The values contained in this table are used in the detection of RA class domains to control the Bridge's priority regeneration function ([Q]:6.9.4) on an RA class domain boundary port, as described in 6.3.2.4. Each table row contains a set of parameters, as detailed in Table 7-7.

Table 7-7—RA Class Domain Boundary Priority Regeneration Table row elements

Name	Data type	Operations supported[a]	References
receivedPriority	unsigned integer [0..7]	RW	7.7.1
regeneratedPriority	unsigned integer [0..7]	RW	7.7.2

[a] R = read only access; RW = Read/Write access.

7.7.1 receivedPriority

This parameter contains a received priority that is associated with an RA class configured in the RA Class Bridge Table (7.4).

7.7.2 regeneratedPriority

This parameter contains the regenerated priority for the given received priority contained in 7.7.1.

7.8 Redundancy Context Table

There is one RAP Redundancy Context Table per RAP Propagator (6.8). Each table row contains a set of parameters that defines a Redundancy Context (6.3.9.2) supported by the Bridge, as detailed in Table 7-8.

7.9 Talker Announce Registration Port Table

There is one Talker Announce Registration Port Table per Bridge Port for a RAP Propagator (6.8). Each table row in the table associated with a Port represents a Talker Announce registration [6.3.4 item b)] on that

Table 7-8—Redundancy Context Table row elements

Name	Data type	Operations supported[a]	References
redundancyContextId	unsigned integer [1..4094]	RW	6.3.9.2, 6.8.4.11
vlanContextList	a list of unsigned integer [1..4094]	RW	6.3.9.2, 6.8.4.11

[a] R = read only access; RW = Read/Write access.

Port, and contains a set of parameters as detailed in Table 7-9. Rows in the table can be created, updated and deleted dynamically as Talker Announce registrations are created, updated and deleted on the Port.

Table 7-9—Talker Announce Registration Port Table row elements

Name	Data type	Operations supported[a]	References
streamId	octet string(size(8))	R	6.5.3.1
streamRank	unsigned integer [0..1]	R	6.5.3.2
accumulatedMaxLatency	unsigned integer	R	6.5.3.3
accumulatedMinLatency	unsigned integer	R	6.5.3.4
destinationMacAddress	MAC address	R	6.5.3.5.1
priority	unsigned integer [0..7]	R	6.5.3.5.2
vid	unsigned integer [1..4094]	R	6.5.3.5.3
talkerTokenBucketTSpec	Boolean	R	7.9.1
talkerTokenBucketTSpecValue	octet string	R	7.9.2
talkerMsrpTSpec	Boolean	R	7.9.3
talkerMsrpTSpecValue	octet string	R	7.9.4
networkTSpec	Boolean	R	7.9.5
networkTSpecValue	octet string	R	7.9.6
redundancyControl	Boolean	R	7.9.7
redundancyControlValue	octet string	R	7.9.8
failureInfo	Boolean	R	7.9.9
failureInfoValue	octet string	R	7.9.10
orgDefinedInfo	Boolean	R	7.9.11
orgDefinedInfoValue	octet string	R	7.9.12
invalid	Boolean	R	6.8.4.1 c)
ingressBlocked	Boolean	R	6.8.4.1 d)

[a] R = read only access; RW = Read/Write access.

7.9.1 talkerTokenBucketTSpec

This parameter contains a Boolean value that indicates whether a Token Bucket TSpec sub-TLV (6.5.3.6) is contained in the TalkerTSpec field (6.5.3.6) of the Talker Announce attribute (TRUE) or not (FALSE).

7.9.2 talkerTokenBucketTSpecValue

When talkerTokenBucketTSpec (7.9.1) is TRUE, this parameter contains an octet string copied from the whole of the Value field of a Token Bucket TSpec sub-TLV (6.5.3.6) contained in the TalkerTSpec field (6.5.3.6) of the Talker Announce attribute.

When talkerTokenBucketTSpec is FALSE, this parameter contains the empty octet string.

7.9.3 talkerMsrpTSpec

This parameter contains a Boolean value that indicates whether an Interval-based TSpec sub-TLV (6.5.3.7) is contained in the TalkerTSpec field (6.5.3.6) of the Talker Announce attribute.

7.9.4 talkerMsrpTSpecValue

When talkerMsrpTSpec (7.9.3) is TRUE, this parameter contains an octet string copied from the whole of the Value field of an Interval-based TSpec sub-TLV (6.5.3.7) contained in the TalkerTSpec field (6.5.3.6) of the Talker Announce attribute.

When talkerMsrpTSpec is FALSE, this parameter contains the empty octet string.

7.9.5 networkTSpec

This parameter contains a Boolean value that indicates whether a Token Bucket TSpec sub-TLV (6.5.3.6) is contained in the NetworkTSpec field (6.5.3.9) of the Talker Announce attribute (TRUE) or not (FALSE).

7.9.6 networkTSpecValue

When networkTSpec (7.9.5) is TRUE, this parameter returns an octet string copied from the whole of the Value field of a Token Bucket TSpec sub-TLV (6.5.3.6) contained in the NetworkTSpec field (6.5.3.9) of the Talker Announce attribute.

When networkTSpec is FALSE, this parameter contains the empty octet string.

7.9.7 redundancyControl

This parameter contains a Boolean value that indicates whether a Redundancy Control sub-TLV (6.5.3.9) is contained in the Talker Announce attribute (TRUE) or not (FALSE).

7.9.8 redundancyControlValue

When redundancyControl (7.9.7) is TRUE, this parameter contains an octet string copied from the whole of the Value field of a Redundancy Control sub-TLV (6.5.3.9) contained in the Talker Announce attribute.

When redundancyControl is FALSE, this parameter contains the empty octet string.

7.9.9 failureInfo

This parameter contains a Boolean value that indicates whether a Failure Information sub-TLV (6.5.3.9.2) is contained in the Talker Announce attribute (TRUE) or not (FALSE).

7.9.10 failureInfoValue

When failureInfo (7.9.9) is TRUE, this parameter contains an octet string copied from the whole of the Value field of a Failure Information sub-TLV (6.5.3.9.2) contained in the Talker Announce attribute.

When failureInfo is FALSE, this parameter contains the empty octet string.

7.9.11 orgDefinedInfo

This parameter contains a Boolean value that indicates whether an Organizationally Defined sub-TLV (6.5.3.10) is contained in the Talker Announce attribute (TRUE) or not (FALSE).

7.9.12 orgDefinedInfoValue

When orgDefinedInfo (7.9.11) is TRUE, this parameter contains an octet string copied from the whole of the Value field of an Organizationally Defined sub-TLV (6.5.3.10) contained in the Talker Announce attribute.

When orgDefinedInfo is FALSE, this parameter contains the empty octet string.

7.10 Talker Announce Declaration Port Table

There is one Talker Announce Declaration Port Table per Bridge Port for a RAP Propagator (6.8). Each table row in the Table associated with a Port represents a Talker Announce declaration [6.3.4 item a)] on that Port, and contains each of the parameters as specified in Table 7-9 except the invalid and ingressBlocked parameters. Rows in the table can be created, updated and deleted dynamically as Talker Announce declarations are created, updated and deleted on the Port.

7.11 Listener Attach Registration Port Table

There is one Listener Attach Registration Port Table per Bridge Port for a RAP Propagator (6.8). Each table row in the Table associated with a Port represents a Listener Attach registration [6.3.5 item b)] on that Port and contains a set of parameters as detailed in Table 7-10. Rows in the table can be created, updated and deleted dynamically as the Listener Attach registrations are created, updated and deleted on the Port.

7.11.1 vlanContextStatus

This parameter contains a Boolean value that indicates whether a VLAN Context Status sub-TLV (6.5.4.4) is contained in the Listener Attach attribute (TRUE) or not (FALSE).

7.11.2 vlanContextStatusValue

When vlanContextStatus (7.11.1) is TRUE, this parameter contains an octet string copied from the whole of the Value field of a VLAN Context Status sub-TLV (6.5.4.4) contained in the Listener Attach attribute.

Table 7-10—Listener Attach Registration Port Table row elements

Name	Data type	Operations supported[a]	References
streamId	octet string(size(8))	R	6.5.4.1
vid	unsigned integer [1..4094]	R	6.5.4.2
listenerAttachStatus	Enumerated	R	6.5.4.3
vlanContextStatus	Boolean	R	7.11.1
vlanContexStatusValue	octet string	R	7.11.2
reservationAge	unsigned integer	R	6.8.4.3 e)
ingressStatus	unsigned integer	R	6.8.4.3 f)

[a] R = read only access; RW = Read/Write access.

7.12 Listener Attach Declaration Port Table

There is one Listener Attach Declaration Port Table per Bridge Port for a RAP Propagator (6.8). Each table row in the Table associated with a Port represents a Listener Attach declaration [6.3.5 item a)] on that Port and contains each of the parameters as specified in Table 7-10 except the reservationAge and ingressStatus parameters. Rows in the table can be created, updated and deleted dynamically as Listener Attach declarations are created, updated and deleted on the Port.

8. YANG data model for RAP

YANG (IETF RFC 7950) is a data modeling language used to model configuration data and state data for remote network management protocols. YANG-based remote network management protocols include the Network Configuration Protocol (NETCONF) (IETF RFC 6242) and RESTCONF (IETF RFC 8040). Each remote network management protocol uses a specific encoding on the wire, such as XML or JSON. A YANG module specifies the organization and rules for the management data, and a mapping from YANG to the specific encoding enables the data to be understood correctly by both client and server.

This clause specifies the YANG data model for RAP.

This clause:

a) Introduces the organization of the data models (8.1)
b) Shows the relationship with other standards (8.2)
c) Provides an overview of the hierarchy of the data models using a representation similar to the Unified Modeling Language (UML) (8.3)
d) Summarizes the structure of the YANG data model (8.4)
e) Reviews security considerations (8.5)
f) Provides a schema tree as an overview of the YANG module (8.6)
g) Specifies the YANG module (8.7)

8.1 The YANG framework

The YANG framework applies hierarchy in the following areas:

a) The Uniform Resource Name (URN), as specified in IEEE Std 802d-2017.
b) The YANG objects form a hierarchy of configuration and operational data structures that define the YANG model.

Clause 7 specifies the information model for management of this standard. The data model for a specific management mechanism is derived from the information model. Since YANG-based protocols are an example of a management mechanism, the YANG data model of this clause is derived from Clause 7.

8.2 Relationship to other YANG modules

The RAP YANG model augments the VLAN Bridge component model as defined in [Q].

8.3 RAP YANG model

This clause uses a UML-like representation to provide an overview of the hierarchy of the RAP YANG data model.

For all figures, data that is imported from outside this standard is shown in white, and data in augments of those YANG modules is shown in gray.

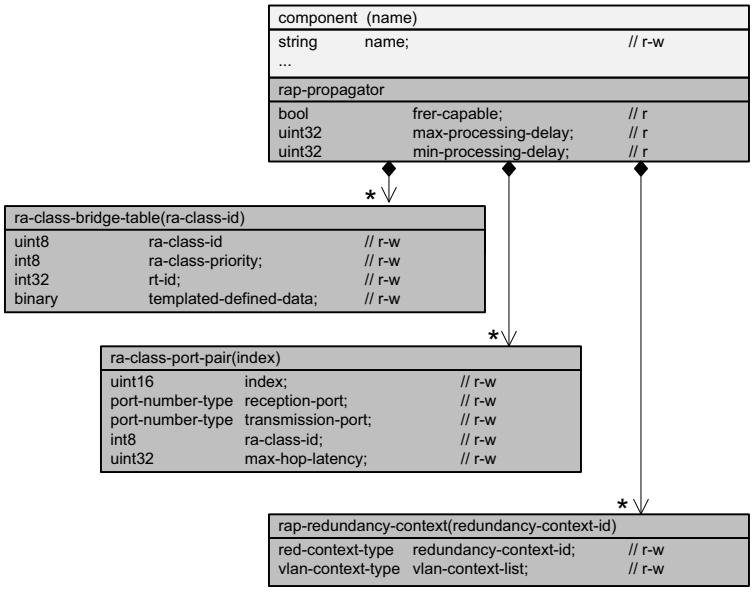


Figure 8-1—Bridge component augmentation

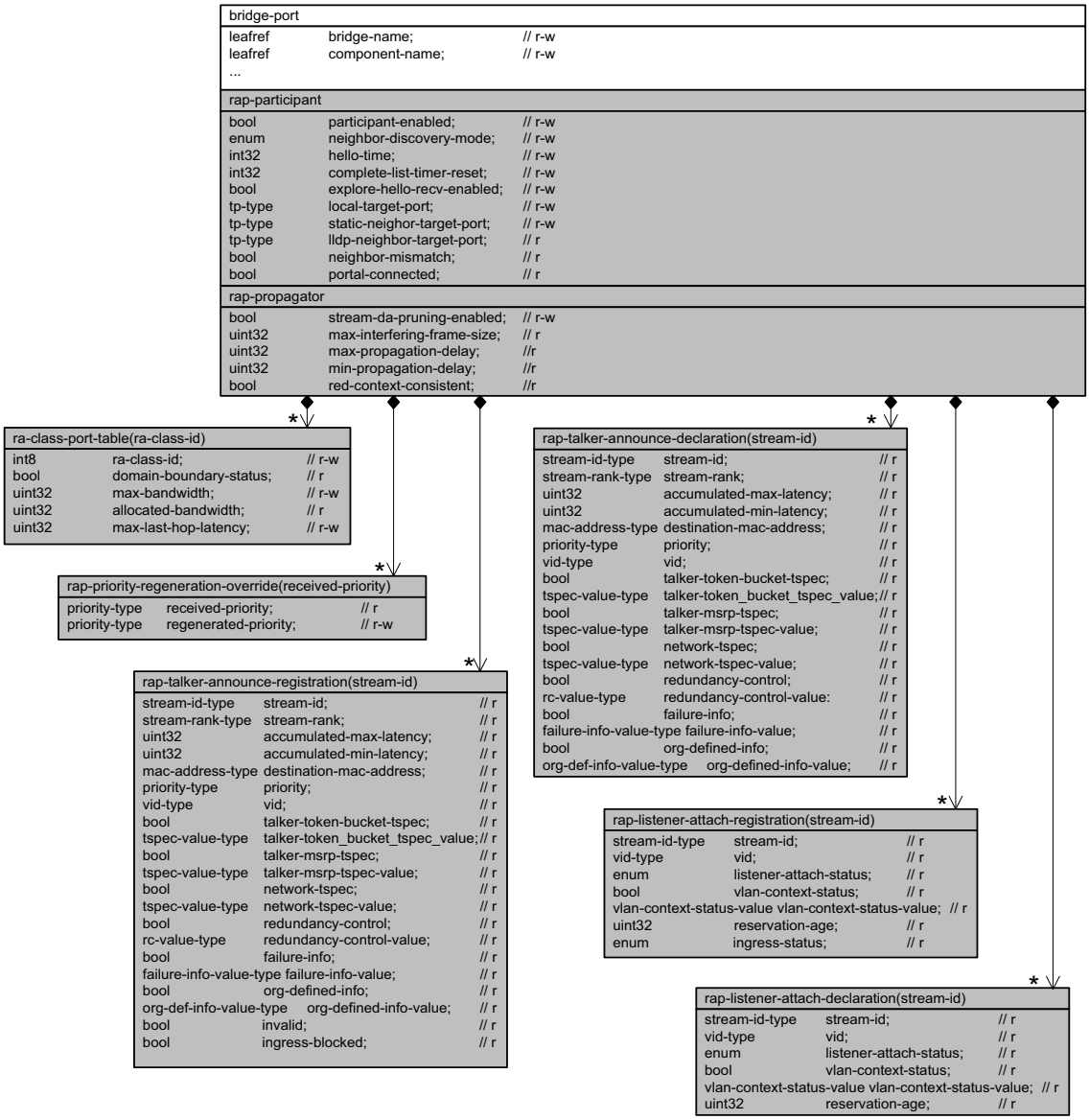


Figure 8-2—Bridge port augmentation

8.4 Structure of the YANG model

The RAP YANG model is provided as a single YANG module.

In the YANG module definitions, if any discrepancy between the “description” text and the corresponding definition in any other part of this standard occurs, the definitions outside this clause take precedence.

8.5 Security Considerations

The YANG module specified in this clause is designed to be accessed via a network configuration protocol, such as NETCONF (IETF RFC 6242 [B13a]) and RESTCONF (IETF RFC 8040). NETCONF and RESTCONF protocols provide the means to secure communication between client and server, using secure transport layers such as Secure Shell (SSH) (IETF RFC 6242) and Transport Layer Security (TLS) (IETF RFC 7589).

It is the responsibility of a system's implementor and administrator to ensure that the protocol entities in the system that support NETCONF, and any other remote configuration protocols that make use of these YANG modules, are properly configured to allow access only to those principals (users) that have legitimate rights to read or write data nodes. This standard does not specify how the credentials of those users are to be stored or validated.

8.6 YANG data scheme tree definitions

The schema tree in this clause is provided as an overview of the RAP YANG module. The symbols and their meaning are specified in YANG Tree Diagrams (IETF RFC 8340).

8.6.1 Tree diagram for ieee802-dot1dd-rap.yang

```
module: ieee802-dot1dd-rap

  augment /dot1q:bridges/dot1q:bridge/dot1q:component:
    +--rw rap-propagator
    |  +--ro frer-capable?           boolean
    |  +--ro max-processing-delay?   uint32
    |  +--ro min-processing-delay?   uint32
    +--rw ra-class-bridge-table* [ra-class-id]
    |  +--rw ra-class-id               uint8
    |  +--rw ra-class-priority?        int8
    |  +--rw rt-id?                    int32
    |  +--rw templated-defined-data?   binary
    +--rw ra-class-port-pair* [index]
    |  +--rw index                 uint16
    |  +--rw reception-port?       dot1qtypes:port-number-type
    |  +--rw transmission-port?    dot1qtypes:port-number-type
    |  +--rw ra-class-id?          int8
    |  +--rw max-hop-latency?      uint32
    +--rw rap-redundancy-context* [redundancy-context-id]
       +--rw redundancy-context-id    uint16
       +--rw vlan-context-list* [vlan-context]
          +--rw vlan-context    uint16
  augment /if:interfaces/if:interface/dot1q:bridge-port:
    +--rw rap-participant
    |  +--rw participant-enabled?          boolean
    |  +--rw neighbor-discovery-mode?      enumeration
    |  +--rw hello-time?                   int32
    |  +--rw complete-list-timer-reset?    int32
    |  +--rw explore-hello-recv-enabled?   boolean
    |  +--rw local-target-port
    |  |  +--rw chassis-id?    ieeetypes:chassis-id-type
    |  |  +--rw port-id?       ieeetypes:port-id-type
    |  |  +--rw ecp-capable?   boolean
    |  |  +--rw tcp-capable?   boolean
    |  |  +--rw tcp-port?      uint16
    |  |  +--rw addr-ip-v4?    inet:ipv4-address
    |  |  +--rw addr-ip-v6?    inet:ipv6-address
    |  +--rw static-neighor-target-port
    |  |  +--rw chassis-id?    ieeetypes:chassis-id-type
    |  |  +--rw port-id?       ieeetypes:port-id-type
    |  |  +--rw ecp-capable?   boolean
    |  |  +--rw tcp-capable?   boolean
    |  |  +--rw tcp-port?      uint16
    |  |  +--rw addr-ip-v4?    inet:ipv4-address
    |  |  +--rw addr-ip-v6?    inet:ipv6-address
    |  +--ro lldp-neighbor-target-port
```

```
|  |  +--ro chassis-id?    ieeetypes:chassis-id-type
|  |  +--ro port-id?       ieeetypes:port-id-type
|  |  +--ro ecp-capable?   boolean
|  |  +--ro tcp-capable?   boolean
|  |  +--ro tcp-port?      uint16
|  |  +--ro addr-ip-v4?    inet:ipv4-address
|  |  +--ro addr-ip-v6?    inet:ipv6-address
|  +--ro neighbor-mismatch?            boolean
|  +--ro portal-connected?             boolean
+--rw rap-propagator
|  +--rw stream-da-pruning-enabled?    boolean
|  +--ro max-interfering-frame-size?   uint32
|  +--ro max-propagation-delay?        uint32
|  +--ro min-propagation-delay?        uint32
|  +--ro red-context-consistent?       boolean
+--rw ra-class-port-table* [ra-class-id]
|  +--rw ra-class-id               uint8
|  +--ro domain-boundary-status?   boolean
|  +--rw max-bandwidth?            uint32
|  +--ro allocated-bandwidth?      uint32
|  +--rw max-last-hop-latency?     uint32
+--rw rap-priority-regeneration-override* [received-priority]
|  +--rw received-priority       dot1qtypes:priority-type
|  +--rw regenerated-priority?   dot1qtypes:priority-type
+--ro rap-talker-announce-registration* [stream-id vid]
|  +--ro stream-id                          tsn:stream-id-type
|  +--ro stream-rank?                       uint8
|  +--ro accumulated-max-latency?           uint32
|  +--ro accumulated-min-latency?           uint32
|  +--ro destination-mac-address?           ieeetypes:mac-address
|  +--ro priority?                          dot1qtypes:priority-type
|  +--ro vid                                uint16
|  +--ro talker-token-bucket-tspec?         boolean
|  +--ro talker-token_bucket_tspec_value?   binary
|  +--ro talker-msrp-tspec?                 boolean
|  +--ro talker-msrp-tspec-value?           binary
|  +--ro network-tspec?                     boolean
|  +--ro network-tspec-value?               binary
|  +--ro redundancy-control?                boolean
|  +--ro redundancy-control-value?          binary
|  +--ro failure-info?                      boolean
|  +--ro failure-info-value?                binary
|  +--ro org-defined-info?                  boolean
|  +--ro org-defined-info-value?            binary
|  +--ro invalid?                           boolean
|  +--ro ingress-blocked?                   boolean
+--ro rap-talker-announce-declaration* [stream-id vid]
|  +--ro stream-id                          tsn:stream-id-type
|  +--ro stream-rank?                       uint8
|  +--ro accumulated-max-latency?           uint32
|  +--ro accumulated-min-latency?           uint32
|  +--ro destination-mac-address?           ieeetypes:mac-address
|  +--ro priority?                          dot1qtypes:priority-type
|  +--ro vid                                uint16
|  +--ro talker-token-bucket-tspec?         boolean
|  +--ro talker-token_bucket_tspec_value?   binary
|  +--ro talker-msrp-tspec?                 boolean
|  +--ro talker-msrp-tspec-value?           binary
|  +--ro network-tspec?                     boolean
|  +--ro network-tspec-value?               binary
|  +--ro redundancy-control?                boolean
|  +--ro redundancy-control-value?          binary
|  +--ro failure-info?                      boolean
|  +--ro failure-info-value?                binary
|  +--ro org-defined-info?                  boolean
|  +--ro org-defined-info-value?            binary
|  +--ro invalid?                           boolean
|  +--ro ingress-blocked?                   boolean
+--ro rap-listener-attach-registration* [stream-id vid]
|  +--ro stream-id                    tsn:stream-id-type
|  +--ro vid                          uint16
|  +--ro listener-attach-status?      enumeration
```

```
   |  +--ro vlan-context-status?         boolean
   |  +--ro vlan-context-status-value?   binary
   |  +--ro reservation-age?             uint32
   |  +--ro ingress-status?              enumeration
   +--ro rap-listener-attach-declaration* [stream-id vid]
      +--ro stream-id                     tsn:stream-id-type
      +--ro vid                           uint16
      +--ro listener-attach-status?       enumeration
      +--ro vlan-context-status?          boolean
      +--ro vlan-context-status-value?    binary
```

8.7 YANG module ieee802-dot1dd-rap

```
module ieee802-dot1dd-rap {
  yang-version 1.1;
  namespace "urn:ieee:std:802.1DD:yang:ieee802-dot1dd-rap";
  prefix rap;

  import ietf-interfaces {
    prefix if;
  }
  import ieee802-dot1q-types {
    prefix dot1qtypes;
  }
  import ieee802-dot1q-bridge {
    prefix dot1q;
  }
  import ieee802-dot1q-tsn-types {
    prefix tsn;
  }
  import ieee802-types {
    prefix ieeetypes;
  }
  import ietf-inet-types {
    prefix inet;
  }

  organization
    "IEEE 802.1 Working Group";
  contact
    "WG-URL: http://www.ieee802.org/1/
     WG-EMail: stds-802-1-l@ieee.org

     Contact: IEEE 802.1 Working Group Chair
     Postal: C/O IEEE 802.1 Working Group
            IEEE Standards Association
            445 Hoes Lane
            Piscataway, NJ 08854
            USA

     E-mail: stds-802-1-chairs@ieee.org";
  description
    "This module provides management of 802.1Q Bridge components that
     support the Resource Allocation Protocol (RAP).

     Copyright (C) IEEE (202x).

     This version of this YANG module is part of IEEE Std 802.1DD;
     see the standard itself for full legal notices.";

  revision 2024-11-13 {
    description
      "Published as part of IEEE Std 802.1Qdd-202x.

       The following reference statement identifies each referenced
       IEEE Standard as updated by applicable amendments.";
    reference
```

```
    "IEEE Std 802.1DD Resource Allocation Protocol:
     IEEE Std 802.1Q Bridges and Bridged Networks:
     IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
     IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qcj-2023,
     IEEE Std 802.1Qdj-2024, IEEE Std 802.1Qdx-2024,
     IEEE Std 802.1Qdy-2024.";
}

grouping target-port {
  leaf chassis-id {
    type ieeetypes:chassis-id-type;
    description
      "Chassis component associated with the local system.";
    reference
      "8.5.2.3 of IEEE Std 802.1AB-2016";
  }
  leaf port-id {
    type ieeetypes:port-id-type;
    description
      "Port component associated with a given port in the local
       system.";
    reference
      "8.5.3.3 of IEEE Std 802.1AB-2016";
  }
  leaf ecp-capable {
    type boolean;
    description
      "A Boolean value indicating whether the target
       port supports the LRP-DT ECP mechanism (TRUE) or not (FALSE).";
  }
  leaf tcp-capable {
    type boolean;
    description
      "A Boolean value indicating whether that the target
       port supports the LRP-DT TCP mechanism (TRUE) or not (FALSE).";
  }
  leaf tcp-port {
    type uint16;
    description
      "A 2-byte TCP port number for the target port.";
    reference
      "C.2.2.6.1 of IEEE Std 802.1CS-2020";
  }
  leaf addr-ip-v4 {
    type inet:ipv4-address;
    description
      "A 4-byte IPv4 address for the target port or NULL.";
    reference
      "Item 1) in C.2.2.6.2 of IEEE Std 802.1CS-2020";
  }
  leaf addr-ip-v6 {
    type inet:ipv6-address;
    description
      "A 16-byte IPv6 address for the target port or NULL.";
    reference
      "Item 2) in C.2.2.6.2 of IEEE Std 802.1CS-2020";
  }
}

grouping rap-talker-announce {
  leaf stream-id {
    type tsn:stream-id-type;
    description
      "An 8-octet field encoding the StreamID element as specified in
       46.2.3.1 of IEEE Std 802.1Q.";
    reference
      "6.5.3.1 of IEEE Std 802.1DD";
  }
  leaf stream-rank {
    type uint8;
    description
      "A 1-octet field encoding a Rank value as specified in
```

```
        46.2.3.2.1 of IEEE Std 802.1Q.";
      reference
        "6.5.3.2 of IEEE Std 802.1DD";
    }
    leaf accumulated-max-latency {
      type uint32;
      description
        "A 4-octet field encoding the AccumulatedLatency element as
         specified in 46.2.5.2 of IEEE Std 802.1Q.";
      reference
        "6.5.3.3 of IEEE Std 802.1DD";
    }
    leaf accumulated-min-latency {
      type uint32;
      description
        "A 4-octet unsigned integer, indicating the minimum latency, in
         nanoseconds, that a single frame of the stream can encounter when
         transmitted from the Talker along a given path to the Port
         declaring this attribute.";
      reference
        "6.5.3.4 of IEEE Std 802.1DD";
    }
    leaf destination-mac-address {
      type ieeetypes:mac-address;
      description
        "A 6-octet destination MAC address of the data frames of the
         stream.";
      reference
        "6.5.3.5.1 of IEEE Std 802.1DD";
    }
    leaf priority {
      type dot1qtypes:priority-type;
      description
        "A 3-bit unsigned integer, indicating the priority to be encoded in
         the PCP field of the VLAN tag, with which the data frames of the
         stream are tagged. This priority value is used by each receiving
         Bridge to associate the stream to a local RA class of the same
         priority.";
      reference
        "6.5.3.5.2 of IEEE Std 802.1DD";
    }
    leaf vid {
      type uint16 {
        range "1..4094";
      }
      description
        "A 12-bit VID. The semantics of this field is dependent on the type
         of a Talker Announcement in which the Talker Announce attribute is
         used, as follows:
         a) In the case of a Single-Context Talker
         Announcement, this field indicates a VID to be encoded in the VLAN
         tag with which the data frames of the stream are tagged, and also
         a single VLAN Context used by the Talker Announce attribute.
         b) In the case of a Multi-Context Talker Announcement, this field
         is set to the numerically smallest VlanContextId value contained
         in a VLAN Context Information sub-TLV.";
      reference
        "6.5.3.5.3 of IEEE Std 802.1DD";
    }
    leaf talker-token-bucket-tspec {
      type boolean;
      description
        "This parameter returns a Boolean value that indicates whether a
         Token Bucket TSpec sub-TLV is contained in the TalkerTSpec field
         of the Talker Announce attribute (TRUE) or not (FALSE).";
      reference
        "7.9.1 of IEEE Std 802.1DD";
    }
    leaf talker-token_bucket_tspec_value {
      type binary;
      description
        "When talkerTokenBucketTSpec is TRUE, this parameter returns an
```

```
        octet string that is copied from the whole of the Value field of a
        Token Bucket TSpec sub-TLV contained in the TalkerTSpec field of
        the Talker Announce attribute.";
      reference
        "7.9.2 of IEEE Std 802.1DD";
    }
    leaf talker-msrp-tspec {
      type boolean;
      description
        "This parameter returns a Boolean value that indicates whether a
         MSRP TSpec sub-TLV is contained in the TalkerTSpec field of the
         Talker Announce attribute.";
      reference
        "7.9.3 of IEEE Std 802.1DD";
    }
    leaf talker-msrp-tspec-value {
      type binary;
      description
        "When talkerMsrpTSpec is TRUE, this parameter returns an octet
         string that is copied from the whole of the Value field of a MSRP
         TSpec sub-TLV contained in the TalkerTSpec field of the Talker
         Announce attribute.";
      reference
        "7.9.4 of IEEE Std 802.1DD";
    }
    leaf network-tspec {
      type boolean;
      description
        "This parameter returns a Boolean value that indicates whether a
         Token Bucket TSpec sub-TLV is contained in the NetworkTSpec field
         of the Talker Announce attribute (TRUE) or not (FALSE).";
      reference
        "7.9.5 of IEEE Std 802.1DD";
    }
    leaf network-tspec-value {
      type binary;
      description
        "When networkTSpec is TRUE, this parameter returns an octet string
         that is copied from the whole of the Value field of a Token Bucket
         TSpec sub-TLV contained in the NetworkTSpec field of the Talker
         Announce attribute.";
      reference
        "7.9.6 of IEEE Std 802.1DD";
    }
    leaf redundancy-control {
      type boolean;
      description
        "This parameter returns a Boolean value that indicates whether a
         Redundancy Control sub-TLV is contained in the Talker Announce
         attribute (TRUE) or not (FALSE).";
      reference
        "7.9.7 of IEEE Std 802.1DD";
    }
    leaf redundancy-control-value {
      type binary;
      description
        "When redundancyControl is TRUE, this parameter returns an octet
         string that is copied from the whole of the Value field of a
         Redundancy Control sub-TLV contained in the Talker Announce
         attribute.";
      reference
        "7.9.8 of IEEE Std 802.1DD";
    }
    leaf failure-info {
      type boolean;
      description
        "This parameter returns a Boolean value that indicates whether a
         Failure Information sub-TLV is contained in the Talker Announce
         attribute (TRUE) or not (FALSE).";
      reference
        "7.9.9 of IEEE Std 802.1DD";
    }
```

```
    leaf failure-info-value {
      type binary;
      description
        "When failureInfo is TRUE, this parameter returns an octet string
         that is copied from the whole of the Value field of a Failure
         Information sub-TLV contained in the Talker Announce attribute.";
      reference
        "7.9.10 of IEEE Std 802.1DD";
    }
    leaf org-defined-info {
      type boolean;
      description
        "This parameter returns a Boolean value that indicates whether an
         Organizationally Defined sub-TLV is contained in the Talker
         Announce attribute (TRUE) or not (FALSE).";
      reference
        "7.9.11 of IEEE Std 802.1DD";
    }
    leaf org-defined-info-value {
      type binary;
      description
        "When orgDefinedInfo is TRUE, this parameter returns an octet
         string that is copied from the whole of the Value field of an
         Organizationally Defined sub-TLV contained in the Talker Announce
         attribute.";
      reference
        "7.9.12 of IEEE Std 802.1DD";
    }
    leaf invalid {
      type boolean;
      description
        "A Boolean indicating whether the Talker Announce registration is
         invalid (TRUE) or not (FALSE), as determined by the isTaInvalid()
         procedure (6.8.5.3)";
      reference
        "item c) in 6.8.4.1 of IEEE Std 802.1DD";
    }
    leaf ingress-blocked {
      type boolean;
      description
        "A Boolean indicating whether the Talker Announce registration is
         ingress blocked (TRUE) or not (FALSE), as determined by the
         isTaIngressBlocked() procedure (6.8.5.4)";
      reference
        "item d) in 6.8.4.1 of IEEE Std 802.1DD";
    }
  }

  grouping rap-listener-attach {
    leaf stream-id {
      type tsn:stream-id-type;
      description
        "An 8-octet field encoding the StreamID element as specified in
         46.2.3.1 of IEEE Std 802.1Q.";
      reference
        "6.5.4.1 of IEEE Std 802.1D";
    }
    leaf vid {
      type uint16 {
        range "1..4094";
      }
      description
        "A 12-bit integer, indicating a VID to be encoded in the VLAN tag
         with which the data frames of the stream are tagged.";
      reference
        "6.5.4.2 of IEEE Std 802.1DD";
    }
    leaf listener-attach-status {
      type enumeration {
        enum attach-ready {
          value 1;
        }
```

```
      enum attach-fail {
        value 2;
      }
      enum attach-partial-fail {
        value 3;
      }
    }
    description
      "An enumeration indicating the listener attach status";
    reference
      "6.5.4.3 of IEEE Std 802.1DD";
  }
  leaf vlan-context-status {
    type boolean;
    description
      "This parameter returns a Boolean value that indicates whether a
       VLAN Context Status sub-TLV is contained in the Listener Attach
       attribute (TRUE) or not (FALSE).";
    reference
      "7.11.1 of IEEE Std 802.1DD";
  }
  leaf vlan-context-status-value {
    type binary;
    description
      "When vlanContextStatus is TRUE, this parameter returns an octet
       string that is copied from the whole of the Value field of a VLAN
       Context Status sub-TLV contained in the Listener Attach
       attribute.";
    reference
      "6.5.4.4 of IEEE Std 802.1DD";
  }
}

augment "/dot1q:bridges/dot1q:bridge/dot1q:component" {
  description
    "Augment Bridge with RAP configuration.";
  reference
    "6 of IEEE Std 802.1DD.";
  container rap-propagator {
    leaf frer-capable {
      type boolean;
      config false;
      description
        "A Boolean value, indicating whether the Bridge is a
         FRER-capable Bridge (TRUE) or not (FALSE).";
      reference
        "6.8.4.11 of IEEE Std 802.1DD.";
    }
    leaf max-processing-delay {
      type uint32;
      config false;
      description
        "An unsigned integer, indicating the maximum delay, in
         nanoseconds, that a frame can experience during the
         forwarding process of the Bridge until it is placed into an
         outbound queue.";
      reference
        "6.8.4.12 of IEEE Std 802.1DD.";
    }
    leaf min-processing-delay {
      type uint32;
      config false;
      description
        "An unsigned integer, indicating the minimum delay, in
         nanoseconds, that a frame can experience during the
         forwarding process of the Bridge until it is placed into an
         outbound queue.";
      reference
        "6.8.4.13 of IEEE Std 802.1DD.";
    }
  }
  list ra-class-bridge-table {
```

```
    key "ra-class-id";
    description
      "Each table row contains a set of parameters that defines a
       single RA class";
    leaf ra-class-id {
      type uint8 {
        range "0 .. 255";
      }
      description
        "The RA class ID is an integer in the range of 0 through 255
         that identifies an RA class supported by an end station or
         Bridge.";
      reference
        "6.3.2.1 of IEEE Std 802.1DD.";
    }
    leaf ra-class-priority {
      type int8 {
        range "0 .. 7";
      }
      description
        "Each RA class supported by an end station or Bridge is
         associated with a unique priority value in the range 0
         through 7, termed RA class priority. The RA class priority
         indicates the received priority of the frames which are to
         be mapped to the traffic class(es) with which that RA class
         is associated.";
      reference
        "6.3.2.2 of IEEE Std 802.1DD.";
    }
    leaf rt-id {
      type int32;
      description
        "An RA class template is identified by an RA Class Template
         Identifier (RTID), which encodes a 3-octet OUI or CID value
         identifying the organization that defines that template,
         followed by a 1-octet index allocated by that
         organization.";
      reference
        "6.3.2.3 of IEEE Std 802.1DD.";
    }
    leaf templated-defined-data {
      type binary;
      description
        "The encoding of this field and the semantics associated
         with its values if any, is specific to the RA class
         template identified by the value contained in rt-id";
      reference
        "6.5.2.2.6 of IEEE Std 802.1DD.";
    }
  }
  list ra-class-port-pair {
    key "index";
    description
      "Each table row corresponds to an RA class configured in the
       RA Class Bridge Table and contains a set of parameters
       associated with a reception Port and a transmission Port";
    leaf index {
      type uint16;
      description
        "The index for the list";
    }
    leaf reception-port {
      type dot1qtypes:port-number-type;
      description
        "The port number of the associated reception Port.";
      reference
        "Item a) in 6.8.4.9 of IEEE Std 802.1DD.";
    }
    leaf transmission-port {
      type dot1qtypes:port-number-type;
      description
        "The port number of the associated transmission Port.";
```

```
      reference
        "Item b) in 6.8.4.9 of IEEE Std 802.1DD.";
    }
    leaf ra-class-id {
      type int8;
      description
        "The RA class ID of the associated local RA class. ";
      reference
        "Item c) in 6.8.4.9 of IEEE Std 802.1DD.";
    }
    leaf max-hop-latency {
      type uint32;
      description
        "An unsigned integer, containing the administratively
         configured maximum latency value, in nanoseconds, that is
         used as an upper bound latency in the latency constraint
         imposed on each stream reserved in the RA class, received
         on the reception Port, and transmitted on the transmission
         Port. The latency is measured from a point located in an
         upstream station connected via a LAN to the reception Port,
         to a point located in the transmission Port, where the
         exact measurement points are specific to and defined by the
         RA class template being used by the RA class.";
      reference
        "Item d) in 6.8.4.9 of IEEE Std 802.1DD.";
    }
  }
  list rap-redundancy-context {
    key "redundancy-context-id";
    description
      "Each table row contains a set of parameters associated with a
       Redundancy Context supported by the Bridge.";
    leaf redundancy-context-id {
      type uint16 {
        range "1..4096";
      }
      description
        "The Redundancy Context ID of the Redundancy Context";
      reference
        "Item a) in 6.8.4.10 of IEEE Std 802.1DD.";
    }
    list vlan-context-list {
      key "vlan-context";
      description
        "The list of VLAN Context IDs associated with the Redundancy
                 Context.";
      reference
        "Item b) in 6.8.4.10 of IEEE Std 802.1DD.";
      leaf vlan-context {
        type uint16 {
          range "1..4096";
        }
      }
    }
  }
}

augment "/if:interfaces/if:interface/dot1q:bridge-port" {
  description
    "Augment Bridge Port with RAP configuration";
  reference
    "6 of IEEE Std 802.1DD.";
  container rap-participant {
    leaf participant-enabled {
      type boolean;
      description
        "A Boolean variable indicating whether the operation of the
         RAP Participant state machine is administratively enabled
         (TRUE) or not (FALSE).";
      reference
        "6.7.4.2 of IEEE Std 802.1DD.";
    }
```

```
    leaf neighbor-discovery-mode {
      type enumeration {
        enum lldp-discovery {
          value 1;
        }
        enum static-configuration {
          value 2;
        }
        enum exploratory-hello {
          value 3;
        }
      }
      description
        "An administratively assigned value, indicating the operation
         mode in which neighbor discovery is performed on the local
         target Port, and taking one of the following enumerated
         values:
         1) LLDP_DISCOVERY: The information about a neighbor target
         port to be passed to the underlying LRP is obtained through
         the exchange of LRP Discovery TLVs (Annex C of IEEE Std
         802.1CS-2020) by use of LLDP, and contained in
         lldpNeighborTargetPort (51.7.4.7.4).
         2) STATIC_CONFIGURATION: The information about a neighbor
         target port to be passed to the underlying LRP is statically
         configured by the management and contained in
         staticNeighborTargetPort (51.7.4.7.3).
         3) EXPLORATORY_HELLO: No neighbor target port information
         needs to be passed to the underlying LRP.";
      reference
        "6.7.4.3 of IEEE Std 802.1DD.";
    }
    leaf hello-time {
      type int32;
      default "30";
      description
        "An administratively assigned integer value, in the range 30
         through 65535, for the Hello Time parameter in a Local Target
         Port request issued by the RAP Participant state machine to
         the underlying LRP.";
      reference
        "6.7.4.4 of IEEE Std 802.1DD.";
    }
    leaf complete-list-timer-reset {
      type int32;
      description
        "An administratively assigned integer value, in the range x
         through y, for the cplCompleteListTimerReset parameter in a
         Local Target Port request issued by the RAP Participant state
         machine to the underlying LRP. The default value is z.";
      reference
        "6.7.4.5 of IEEE Std 802.1DD.";
    }
    leaf explore-hello-recv-enabled {
      type boolean;
      description
        "An administratively assigned Boolean value for the
         imPplExploreRecv parameter in a Neighbor Target Port request
         issued by the RAP Participant state machine to the underlying
         LRP.";
      reference
        "6.7.4.6 of IEEE Std 802.1DD.";
    }
    container local-target-port {
      uses target-port;
      description
        "Contains the administratively configured parameters of the
         local target port.";
      reference
        "6.7.4.7.2 of IEEE Std 802.1DD.";
    }
    container static-neighor-target-port {
      uses target-port;
```

```
      description
        "Contains the administratively configured parameters of a
         neighbor target port to which the local target port is to be
         connected.";
      reference
        "6.7.4.7.3 of IEEE Std 802.1DD.";
    }
    container lldp-neighbor-target-port {
      uses target-port;
      config false;
      description
        "Contains the configuration parameters of a neighbor target
         port discovered by LLDP.";
      reference
        "6.7.4.7.4 of IEEE Std 802.1DD.";
    }
    leaf neighbor-mismatch {
      type boolean;
      config false;
      description
        "A Boolean variable, set TRUE when detecting a mismatch between
         the local target port and the neighbor target port to be
         connected.";
      reference
        "6.7.4.8 of IEEE Std 802.1DD.";
    }
    leaf portal-connected {
      type boolean;
      config false;
      description
        "A Boolean value indicating whether a Portal association for the
         Portal as indicated in portal-id (51.7.4.11) has been
         established by the underlying LRP (TRUE) or not (FALSE).";
      reference
        "6.7.4.12 of IEEE Std 802.1DD.";
    }
  }
  container rap-propagator {
    leaf stream-da-pruning-enabled {
      type boolean;
      description
        "A Boolean indicating whether Stream DA Pruning (51.3.4.1.2)
         is administratively enabled (TRUE) or disabled (FALSE) on the
         Port.";
      reference
        "Item b) in 6.8.4.7 of IEEE Std 802.1DD.";
    }
    leaf max-interfering-frame-size {
      type uint32;
      config false;
      description
        "An unsigned integer, indicating the maximum frame size, in
         bytes, including media-dependent overhead (12.4.2.2), that is
         allowed to be transmitted through the Port. The value of this
         parameter is determined by the operation of the underlying
         MAC Service.";
      reference
        "Item e) in 6.8.4.7 of IEEE Std 802.1DD.";
    }
    leaf max-propagation-delay {
      type uint32;
      config false;
      description
        "An unsigned integer, indicating the maximum latency, in
         nanoseconds, a frame can experience when transmitted from the
         underlying physical medium on the Port to a reception port
         connected via a LAN to the Port.";
      reference
        "Item g) in 6.8.4.7 of IEEE Std 802.1DD.";
    }
    leaf min-propagation-delay {
      type uint32;
```

```
      config false;
      description
        "An unsigned integer, indicating the minimum latency, in
         nanoseconds, a frame can experience when transmitted from the
         underlying physical medium on the Port to a reception port
         connected via a LAN to the Port.";
      reference
        "Item h) in 6.8.4.7 of IEEE Std 802.1DD.";
    }
    leaf red-context-consistent {
      type boolean;
      config false;
      description
        "A Boolean indicating whether the Redundancy Context
         configuration in this Bridge is consistent with that in a
         neighboring station on the Port (TRUE) or not (FALSE). ";
      reference
        "Item i) in 6.8.4.7 of IEEE Std 802.1DD.";
    }
  }
  list ra-class-port-table {
    key "ra-class-id";
    description
      "Each table row contains a set of parameters that defines a
       single RA class";
    leaf ra-class-id {
      type uint8 {
        range "0 .. 255";
      }
      description
        "The RA class ID of the associated local RA class.";
      reference
        "Item b) in 6.8.4.8 of IEEE Std 802.1DD.";
    }
    leaf domain-boundary-status {
      type boolean;
      config false;
      description
        "A Boolean indicating whether the Port is a domain boundary
         port for the RA class (TRUE) or not (FALSE).";
      reference
        "Item c) in 6.8.4.8 of IEEE Std 802.1DD.";
    }
    leaf max-bandwidth {
      type uint32;
      description
        "An unsigned integer, indicating the maximum amount of
         bandwidth that can be allocated to the streams reserved in
         the RA class on the Port. The bandwidth value is represented
         as a percentage of the portTransmitRate value on that Port
         and expressed as a fixed-point number scaled by a factor of
         1,000,000; i.e., 100,000,000 (the maximum value)represents
         100%.";
      reference
        "Item d) in 6.8.4.8 of IEEE Std 802.1DD.";
    }
    leaf allocated-bandwidth {
      type uint32;
      config false;
      description
        "An unsigned integer, indicating the amount of bandwidth that has
         been allocated to the streams reserved in the RA class on the
         Port. The bandwidth value is represented as a percentage of the
         portTransmitRate value on that Port and expressed as a fixed-
         point number scaled by a factor of 1,000,000; i.e., 100,000,000
         (the maximum value) represents 100%.";
      reference
        "Item e) in 6.8.4.8 of IEEE Std 802.1DD.";
    }
    leaf max-last-hop-latency {
      type uint32;
      description
```

```
        "The value to be contained in the MaxLastHopLatency field of an
         RA Class Descriptor sub-TLV associated with the RA class in the
         RA attribute declared on the Port.";
      reference
        "Item f) in 6.8.4.8 of IEEE Std 802.1DD.";
    }
  }
  list rap-priority-regeneration-override {
    key "received-priority";
    description
      "Each table row contains a set of parameters for a received
       priority value that is associated with an RA class ";
    leaf received-priority {
      type dot1qtypes:priority-type;
      description
        "Received priority value.";
      reference
        "6.9.4 of IEEE Std 802.1Q";
    }
    leaf regenerated-priority {
      type dot1qtypes:priority-type;
      description
        "Priority regeneration value.";
      reference
        "6.9.4 of IEEE Std 802.1Q";
    }
  }
  list rap-talker-announce-registration {
    key "stream-id vid";
    config false;
    description
      "Each table row in the table associated with a Port corresponds
       to a registration of the Talker Announce attribute on that Port,
       and contains a set of parameters";
    uses rap-talker-announce;
  }
  list rap-talker-announce-declaration {
    key "stream-id vid";
    config false;
    description
      "Each table row in the Table associated with a Port corresponds
       to a declaration of the Talker Announce attribute on that Port,
       and contains the same set of parameters as in the Talker Announce
       Registration";
    uses rap-talker-announce;
  }
  list rap-listener-attach-registration {
    key "stream-id vid";
    config false;
    description
      "Each table row in the Table associated with a Port corresponds to
       a registration of the Listener Attach attribute on that Port and
       contains a set of parameters";
    uses rap-listener-attach;
    leaf reservation-age {
      type uint32;
      description
        "A 32-bit unsigned integer, indicating the time, in seconds,
         since a reservation associated with the Listener Attach
         registration was successfully made, and set to zero when the
         reservation is removed.";
      reference
        "Item e) in 6.8.4.3 of IEEE Std 802.1DD";
    }
    leaf ingress-status {
      type enumeration {
        enum not-propagated {
          value 0;
        }
        enum attach-ready {
          value 1;
        }
```

```
          enum attach-fail {
            value 2;
          }
          enum attach-partial-fail {
            value 3;
          }
        }
        description
          "The ingress status of the Listener Attach registration.";
        reference
          "item f) in 6.8.4.3 of IEEE Std 802.1DD";
      }
    }
    list rap-listener-attach-declaration {
      key "stream-id vid";
      config false;
      description
        "Each table row in the Table associated with a Port corresponds to
         a declaration of the Listener Attach attribute on that Port and
         contains the same set of parameters except the reservationAge
         parameter as in the Listener Attach Registration";
      uses rap-listener-attach;
    }
  }
}
```

Annex A

(normative)

PICS proforma—<RAP implementations>[11]

A.1 Introduction

The supplier of a protocol implementation that is claimed to conform to this standard shall complete the following Protocol Implementation Conformance Statement (PICS) proforma.

A completed PICS proforma is the PICS for the implementation in question. The PICS is a statement of which capabilities and options of the protocol have been implemented. The PICS can have a number of uses, including use

a) By the protocol implementer, as a checklist to reduce the risk of failure to conform to the standard through oversight.
b) By the supplier and acquirer—or potential acquirer—of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard PICS proforma.
c) By the user—or potential user—of the implementation, as a basis for initially checking the possibility of interworking with another implementation (note that, while interworking can never be guaranteed, failure to interwork can often be predicted from incompatible PICSs).
d) By a protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.

A.2 Abbreviations and special symbols

A.2.1 Status symbols

M	mandatory
O	optional
O.n	optional, but support of at least one of the group of options labeled by the same numeral n is required
X	prohibited
pred:	conditional-item symbol, including predicate identification: see A.3.4
¬	logical negation, applied to a conditional item’s predicate

A.2.2 General abbreviations

N/A	not applicable
PICS	Protocol Implementation Conformance Statement

[11] *Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

A.3 Instructions for completing the PICS proforma

A.3.1 General structure of the PICS proforma

The first part of the PICS proforma, implementation identification and protocol summary, is to be completed as indicated with the information necessary to identify fully both the supplier and the implementation.

The main part of the PICS proforma is a fixed-format questionnaire, divided into several subclauses, each containing a number of individual items. Answers to the questionnaire items are to be provided in the rightmost column, either by simply marking an answer to indicate a restricted choice (usually Yes or No) or by entering a value or a set or range of values. (Note that there are some items where two or more choices from a set of possible answers can apply; all relevant choices are to be marked.)

Each item is identified by an item reference in the first column. The second column contains the question to be answered; the third column records the status of the item—whether support is mandatory, optional, or conditional: see also A.3.4. The fourth column contains the reference or references to the material that specifies the item in the main body of this standard, and the fifth column provides the space for the answers.

A supplier may also provide (or be required to provide) further information, categorized as either Additional Information or Exception Information. When present, each kind of further information is to be provided in a further subclause of items labeled Ai or Xi, respectively, for cross-referencing purposes, where i is any unambiguous identification for the item (e.g., simply a numeral). There are no other restrictions on its format and presentation.

A completed PICS proforma, including any Additional Information and Exception Information, is the Protocol Implementation Conformation Statement for the implementation in question.

NOTE—Where an implementation is capable of being configured in more than one way, a single PICS may be able to describe all such configurations. However, the supplier has the choice of providing more than one PICS, each covering some subset of the implementation’s configuration capabilities, in case that makes for easier and clearer presentation of the information.

A.3.2 Additional information

Items of Additional Information allow a supplier to provide further information intended to assist the interpretation of the PICS. It is not intended or expected that a large quantity will be supplied, and a PICS can be considered complete without any such information. Examples might be an outline of the ways in which a (single) implementation can be set up to operate in a variety of environments and configurations, or information about aspects of the implementation that are outside the scope of this standard but that have a bearing on the answers to some items.

References to items of Additional Information may be entered next to any answer in the questionnaire and may be included in items of Exception Information.

A.3.3 Exception information

It may occasionally happen that a supplier will wish to answer an item with mandatory status (after any conditions have been applied) in a way that conflicts with the indicated requirement. No preprinted answer will be found in the Support column for this item. Instead, the supplier shall write the missing answer into

the Support column, together with an *Xi* reference to an item of Exception Information, and shall provide the appropriate rationale in the Exception item itself.

An implementation for which an Exception item is required in this way does not conform to this standard.

NOTE—A possible reason for the situation described previously is that a defect in this standard has been reported, a correction for which is expected to change the requirement not met by the implementation.

A.3.4 Conditional status

A.3.4.1 Conditional items

The PICS proforma contains a number of conditional items. These are items for which both the applicability of the item itself, and its status if it does apply—mandatory or optional—are dependent on whether certain other items are supported.

Where a group of items is subject to the same condition for applicability, a separate preliminary question about the condition appears at the head of the group, with an instruction to skip to a later point in the questionnaire if the "Not Applicable" answer is selected. Otherwise, individual conditional items are indicated by a conditional symbol in the Status column.

A conditional symbol is of the form "**pred:** S" where **pred** is a predicate as described in A.3.4.2 below, and S is a status symbol, M or O.

If the value of the predicate is true (see A.3.4.2), the conditional item is applicable, and its status is indicated by the status symbol following the predicate: The answer column is to be marked in the usual way. If the value of the predicate is false, the "Not Applicable" (N/A) answer is to be marked.

A.3.4.2 Predicates

A predicate is one of the following:

a) An item-reference for an item in the PICS proforma: The value of the predicate is true if the item is marked as supported and is false otherwise.

b) A predicate-name, for a predicate defined as a boolean expression constructed by combining item-references using the boolean operator OR: The value of the predicate is true if one or more of the items is marked as supported.

c) The logical negation symbol "¬" prefixed to an item-reference or predicate-name: The value of the predicate is true if the value of the predicate formed by omitting the "¬" symbol is false, and vice versa.

Each item whose reference is used in a predicate or predicate definition, or in a preliminary question for grouped conditional items, is indicated by an asterisk in the Item column.

A.4 PICS proforma—<RAP implementations>

A.4.1 Implementation identification

Supplier	
Contact point for queries about the PICS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification, e.g., name(s) and version(s) of machines and/or operating system names	
NOTE 1—Only the first three items are required for all implementations; other information may be completed as appropriate in meeting the requirement for full identification. NOTE 2—The terms “Name” and “Version” should be interpreted appropriately to correspond with a supplier’s terminology (e.g., Type, Series, Model).	

A.4.2 Protocol summary

Identification of protocol specification	IEEE Std 802.1DD, Resource Allocation Protocol			
Identification of amendments and corrigenda to the PICS proforma that have been completed as part of the PICS	Amd.	:	Corr.	: :
	Amd.	:	Corr.	:
Have any Exception items been required? (See A.3.3: the answer “Yes” means that the implementation is not conformant).	No [ ]		Yes [ ]	

Date of Statement	

A.5 Major capabilities

Item	Feature	Status	References	Support
RRI	Is a RAP instance implemented as a RAP Relay Instance?	O	5.4	Yes [ ] No [ ]
REI	Is a RAP instance implemented as a RAP Relay Instance?	O	5.5	Yes [ ] No [ ]
RMGI	Is a RAP instance implemented as a RAP/MSRP Gateway instance?	O	5.6	Yes [ ] No [ ]
RNB	RAP Native Bridge implemented?	O	5.7	Yes [ ] No [ ]
RCB	RAP Controlled Bridge implemented?	O	5.8	Yes [ ] No [ ]
RBP	RAP Bridge Proxy implemented?	O	5.9	Yes [ ] No [ ]
FRB	FRER-capable RAP Bridge implemented	O	5.10	Yes [ ] No [ ]
RNE	RAP Native end station implemented?	O	5.11	Yes [ ] No [ ]
RCE	RAP Controlled end station implemented?	O	5.12	Yes [ ] No [ ]
REP	RAP End Station Proxy implemented?	O	5.13	Yes [ ] No [ ]
FRT	FRER-capable RAP Talker implemented?	O	5.14	Yes [ ] No [ ]
FRL	FRER-capable RAP Listener implemented?	O	5.15	Yes [ ] No [ ]
RMG	RAP/MSRP Gateway implemented?	O	5.16	Yes [ ] No [ ]

A.6 RAP Relay instance capabilities

Item	Feature	Status	References	Support
RRI-1	Does the implementation support the RPSI and the RAP Participant state machine?	RRI:M	5.4 item a), 6.7	Yes [ ]
RRI-2	Does the implementation support the RAP Propagator state machine?	RRI:M	5.4 item b), 6.8	Yes [ ]
RRI-3	Does the implementation support the Application Information TLV?	RRI:M	5.4 item c), 6.4.4	Yes [ ]
RRI-4	Does the implementation support the RAP attributes and their TLV encodings?	RRI:M	5.4 item d), 6.5	Yes [ ]
RRI-5	Does the implementation support the RAP management?	RRI:M	5.4 item e), Clause 7	Yes [ ]

A.7 RAP End instance capabilities

Item	Feature	Status	References	Support
REI-1	Does the implementation support the RPSI and the RAP Participant state machine?	REI:M	5.5 item a), 6.7	Yes [ ]
REI-2	Does the implementation support the RESI and the RAP Endpoint procedures?	REI:M	5.5 item b), 6.6	Yes [ ]
REI-3	Does the implementation support the Application Information TLV?	REI:M	5.5 item c), 6.4.4	Yes [ ]
REI-4	Does the implementation support the RAP attributes and their TLV encodings?	REI:M	5.5 item d), 6.5	Yes [ ]
REI-5	Does the implementation support the RAP management?	REI:M	5.5 item e), Clause 7	Yes [ ]

A.8 RAP/MSRP Gateway instance capabilities

Item	Feature	Status	References	Support
RMGI-1	Does the implementation support the RPSI and the RAP Participant state machine?	RMGI:M	5.6 item a), 6.7	Yes [ ]
RMGI-2	Does the implementation support the RAP Propagator state machine?	RMGI:M	5.6 item b), 6.8	Yes [ ]
RMGI-3	Does the implementation support the RAP/MSRP Converter?	RMGI:M	5.6 item c), 6.9	Yes [ ]
RMGI-4	Does the implementation support the Application Information TLV?	RMGI:M	5.6 item d), 6.4.4	Yes [ ]
RMGI-5	Does the implementation support the RAP attributes and their TLV encodings?	RMGI:M	5.6 item e), 6.5	Yes [ ]
RMGI-6	Does the implementation support the RAP management?	RMGI:M	5.6 item f), Clause 7	Yes [ ]

A.9 RAP Native Bridge capabilities

Item	Feature	Status	References	Support
RNB-1	Does the implementation include a single conformant VLAN Bridge component?	RNB:M	5.7 item a), [Q]:5.4	Yes [ ]
RNB-2	Does the implementation conform to the required and optional behaviors for a Native relay system?	RNB:M	5.7 item b), [CS]:5.6, [CS]:5.7	Yes [ ]
RNB-3	Does the implementation support Edge Control Protocol on each Port and associated management entities?	RNB:M	5.7 item c), [Q]:43, [Q]:12.27	Yes [ ]
RNB-4	Does the implementation conform to the requirements for use of ECP as the LRP-DT data transport mechanism?	RNB:M	5.7 item d), [CS]:6.9.1, 6.4.3	Yes [ ]
RNB-5	Does the implementation include a single conformant RAP Relay instance?	RNB:M	5.7 item e), 5.4	Yes [ ]

A.10 RAP Controlled Bridge capabilities

Item	Feature	Status	References	Support
RCB-1	Does the implementation include a single conformant VLAN Bridge component?	RCB:M	5.8 item a), [Q]:5.4	Yes [ ]
RCB-2	Does the implementation conform to the optional behaviors for a Controlled system?	RCB:M	5.8 item b), [CS]:5.10	Yes [ ]

A.11 RAP Bridge Proxy capabilities

Item	Feature	Status	References	Support
RBP-1	Does the implementation conform to the required and optional behaviors for a Proxy system?	RBP:M	5.9 item a), [CS]:5.8, [CS]:5.9	Yes [ ]
RBP-2	Does the implementation include, for each RAP Controlled Bridge it proxies for, one conformant RAP Relay instance?	RBP:M	5.9 item b), 5.4	Yes [ ]

A.12 FRER-capable RAP Bridge capabilities

Item	Feature	Status	References	Support
FRB-1	Does the implementation conform to the requirements for either a RAP Native Bridge or a RAP Controlled Bridge?	FRB:M	5.10, 5.7, 5.8	Yes [ ]
FRB-2	Does the implementation conform to the required and optional behaviors for a FRER C-component?	FRB:M	5.10 item a), [CB]:5.15	Yes [ ]
FRB-3	Does the implementation support Active Destination MAC and VLAN Stream identification function?	FRB:M	5.10 item b), [CB]:6.6	Yes [ ]

A.13 RAP Native end station capabilities

Item	Feature	Status	References	Support
RNE-1	Does the implementation conform to the required behaviors for a Native end system?	RNE:M	5.11 item a), [CS]:5.4	Yes [ ]
RNE-2	Does the implementation support the operation of the Applicant state machine and the Registrar state machine?	RNE:M	5.11 item b), [CS]:8.3, [CS]:8.4	Yes [ ]
RNE-3	Does the implementation support Edge Control Protocol and associated management entities?	RNE:M	5.11 item c), [Q]:43, [Q]:12.27	Yes [ ]
RNE-4	Does the implementation conform to the requirements for use of ECP as the LRP-DT data transport mechanism?	RNE:M	5.11 item d), [CS]:6.9.1, 6.4.3	Yes [ ]
RNE-5	Does the implementation include a single conformant RAP End instance?	RNE:M	5.11 item e), 5.5	Yes [ ]

A.14 RAP Controlled end station capabilities

Item	Feature	Status	References	Support
RCE-1	Does the implementation conform to the optional behaviors for a Controlled system?	RCE:M	5.12 item a), [CS]:5.10	Yes [ ]

A.15 RAP End Station Proxy capabilities

Item	Feature	Status	References	Support
REP-1	Does the implementation conform to the required and optional behaviors for a Proxy system?	REP:M	5.13 item a), [CS]:5.8, [CS]:5.9	Yes [ ]
REP-2	Does the implementation include, for each RAP Controlled end station it proxies for, one conformant RAP End instance?	REP:M	5.13 item b), 5.5	Yes [ ]

A.16 FRER-capable RAP Talker capabilities

Item	Feature	Status	References	Support
FRT-1	Does the implementation conform to the requirements for either a RAP Native end station or a RAP Controlled end station?	FRT:M	5.14, 5.11, 5.12	Yes [ ]
FRT-2	Does the implementation conform to the required, recommended and optional behaviors for a Talker end system?	FRT:M	5.14 item a), [CB]:5.6, [CB]:5.7, [CB]:5.8	Yes [ ]

A.17 FRER-capable RAP Listener capabilities

Item	Feature	Status	References	Support
FRL-1	Does the implementation conform to the requirements for either a RAP Native end station or a RAP Controlled end station?	FRL:M	5.15, 5.11, 5.12	Yes [ ]
FRL-2	Does the implementation conform to the required, recommended and optional behaviors for a Talker end system?	FRL:M	5.15 item a), [CB]:5.9, [CB]:5.10, [CB]:5.11	Yes [ ]

A.18 RAP/MSRP Gateway capabilities

Item	Feature	Status	References	Support
RMG-1	Does the implementation include a single conformant VLAN Bridge component?	RMG:M	5.16 item a), [Q]:5.4	Yes [ ]
RMG-2	Does the implementation include a single conformant RAP/MSRP Gateway instance?	RMG:M	5.16 item b), 5.6	Yes [ ]
RMG-3	On each Port that supports the operation of RAP, dose the implementation conform to the required and optional behaviors for a Native relay system?	RMG:M	5.16 item c).1), [CS]:5.6, [CS]:5.7	Yes [ ]
RMG-4	On each Port that supports the operation of RAP, dose the implementation support Edge Control Protocol and associated management entities?	RMG:M	5.16 item c).2), [Q]:43, [Q]:12.27	Yes [ ]
RMG-5	On each Port that supports the operation of RAP, dose the implementation conform to the requirements for use of ECP as the LRP-DT data transport mechanism?	RMG:M	5.16 item c).3), [CS]:6.9.1, 6.4.3	Yes [ ]
RMG-6	On each Port that supports the operation of MSRP, dose the implementation include a conformant MSRP implementation for an end station?	RMG:M	5.16 item d), [Q]:5.18.3	Yes [ ]

Annex B

(informative)

Resource allocation examples

This Annex provides some examples of resource allocation using the Resource Allocation Protocol (RAP) specified in this standard. These examples are chosen only for the purpose of illustrating the operation of resource allocation under different scenarios, and are not intended to constraint the range of applicability of RAP in terms of all of the possible usage scenarios.

Note that, unless otherwise stated, the examples in this Annex are based on the following assumptions:

a) In Talker Announce propagation (6.3.4.1), no further conditions, other than those related to the VLAN Context rule as defined in 6.3.4.1.1, that would prevent a Talker Announce registration from being propagated to other Port(s), are met.
b) Resource allocation is successful without encountering any problems that would cause Talker Announcement or Listener Attachment to fail (except the example in B.5)

B.1 Single-path stream example

This subclause gives an example of resource allocation for transmission of a stream along a single path between the Talker and each Listener. Figure B-1 shows the VLAN topology configured in the network for use as a VLAN Context in the resource allocation for the stream.

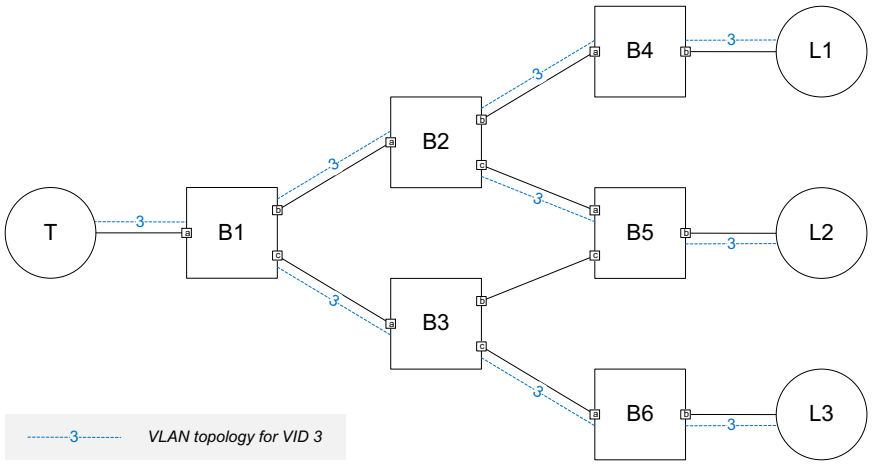


Figure B-1—Single-path stream example: topology and VLAN configuration

Figure B-2 illustrates the propagation of a Single-Context Talker Announcement [item a) in 6.3.4.1.1] initiated by the Talker within the VLAN Context. Note that the Talker Announcement does not go through the link between B3 and B5, as this link is not part of the VLAN Context in use.

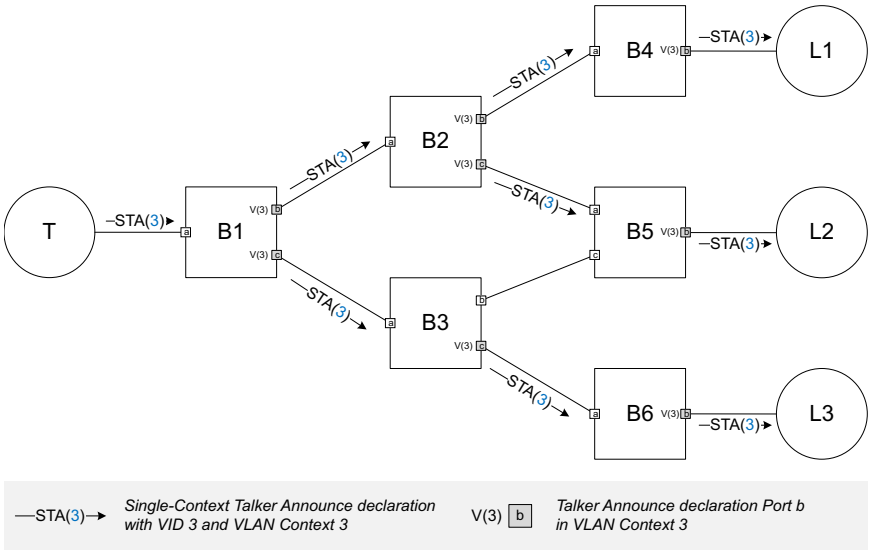


Figure B-2—Single-path stream example: Talker Announcement

As illustrated in Figure B-3, three Listeners participate in the Single-Context Listener Attachment [item a) in 6.3.5.1], which is propagated along the path previously traversed by the associated Talker Announcement in reversed direction and is subject to Listener Attach merge (6.3.5.3) on both Port B2.a and Port B1.a.

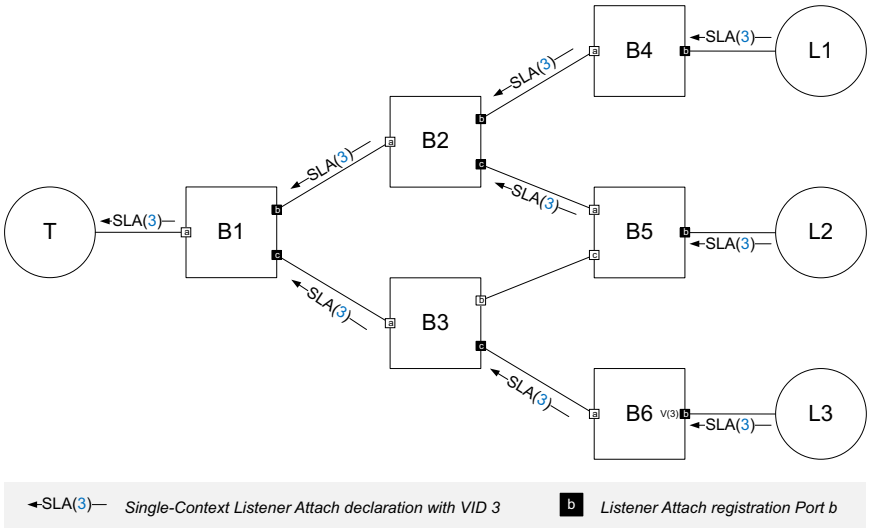


Figure B-3—Single-path stream example: Listener Attachment

Figure B-4 shows the single path established for transmission of the stream from the Talker to each Listener after successful resource allocation.

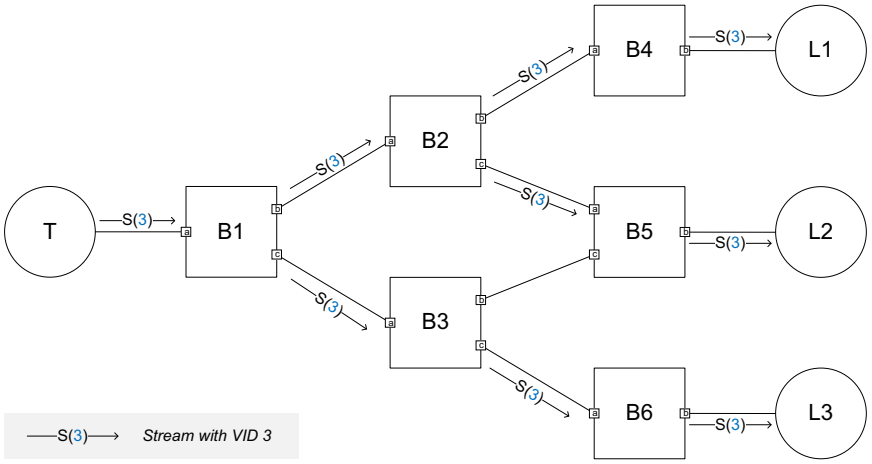


Figure B-4—Single-path stream example: established stream

B.2 Multi-path stream example 1

This subclause gives an example of resource allocation for transmission of a Compound Stream in a network analogous to the one described in [CB]:C.1. As shown in Figure B-5, three VLAN topologies representing an instance of redundant trees constitute a Redundancy Context used in the resource allocation. In this example, FRER functionality is provided only in three FRER-capable end stations.

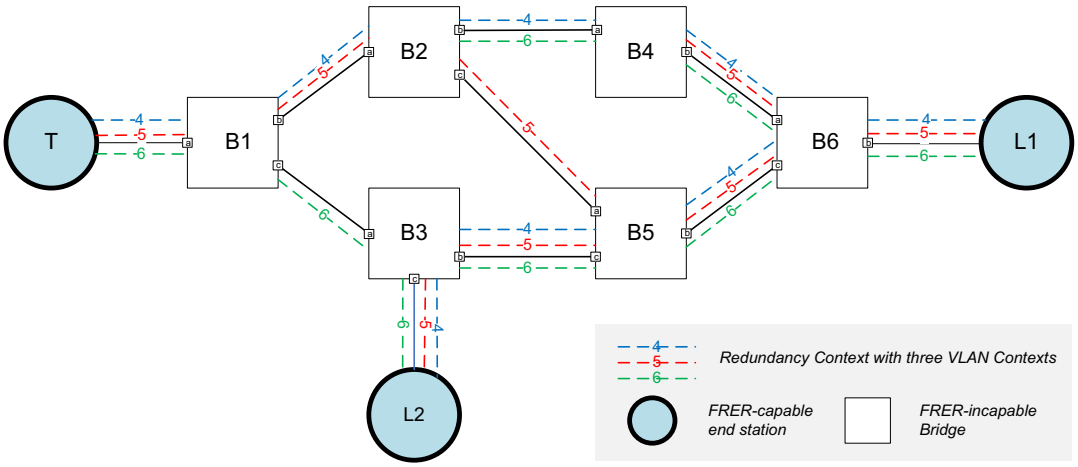


Figure B-5—Multi-path stream example 1: topology and VLAN configuration

As illustrated in Figure B-6, the FRER-capable Talker initiates a Multi-Context Talker Announcement [item b) in 6.3.4.1.1] with three separate Talker Announce declarations, each using a different VID and VLAN Context in the Redundancy Context to represent a Member Stream split from the Compound Stream. Since there are no FRER-capable Bridges in the network, each of these Talker Announce declarations is propagated and processed by the Bridges in the network independently.

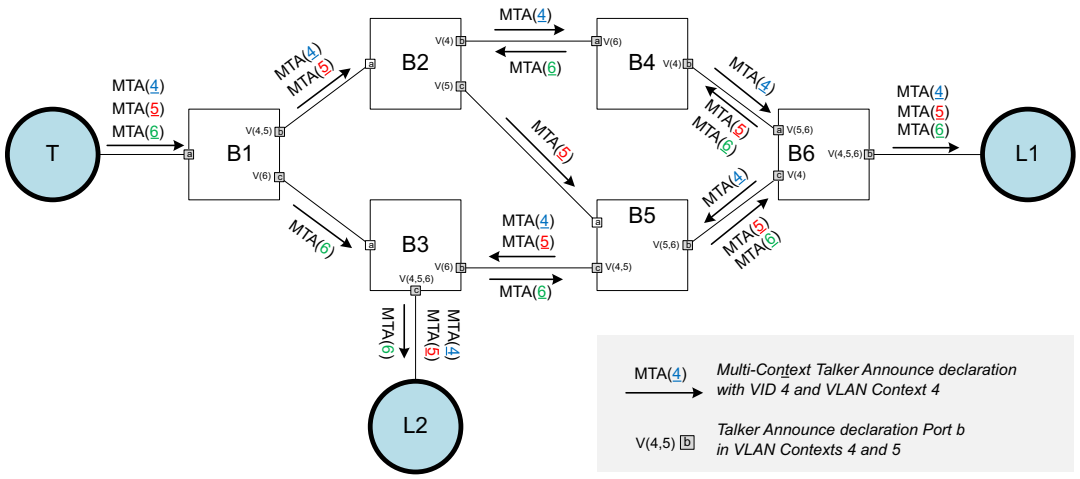


Figure B-6—Multi-path stream example 1: Talker Announcement

Figure B-7 illustrates the Multi-Context Listener Attachment associated with the Multi-Context Talker Announcement shown in Figure B-6. Each Listener makes three Listener Attach declarations, each with the same VID and VLAN Context as received in one of the three Talker Announce registrations. Each Listener Attach declaration made by a Listener is propagated along the path formed by the set of Bridge Ports that contain the associated Taker Announce registration. Listener Attach merge (6.3.5.3) occurs on the Bridge Ports B5.a and B6.a where two Listener Attach registrations propagated from the different Listeners in the same VLAN Context are merged into a joint Listener Attach declaration, indicating the need for the Bridge to multicast the Member Stream received on that Port.

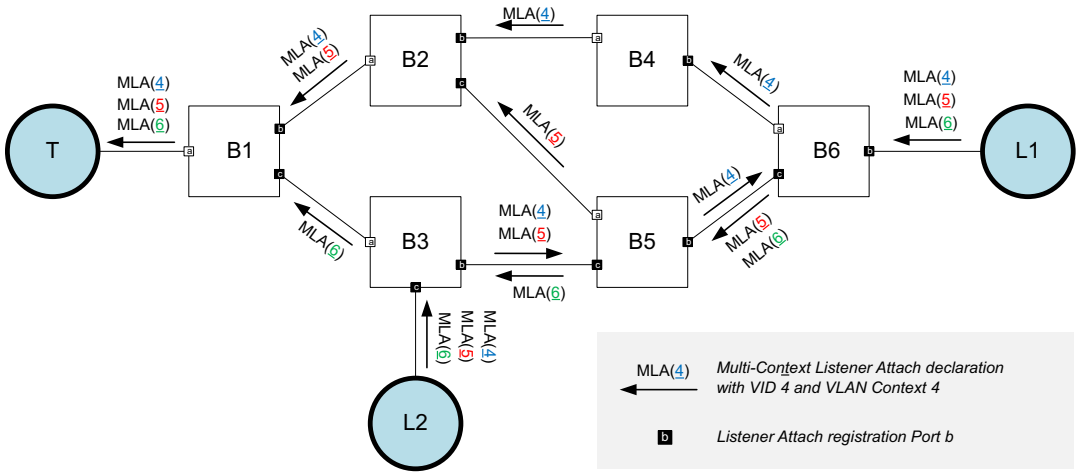


Figure B-7—Multi-path stream example 1: Listener Attachment

Figure B-8 shows the paths established and the FRER functions configured for redundant transmission of the Compound Stream with three Member Streams after successful resource allocation.

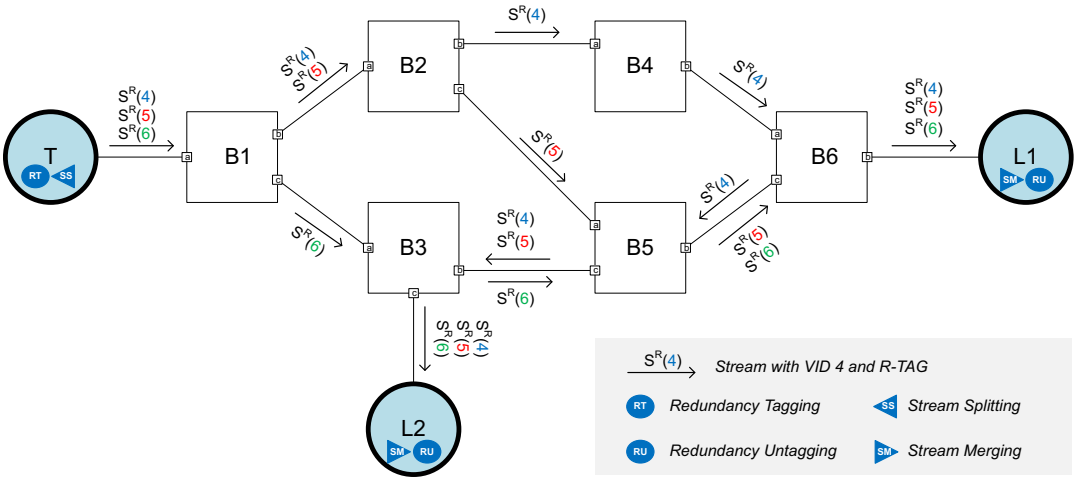


Figure B-8—Multi-path stream example 1: established stream

B.3 Multi-path stream example 2

This subclause gives an example of resource allocation for transmission of a Compound Stream in a network analogous to the one described in [CB]:C.2. As shown in Figure B-9, this example has the same physical topology as that in Figure B-5, but different placement of FRER-capable stations and VLAN configuration. As both the Talker and Listener are not FRER-capable, three FRER-capable Bridges B1, B5 and B6 are placed at the edge of the network as proxies to offer FRER functionality to the end stations. Additionally, another FRER-capable Bridge B2 is placed within the network to split the VLAN topologies, in conjunction with B1, into three disjoint paths. Note that the reason for using a different VLAN configuration in this example is to ensure the VLAN membership configuration in each FRER-capable Bridge where stream splitting is expected, e.g., B1 and B2, meet the condition described in b).) of 6.3.9.4.

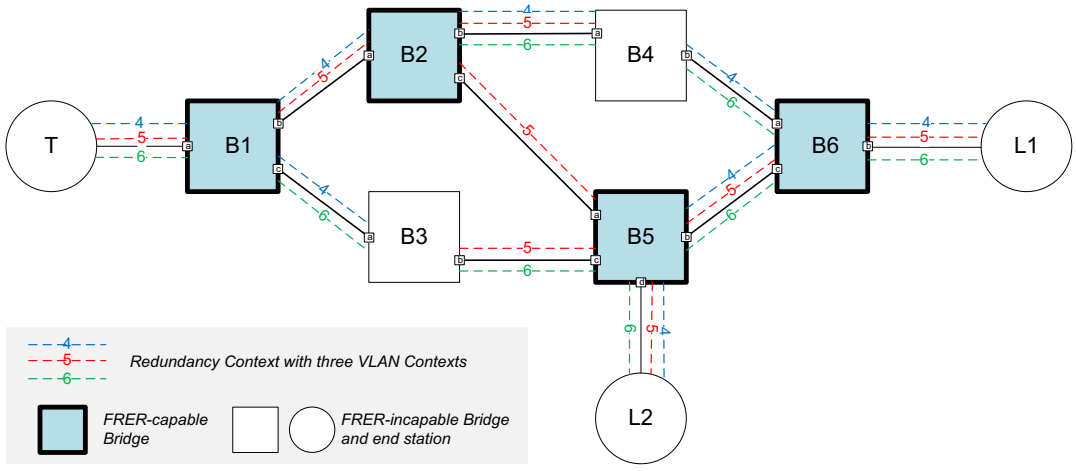


Figure B-9—Multi-path stream example 2: topology and VLAN configuration

As illustrated in Figure B-10, the Talker initiates a Multi-Context Talker Announcement by making a Talker Announce declaration for all of the three VLAN Contexts and with RTagStatus (6.5.3.9.1) set FALSE. The RTagStatus flag of the Talker Announcement is set TRUE by B1, which is responsible for redundancy tagging in accordance with the rule described in a) of 6.3.9.3. Both B1 and B2 are responsible for applying stream splitting to the stream, as all the conditions described in b) of 6.3.9.4 are met. The Talker Announcement is subject to Talker Announce merge (6.3.4.3) on the Bridge Ports B5.b, B5.d and B6.b, each corresponding to a potential stream merging port for the stream, in accordance with the rule described in a) of 6.3.9.5. In addition, the RTagStatus flag of the Talker Announcement is set FALSE on B5.d and B6.b, as the conditions for redundancy untagging are met on these two Ports, in accordance with the rule described in b) of 6.3.9.3.

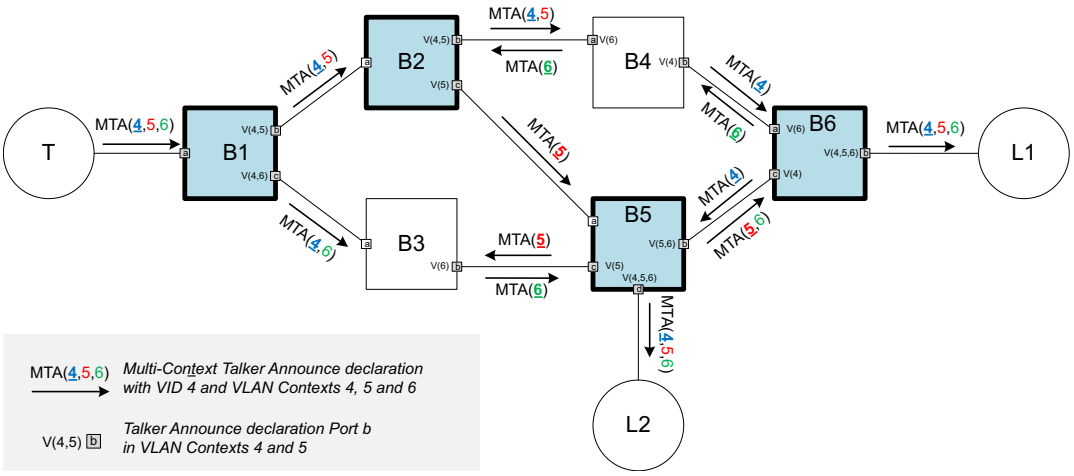


Figure B-10—Multi-path stream example 2: Talker Announcement

Figure B-11 illustrates the Multi-Context Listener Attachment associated with the Multi-Context Talker Announcement shown in Figure B-10. Each Listener makes a single Listener Attach declaration with the same VLAN Contexts as received in the Talker Announce registration and a desired VID chosen from among these VLAN Context IDs. Similar to that in Figure B-7, the Listener Attach declaration on both B5.a and B6.a is merged from the two Listener Attach registrations propagated from different Listeners in the same VLAN Context. Additionally in this example, Listener Attach merge (6.3.5.3) occurs on the Bridge Ports B2.a and B1.a where two Listener Attach registrations containing the split VLAN Contexts are merged into a single Listener Attach declaration with joint VLAN Contexts, as a reserve operation of splitting the Talk Announcement previously performed on the same Port. Note that the Listener Attach declaration on B2.a and B1.a takes a VID that is contained in the VLAN Context(s) of both of the Listener Attach registrations merging into that Listener Attach declaration, because the multicast forwarding mechanism used in stream splitting by FRER-capable Bridges requires stream transmission Ports to be in the member set for a common VID.

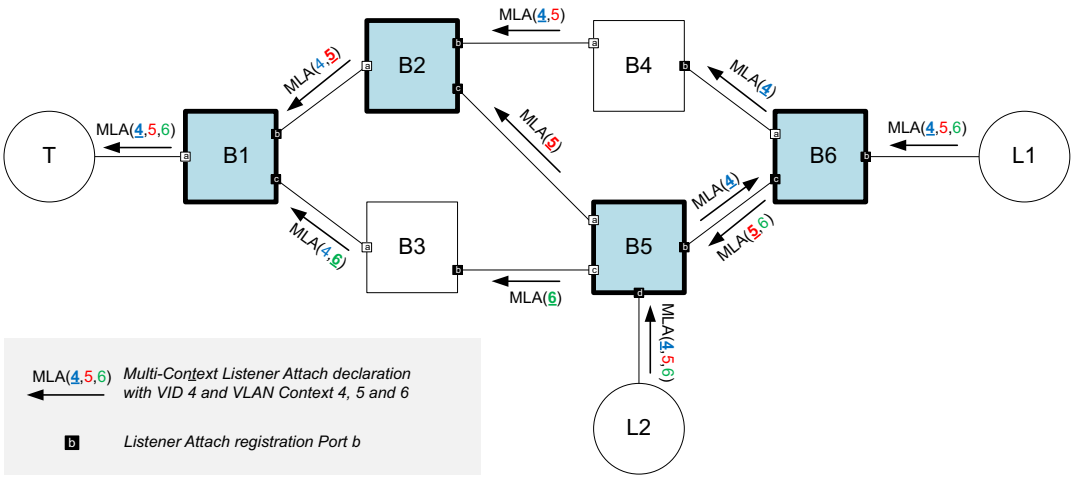


Figure B-11—Multi-path stream example 2: Listener Attachment

Figure B-12 shows the paths established and the FRER functions configured for redundant transmission of the Compound Stream with three Member Streams after successful resource allocation.

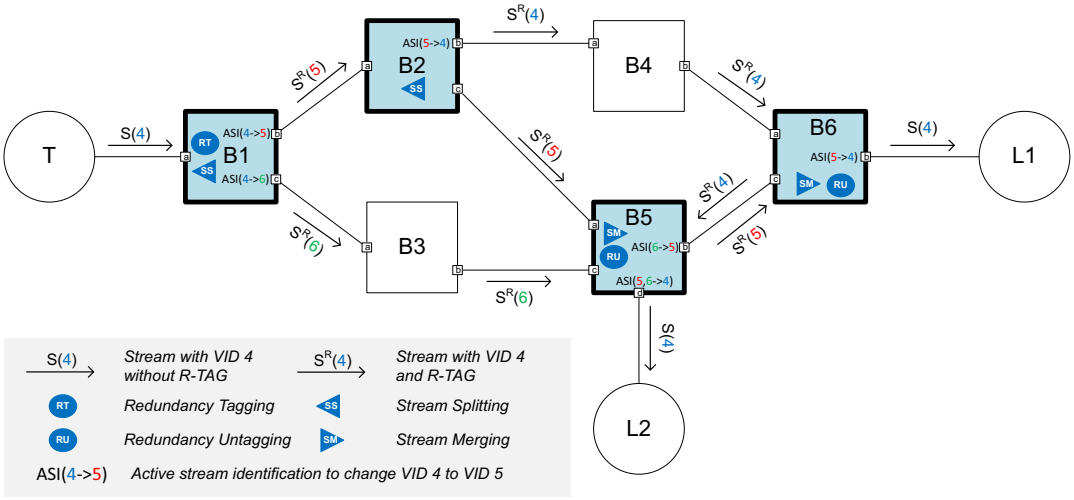


Figure B-12—Multi-path stream example 2: established stream

B.4 Multi-path stream example 3

This subclause gives an example of resource allocation for transmission of a Compound Stream in a network analogous to the one described in [CB]:C.3. As shown in Figure B-13, FRER functionality is provided in each end station and some of the Bridges. Four FRER-capable Bridges are connected as a ring, each acting as both a stream splitting point and a stream merging point, to provide protection against multiple simultaneous failures. A Redundancy Context with two VLAN topologies is configured in the network. In order to meet the stream splitting condition described in b).) of 6.3.9.4, each FRER-capable Bridge has one VLAN whose member set includes all the Bridge Ports. As a consequence, each of those Bridge Ports marked as VLAN topology termination point needs be excluded from the member set for the indicated

VLAN and to have ingress filtering ([Q]:8.6.2) enabled for that VLAN, in order to ensure loop-free Talker Announce propagation.

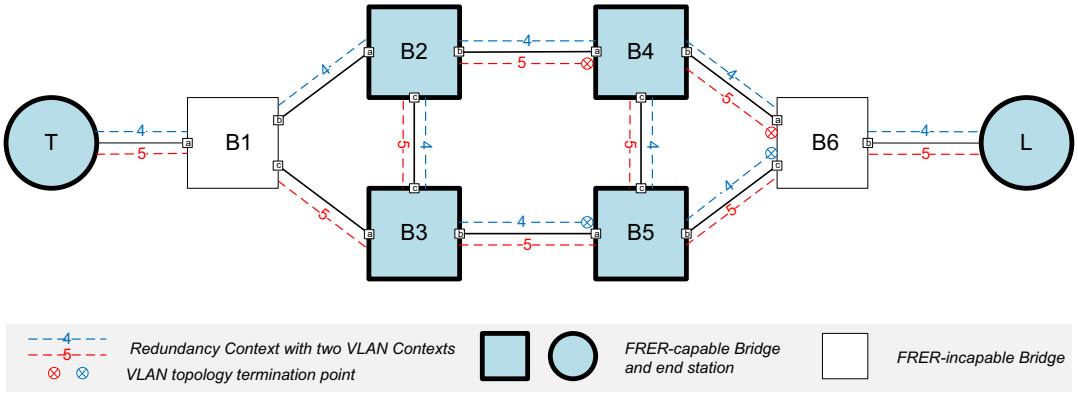


Figure B-13—Multi-path stream example 3: topology and VLAN configuration

As illustrated in Figure B-14, two Talker Announce declarations initiated by the FRER-capable Talker, each with a distinct VLAN Context, are propagated by B1 separately. According to the VLAN configuration, each FRER-capable Bridge propagates the Talker Announce registration on Port a with a single VLAN Context to the other two Ports, and meanwhile merges on Port b two Talker Announce registrations propagated from the other two Ports into a joint Talker Announce declaration with both VLAN contexts. Note that on each Bridge Port that is excluded from the member set for one VLAN Context, ingress blocking (6.3.4.1.3) ensures that the Talker Announce registration received with two VLAN Contexts is propagated only with the other VLAN Context for which the Port in the member set.

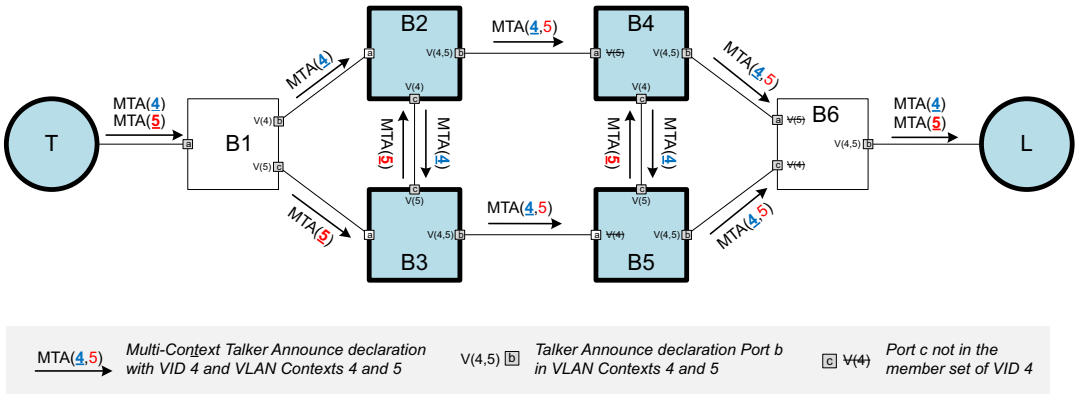


Figure B-14—Multi-path stream example 3: Talker Announcement

Figure B-15 illustrates the Multi-Context Listener Attachment associated with the Multi-Context Talker Announcement shown in Figure B-14. The FRER-capable Listener makes two Listener Attach declarations, each with a different VLAN Context. Listener Attach merge occurs on Port a of each FRER-capable Bridge, resulting in a joint Listener Attach declaration with the same VLAN Context as that contained in the

associated Talker Announce registration on the same Port. Finally, B1 passes two Listener Attach declarations, each with a different VLAN Context, to the Talker.

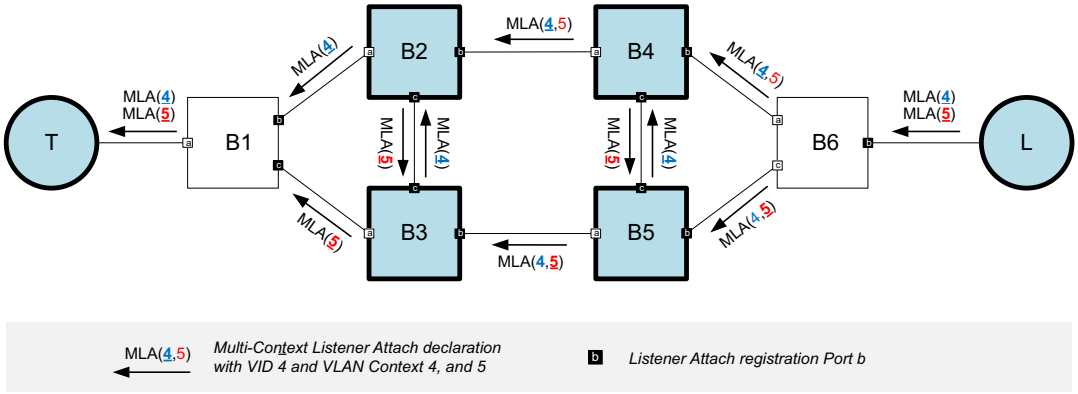


Figure B-15—Multi-path stream example 3: Listener Attachment

Figure B-16 shows the paths established and the required FRER functions configured for redundant transmission of the Compound Stream with two Member Streams after successful resource allocation

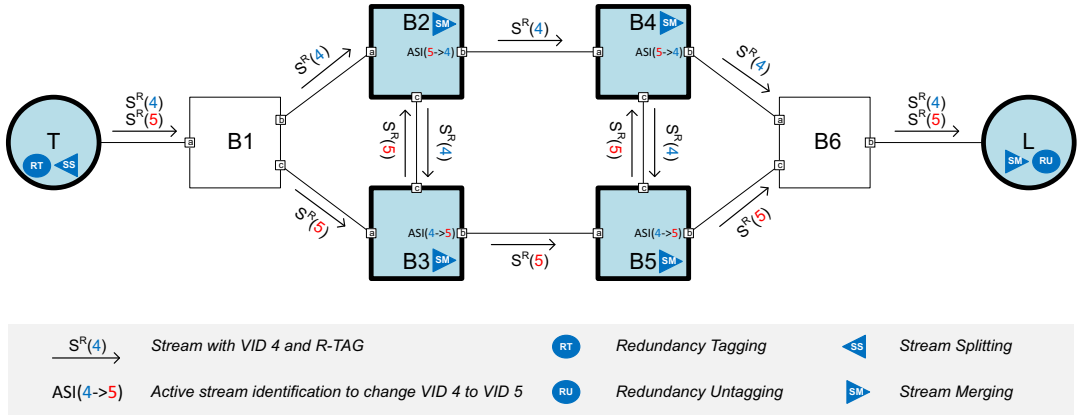


Figure B-16—Multi-path stream example 3: established stream

B.5 Example of status notification and failure diagnosis

This subclause uses the example network of B.4 to illustrate the status notification and failure diagnosis mechanisms in resource allocation. As shown in Figure B-17, it is assumed that the Multi-Context Talker Announcement encounters resource problems on three Bridge Ports.

The problem on Port B1.b causes the Talker Announce declaration on that Port to change to the Announce Fail status [item b) in 6.3.4.2] and to attach the failure information about the problem on that Port to VLAN Context 4. The failure information generated on B1.b remains attached to VLAN Context 4 in the subsequent propagation of the Talker Announcement across the network toward the Listener, regardless of the situations encountered on any downstream Port. Similarly, the problems on B3.b and B4.b lead to Announce Fail in the Talker Announce declarations on those Ports. The failure information of VLAN Context 5 is generated on B3.b, and is later merged along with that of VLAN Context 4 to the Talker Announce declarations on B4.b and B5.b. The reason for the Announce Success status on both B2.b and B5.b is that the Talker Announce declaration merges one Talker Announce registration propagated as

Announce Success, meaning that a reservation can be made for a stream on a merged path as long as at least one of the paths merging into that merged path is available for reservation.

In the end, the Listener receives two Talker Announce declarations, which indicate that at least one path in the network can be reserved for transmission of the Compound Stream as indicated by the Announce Success status of one Talker Announce registration, even though there are problems detected at the locations associated with both VLAN topologies as indicated in the supplied failure information.

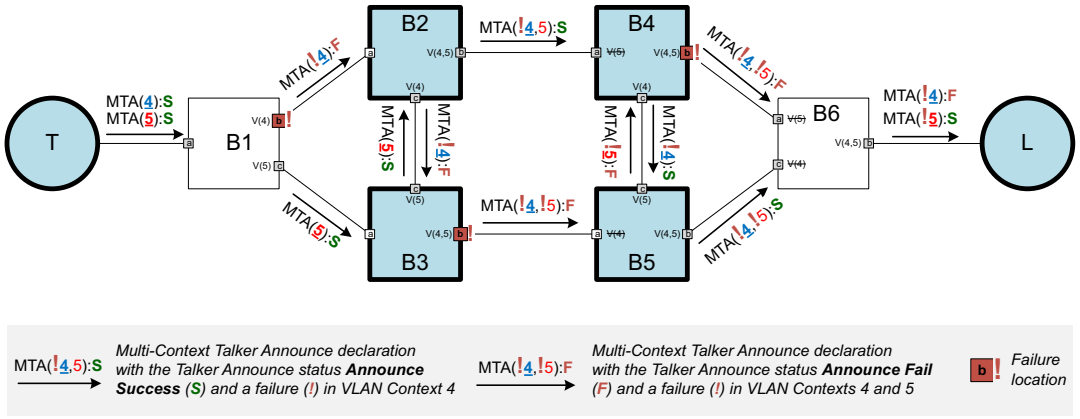


Figure B-17—Example Talker Announce status and failure information

Figure B-18 illustrates the Multi-Context Listener Attachment associated with the Talker Announcement shown in Figure B-17. The Listener makes two Listener Attach declarations with Attach Ready contained in the Listener Attach status (6.3.5.4), indicating it is ready to receive both Member Streams. As the Listener Attachment is propagated along each stream path back to the Talker, a corresponding reservation is made on each Port for a Listener Attach registration with the Attach Ready status and its associated Talker Announce declaration with the Announce Success status. A Listener Attach registration for which a reservation cannot be made is propagated as Attach Fail. On each Bridge Port where Listener Attach merge occurs in this example, i.e., Port a or each FRER-capable Bridge, the Listener Attach declaration is made with Attach Ready because one of the two Listener Attach registrations being merged on that Port is propagated as Attach Ready. Note that, on those Ports with a terminated VLAN, such as B6.a and B6.c, the Listener Attach declaration still contains the terminated VLAN (e.g., VLAN Context 5 on B6.a) in order to match the VLAN Contexts contained in the associated Talker Announce registration on the same Port, but needs to set the corresponding Path status to Unresponsive Path to indicate that Listener Attach declaration does not originate from that VLAN Context.

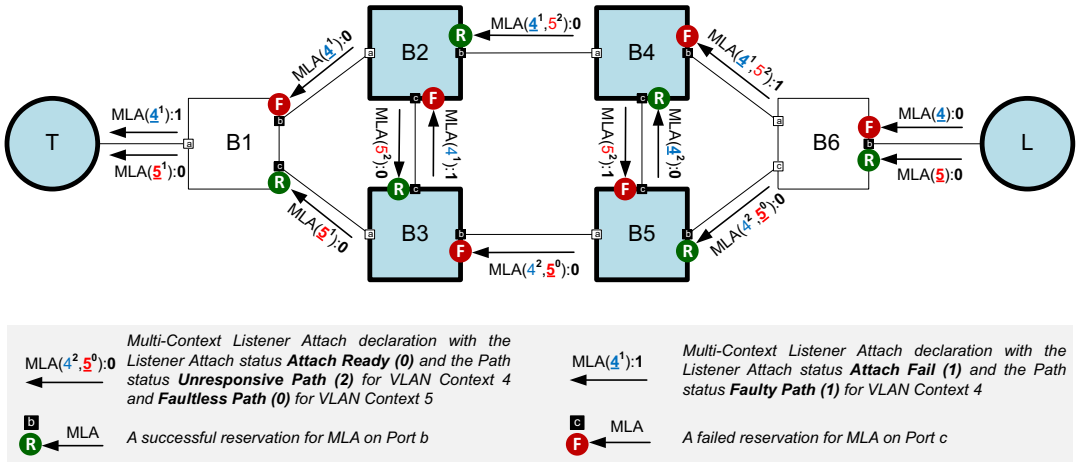


Figure B-18—Example Listener Attach status and Path status

Figure B-19 shows the single path established and the FRER functions configured for the Compound Stream in the case of three problems in the network.

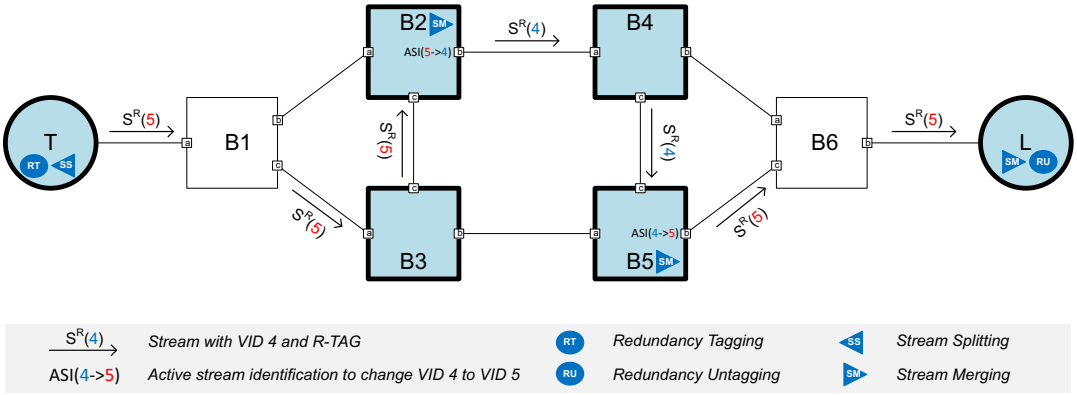


Figure B-19—Example partially established Compound Stream

Annex C

(informative)

Bibliography

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[12] IEEE publications are available from The Institute of Electrical and Electronics Engineers (https://standards.ieee.org/).
[13] The IEEE standards or products referred to in this annex are trademarks of The Institute of Electrical and Electronics Engineers, Inc.

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